Supporting Information

Shared architecture, disease-context-dependent direction: a harmonised cross-disease computational analysis of translation-related transcriptional programs in hepatocellular carcinoma and heart failure

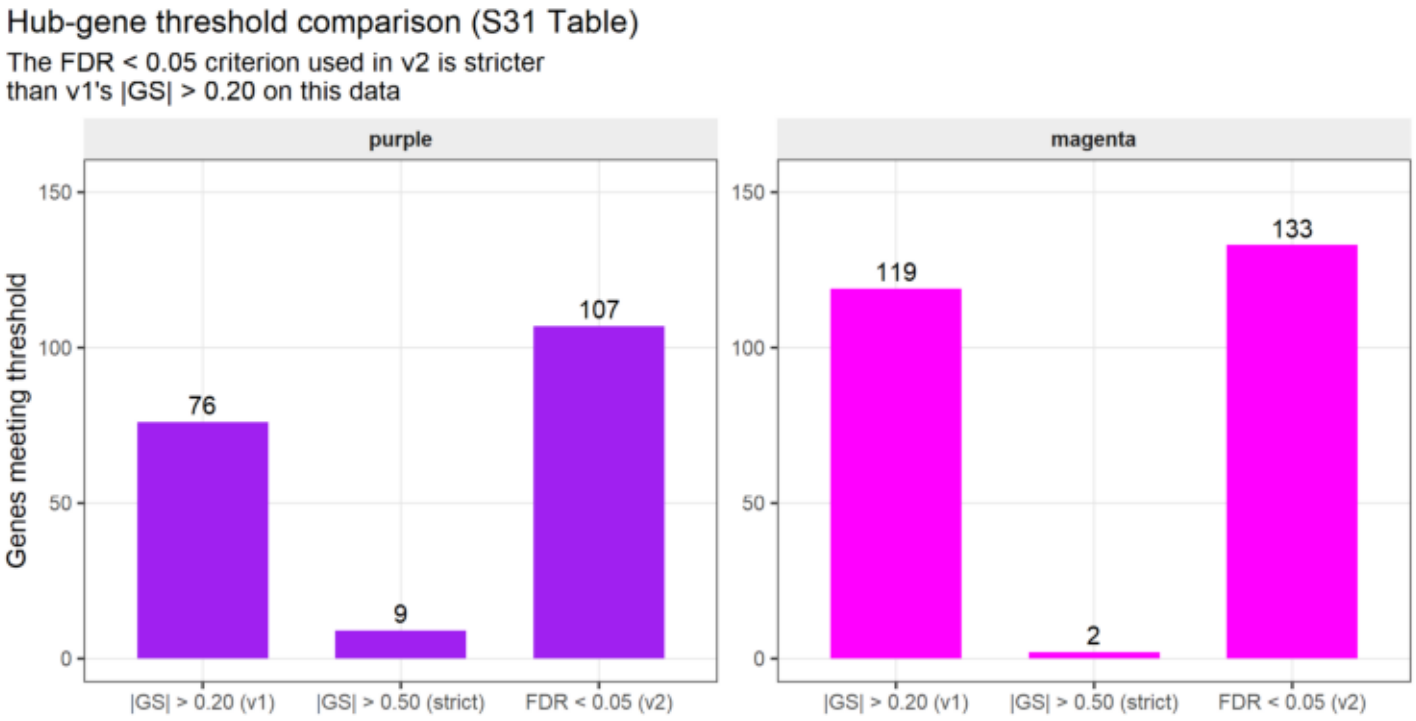
5 supplementary figures · 44 supplementary tables

Every figure and table in this supplement is generated by the frozen pipeline under Code/Revision_v2/ from public data. The configuration hash recorded in S1 Table identifies the exact settings used.

Every table is also supplied complete as a machine-readable CSV alongside this document, and the CSV, not this PDF, is the authoritative full text. Tables longer than 200 rows are printed as a bounded preview of the first 60 and state their true row count and the CSV's filename. Long cell values are shortened with "..." in the printed preview only; the CSV holds them in full.

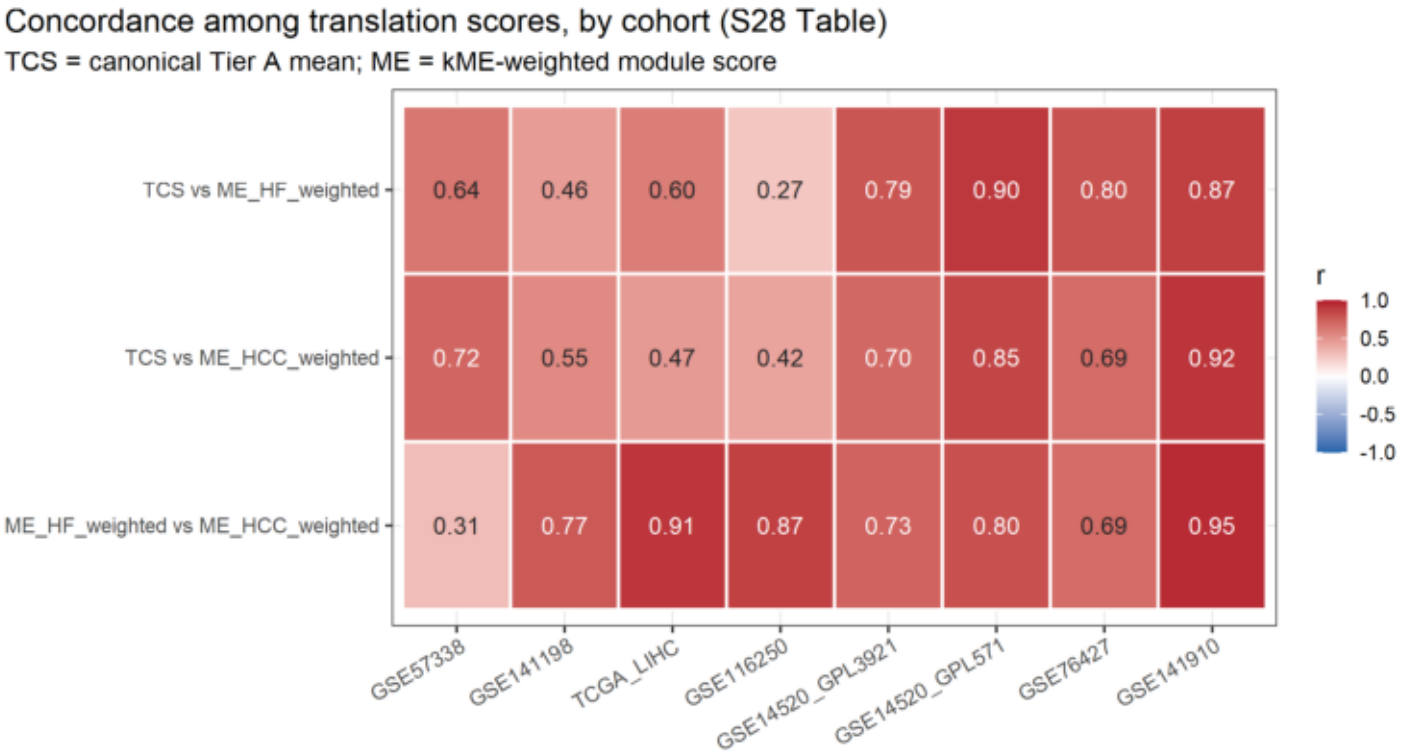
S1 Fig.

Hub-gene threshold comparison for the two designated translation modules (S31 Table). Gene counts meeting v1's |GS| > 0.20, a stricter |GS| > 0.50, and the FDR < 0.05 criterion adopted in v2. The criteria are nested on this data — every gene passing |GS| > 0.50 also passes FDR < 0.05 — and the counts are 76, 9 and 107 for HF purple and 119, 2 and 133 for HCC magenta. |GS| > 0.50 is therefore the strictest of the three and FDR < 0.05 the most permissive; the FDR criterion was adopted for its calibration to sample size rather than for added stringency.



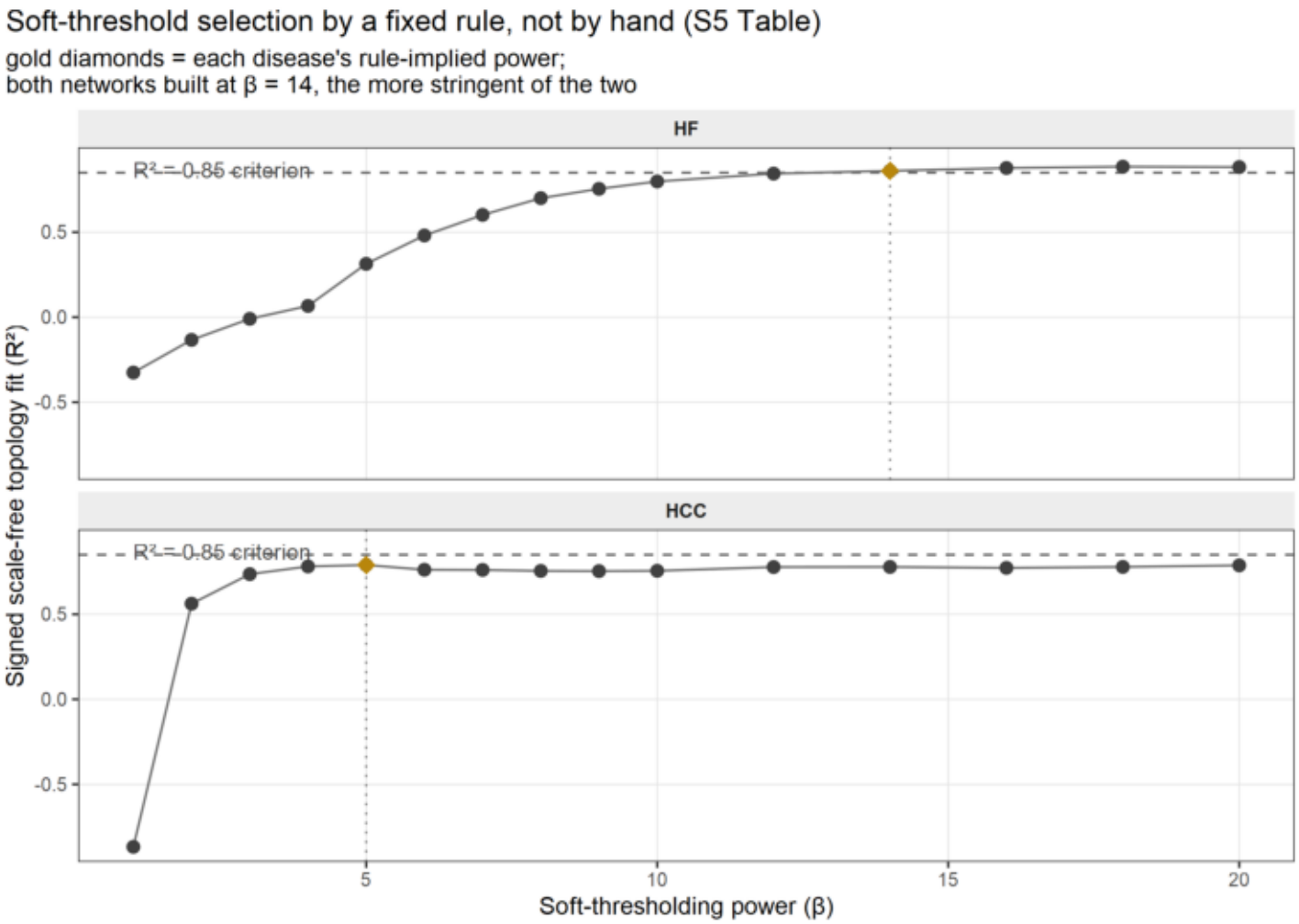
S2 Fig.

Cross-cohort effect-size concordance for the translation score. Module eigengene scores for the HF purple and HCC magenta modules, and the canonical translation score, computed in all eight cohorts and compared pairwise.



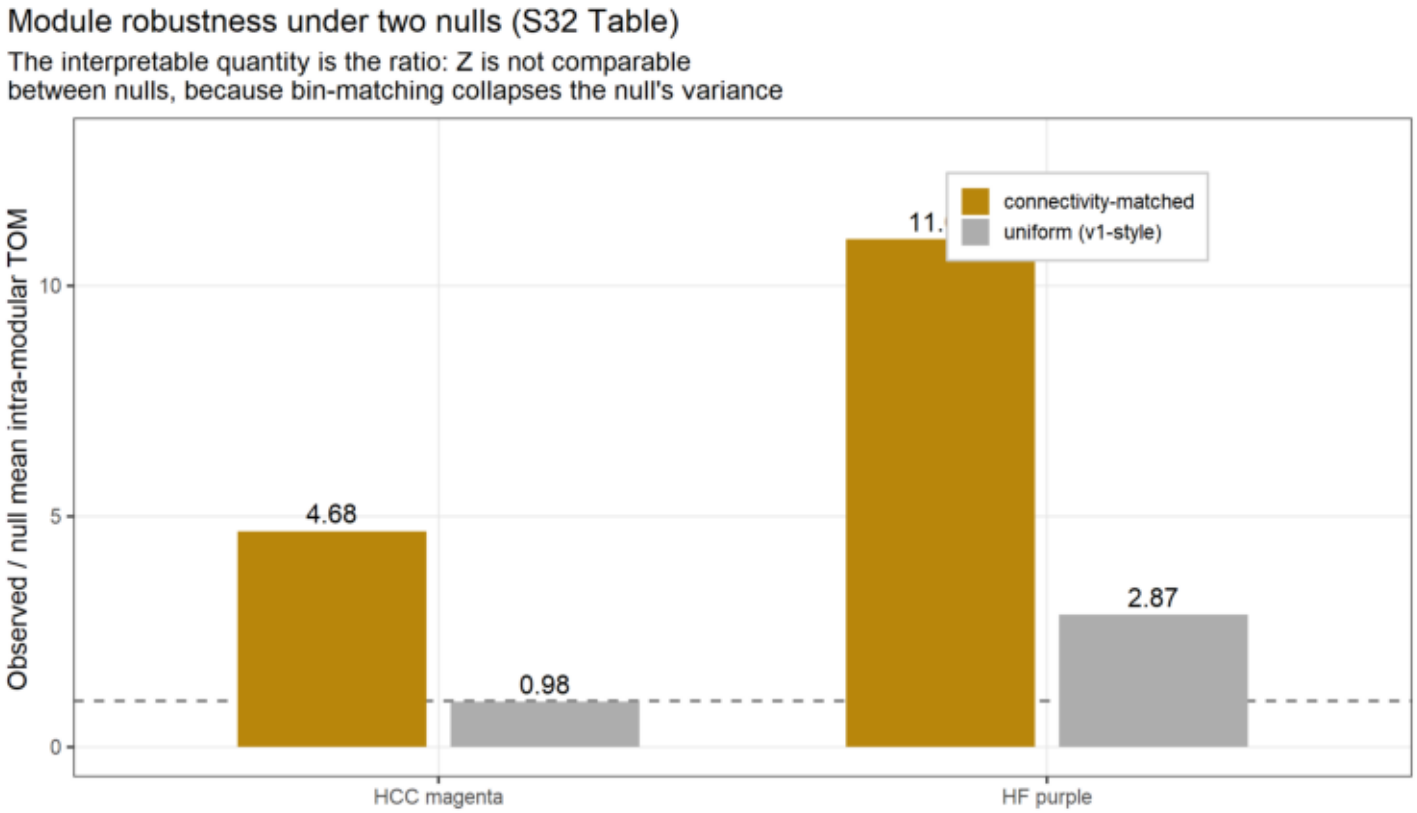
S3 Fig.

Soft-threshold selection. Scale-free topology fit and mean connectivity versus candidate power for the HF network (top) and the HCC network (bottom), both built on disease samples only on the common universe U. The rule-implied power is marked for each disease and the common power β = 14 used for both networks is marked in both panels.



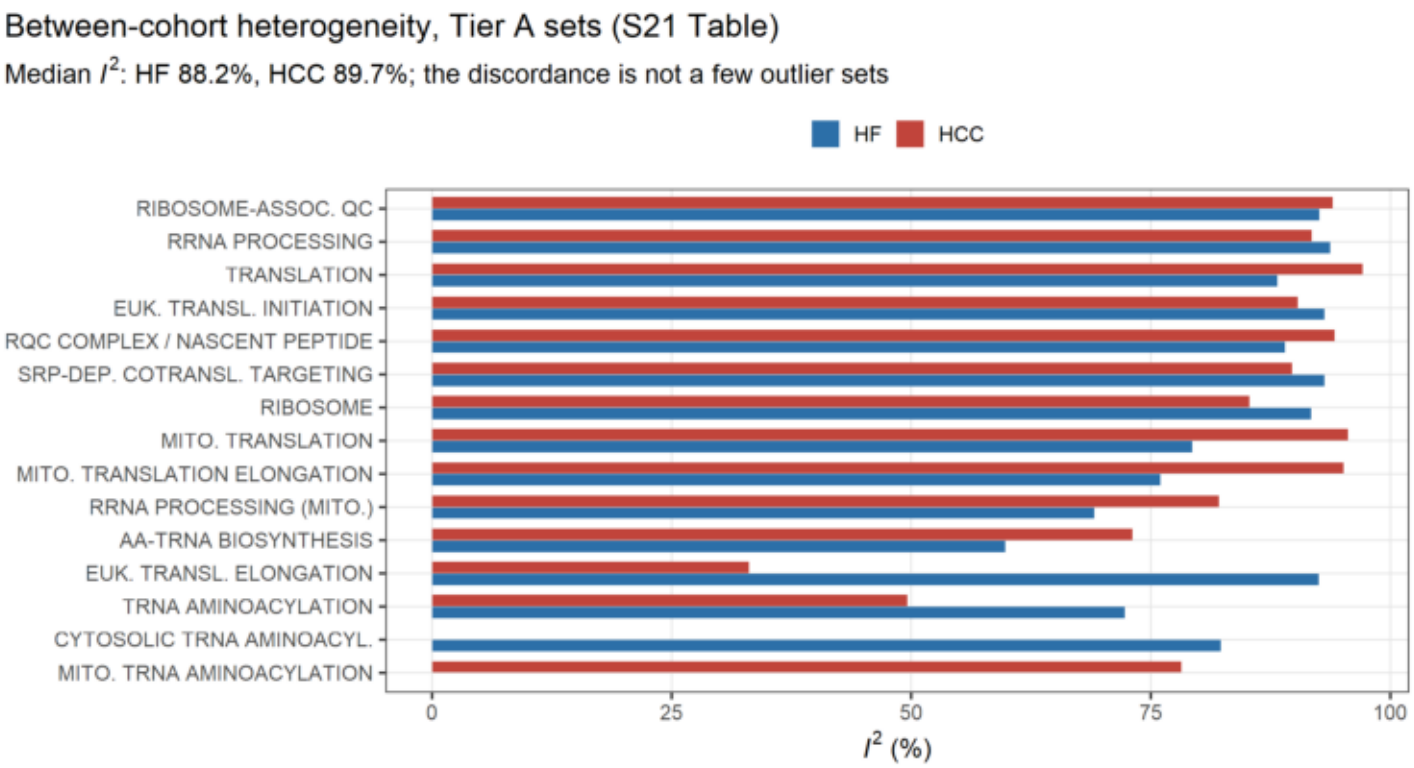
S4 Fig.

Module-internal robustness of the designated translation modules. Permutation distributions of the mean intra-modular topological overlap under the connectivity-matched null (primary) and the uniform null (secondary), for HF purple and HCC magenta; jackknife distributions of mean intra-modular connectivity.



S5 Fig.

Between-cohort heterogeneity. I^2 and Cochran's Q P-values for the 15 Tier A sets within each disease, with pooled effect sizes and confidence intervals.



S1 Table

Gene universe loss chain. Every filtering stage from platform feature to inclusion in the common universe U, separately for GPL11532 (HF array) and GSE141198 (HCC RNA-seq), reporting the number of features surviving at each step. The configuration hash for the frozen settings used throughout is recorded in the header.

platform	stage	n
GPL11532 (HF array)	features on platform (matrix rows)	33297
GPL11532 (HF array)	detected by expression filter	16632
GPL11532 (HF array)	mapped to a gene symbol	13229
GPL11532 (HF array)	unique symbols after collapsing	12090
GPL11532 (HF array)	present in common universe U	10213
GSE141198 (HCC RNA-seq)	features on platform (symbol-...	15036
GSE141198 (HCC RNA-seq)	detected by expression filter	13178
GSE141198 (HCC RNA-seq)	mapped to a gene symbol	13178
GSE141198 (HCC RNA-seq)	unique symbols after collapsing	13178
GSE141198 (HCC RNA-seq)	present in common universe U	10213

S2 Table (page 1 of 4)

A priori gene set definitions and coverage. Every scored gene set with its tier assignment (A, B, or C) and its measured gene count in each platform and in the common universe.

tier	set	n.genes.in msigdb	n.in.HF platform	n.in.HCC platform	n.in.common.U	frac.in.U
TierA	KEGG_RIBOSOME	88	81	85	80	0.909
TierA	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	41	40	41	40	0.976
TierA	REACTOME_TRANSLATION	414	385	406	382	0.923
TierA	...	140	132	136	130	0.929
TierA	...	115	108	110	105	0.913
TierA	REACTOME_TRNA_AMINOACYLATION	50	49	50	49	0.98
TierA	...	30	29	30	29	0.967
TierA	REACTOME_RRNA_PROCESSING	248	226	240	223	0.899
TierA	...	133	126	129	124	0.932
TierA	...	167	160	163	158	0.946
TierA	...	108	103	105	101	0.935
TierA	REACTOME_MITOCHONDRIAL_TRANSLATION	124	104	121	104	0.839
TierA	...	99	92	96	92	0.929
TierA	...	23	23	23	23	1
TierA	...	20	18	18	18	0.9
TierB	KEGG_RIBOSOME	88	81	85	80	0.909
TierB	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	41	40	41	40	0.976
TierB	REACTOME_TRANSLATION	414	385	406	382	0.923
TierB	...	140	132	136	130	0.929
TierB	...	115	108	110	105	0.913
TierB	REACTOME_TRNA_AMINOACYLATION	50	49	50	49	0.98
TierB	...	30	29	30	29	0.967

S2 Table (page 2 of 4)

A priori gene set definitions and coverage. Every scored gene set with its tier assignment (A, B, or C) and its measured gene count in each platform and in the common universe.

tier	set	n.genes.in msigdb	n.in.HF platform	n.in.HCC platform	n.in.common.U	frac.in.U
TierB	REACTOME_RRNA_PROCESSING	248	226	240	223	0.899
TierB	...	133	126	129	124	0.932
TierB	...	167	160	163	158	0.946
TierB	...	108	103	105	101	0.935
TierB	REACTOME_MITOCHONDRIAL_TRANSLATION	124	104	121	104	0.839
TierB	...	99	92	96	92	0.929
TierB	...	23	23	23	23	1
TierB	...	20	18	18	18	0.9
TierB	REACTOME_TRNA_PROCESSING	136	111	131	110	0.809
TierB	...	76	63	75	62	0.816
TierB	...	49	39	47	39	0.796
TierB	...	80	78	78	77	0.963
TierB	...	11	11	11	11	1
TierB	...	7	5	5	5	0.714
TierC_v1	...	47	27	25	20	0.426
TierC_v1	HALLMARK_MYC_TARGETS_V1	200	188	199	187	0.935
TierC_v1	...	30	29	30	29	0.967
TierC_v1	REACTOME_TRNA_AMINOACYLATION	50	49	50	49	0.98
TierC_v1	...	108	103	105	101	0.935
TierC_v1	...	167	160	163	158	0.946
TierC_v1	...	103	99	99	96	0.932
TierC_v1	...	95	84	90	83	0.874

S2 Table (page 3 of 4)

A priori gene set definitions and coverage. Every scored gene set with its tier assignment (A, B, or C) and its measured gene count in each platform and in the common universe.

tier	set	n.genes.in msigdb	n.in.HF platform	n.in.HCC platform	n.in.common.U	frac.in.U
TierC_v1	...	23	23	23	23	1
TierC_v1	HALLMARK_MYC_TARGETS_V2	58	53	56	51	0.879
TierC_v1	...	115	108	110	105	0.913
TierC_v1	...	133	126	129	124	0.932
TierC_v1	REACTOME_RRNA_PROCESSING	248	226	240	223	0.899
TierC_v1	...	71	63	67	60	0.845
TierC_v1	...	40	38	39	37	0.925
TierC_v1	...	64	61	60	59	0.922
TierC_v1	...	140	132	136	130	0.929
TierC_v1	REACTOME_TRANSLATION	414	385	406	382	0.923
TierC_v1	...	54	52	53	52	0.963
TierC_v1	HALLMARK_MTORC1_SIGNALING	200	180	195	176	0.88
TierC_v1	...	23	23	23	23	1
TierC_v1	...	15	14	13	12	0.8
TierC_v1	...	15	9	10	9	0.6
TierC_v1	...	108	45	49	38	0.352
TierC_v1	...	67	48	52	47	0.701
TierC_v1	...	10	9	8	8	0.8
TierC_v1	...	30	30	28	28	0.933
TierC_v1	...	10	10	10	10	1
TierC_v1	...	99	92	96	92	0.929
TierC_v1	...	20	18	18	18	0.9

S2 Table (page 4 of 4)

A priori gene set definitions and coverage. Every scored gene set with its tier assignment (A, B, or C) and its measured gene count in each platform and in the common universe.

tier	set	n.genes.in msigdb	n.in.HF platform	n.in.HCC platform	n.in.common.U	frac.in.U
TierC_v1	...	9	8	8	8	0.889
TierC_v1	REACTOME_MITOCHONDRIAL_TRANSLATION	124	104	121	104	0.839
TierC_v1	...	14	9	9	9	0.643

S3 Table (page 1 of 3)

Tier A gene loss. Members of the 15 Tier A sets that are absent from the common universe U, listed per set with the reason for exclusion, so that the effect of the universe definition on the translation sets is auditable.

gene	in.HF platform	in.HCC platform	in.common.U	loss.reason
FAU	True	True	True	retained
MRPL13	False	True	False	lost on HF array (GPL11532)
RPL10	True	True	True	retained
RPL10A	True	True	True	retained
RPL10L	False	False	False	absent from both platforms
RPL11	True	True	True	retained
RPL12	True	True	True	retained
RPL13	True	True	True	retained
RPL13A	True	True	True	retained
RPL14	True	True	True	retained
RPL15	True	True	True	retained
RPL17	True	True	True	retained
RPL18	True	True	True	retained
RPL18A	True	True	True	retained
RPL19	True	True	True	retained
RPL21	True	True	True	retained
RPL22	False	True	False	lost on HF array (GPL11532)
RPL22L1	False	True	False	lost on HF array (GPL11532)
RPL23	True	True	True	retained
RPL23A	True	True	True	retained
RPL24	True	True	True	retained
RPL26	True	True	True	retained

S3 Table (page 2 of 3)

Tier A gene loss. Members of the 15 Tier A sets that are absent from the common universe U, listed per set with the reason for exclusion, so that the effect of the universe definition on the translation sets is auditable.

gene	in.HF platform	in.HCC platform	in.common.U	loss.reason
RPL26L1	True	True	True	retained
RPL27	True	True	True	retained
RPL27A	True	True	True	retained
RPL28	True	True	True	retained
RPL29	True	True	True	retained
RPL3	True	True	True	retained
RPL30	True	True	True	retained
RPL31	True	True	True	retained
RPL32	True	True	True	retained
RPL34	False	True	False	lost on HF array (GPL11532)
RPL35	True	True	True	retained
RPL35A	True	True	True	retained
RPL36	True	True	True	retained
RPL36A	True	True	True	retained
RPL36AL	True	True	True	retained
RPL37	True	True	True	retained
RPL37A	True	True	True	retained
RPL38	True	True	True	retained
RPL39	True	True	True	retained
RPL3L	True	False	False	lost in HCC RNA-seq
RPL4	True	True	True	retained
RPL41	True	True	True	retained

S3 Table (page 3 of 3)

Tier A gene loss. Members of the 15 Tier A sets that are absent from the common universe U, listed per set with the reason for exclusion, so that the effect of the universe definition on the translation sets is auditable.

Showing the first 60 of 497 rows. The complete table is provided as the machine-readable file Table_S3_tierA_gene_loss.csv.

gene	in.HF platform	in.HCC platform	in.common.U	loss.reason
RPL5	True	True	True	retained
RPL6	True	True	True	retained
RPL7	True	True	True	retained
RPL7A	True	True	True	retained
RPL8	True	True	True	retained
RPL9	True	True	True	retained
RPLP0	True	True	True	retained
RPLP1	True	True	True	retained
RPLP2	True	True	True	retained
RPS10	True	True	True	retained
RPS11	True	True	True	retained
RPS12	True	True	True	retained
RPS13	True	True	True	retained
RPS15	True	True	True	retained
RPS15A	True	True	True	retained
RPS16	True	True	True	retained

S4 Table (page 1 of 3)

Module inventory. Every module detected in each harmonised network, with its gene count, percentage of the universe, and the shared soft-threshold power.

disease	module	n.genes	power	pct.of.genes
HF	grey	2210	14	21.6
HF	turquoise	2132	14	20.9
HF	blue	1525	14	14.9
HF	brown	1516	14	14.8
HF	yellow	496	14	4.9
HF	green	425	14	4.2
HF	red	189	14	1.9
HF	black	184	14	1.8
HF	pink	173	14	1.7
HF	magenta	167	14	1.6
HF	purple	153	14	1.5
HF	greenyellow	150	14	1.5
HF	tan	146	14	1.4
HF	salmon	105	14	1
HF	cyan	103	14	1
HF	midnightblue	93	14	0.9
HF	lightcyan	73	14	0.7
HF	grey60	70	14	0.7
HF	lightgreen	70	14	0.7
HF	lightyellow	58	14	0.6
HF	royalblue	52	14	0.5
HF	darkred	43	14	0.4

S4 Table (page 2 of 3)

Module inventory. Every module detected in each harmonised network, with its gene count, percentage of the universe, and the shared soft-threshold power.

disease	module	n.genes	power	pct.of.genes
HF	darkgreen	41	14	0.4
HF	darkturquoise	39	14	0.4
HCC	turquoise	3771	14	36.9
HCC	blue	1948	14	19.1
HCC	grey	1445	14	14.1
HCC	brown	389	14	3.8
HCC	yellow	347	14	3.4
HCC	green	323	14	3.2
HCC	red	302	14	3
HCC	black	255	14	2.5
HCC	pink	254	14	2.5
HCC	magenta	146	14	1.4
HCC	purple	145	14	1.4
HCC	greenyellow	130	14	1.3
HCC	tan	117	14	1.1
HCC	salmon	110	14	1.1
HCC	cyan	99	14	1
HCC	midnightblue	96	14	0.9
HCC	lightcyan	90	14	0.9
HCC	grey60	78	14	0.8
HCC	lightgreen	68	14	0.7
HCC	lightyellow	57	14	0.6

S4 Table (page 3 of 3)

Module inventory. Every module detected in each harmonised network, with its gene count, percentage of the universe, and the shared soft-threshold power.

disease	module	n.genes	power	pct.of.genes
HCC	royalblue	43	14	0.4

S5 Table (page 1 of 2)

Soft-threshold fit. Signed scale-free topology fit at each candidate power for both diseases, with chosen_own, the power that disease's own soft-threshold rule selects, and chosen_common, the power both networks are actually built at. The rule takes the lowest power reaching the R^2 ≥ 0.85 cut and falls back to the best-fitting power when no candidate does, so chosen_own flags the rule's pick rather than the criterion: no HCC candidate reaches the cut, and its pick is the fallback (β = 5, R^2 = 0.790), while the common power is the stricter heart-failure pick (β = 14).

disease	power	fit	chosen.own	chosen.common
HF	1	-0.3254	False	False
HF	2	-0.1333	False	False
HF	3	-0.009481	False	False
HF	4	0.06626	False	False
HF	5	0.3137	False	False
HF	6	0.4806	False	False
HF	7	0.6016	False	False
HF	8	0.7002	False	False
HF	9	0.7548	False	False
HF	10	0.7983	False	False
HF	12	0.8443	False	False
HF	14	0.8607	True	True
HF	16	0.8774	False	False
HF	18	0.8852	False	False
HF	20	0.8828	False	False
HCC	1	-0.8678	False	False
HCC	2	0.5622	False	False
HCC	3	0.736	False	False
HCC	4	0.7813	False	False
HCC	5	0.7898	True	False
HCC	6	0.7618	False	False
HCC	7	0.7606	False	False

S5 Table (page 2 of 2)

Soft-threshold fit. Signed scale-free topology fit at each candidate power for both diseases, with chosen_own, the power that disease's own soft-threshold rule selects, and chosen_common, the power both networks are actually built at. The rule takes the lowest power reaching the R^2 ≥ 0.85 cut and falls back to the best-fitting power when no candidate does, so chosen_own flags the rule's pick rather than the criterion: no HCC candidate reaches the cut, and its pick is the fallback (β = 5, R^2 = 0.790), while the common power is the stricter heart-failure pick (β = 14).

disease	power	fit	chosen.own	chosen.common
HCC	8	0.755	False	False
HCC	9	0.7539	False	False
HCC	10	0.7555	False	False
HCC	12	0.7777	False	False
HCC	14	0.7783	False	True
HCC	16	0.7728	False	False
HCC	18	0.7784	False	False
HCC	20	0.7875	False	False

S6 Table (page 1 of 4)

Module–clinical-trait relationships in the HF network. Correlation of every HF module eigengene with ischaemic aetiology, sex, and age across the 177 failing ventricles. Disease status itself is not a trait here: the network was built on disease samples only, so all samples are failing myocardium and there is no within-network control group to correlate against. No module survives FDR correction against any of the three traits.

module	trait	r	p	fdr
MEpurple	ICM	-0.1002	0.1845	0.641
MElightgreen	ICM	0.1765	0.01876	0.5953
MEtan	ICM	0.2033	0.006651	0.4589
MEgreen	ICM	0.09785	0.1951	0.641
MEblack	ICM	0.1659	0.02729	0.5953
MEmidnightblue	ICM	0.02303	0.7609	0.8669
MEroyalblue	ICM	0.06373	0.3994	0.8266
MEbrown	ICM	0.06817	0.3673	0.792
MEmagenta	ICM	0.1081	0.1523	0.641
MEdarkgreen	ICM	-0.1003	0.1843	0.641
MEred	ICM	-0.1485	0.04854	0.5953
MEsalmon	ICM	-0.07541	0.3185	0.7325
MEblue	ICM	-0.03795	0.616	0.8571
MElightcyan	ICM	0.1316	0.08071	0.5953
MEpink	ICM	0.1198	0.1121	0.5953
MEturquoise	ICM	0.03653	0.6293	0.8571
MEcyan	ICM	0.0315	0.6773	0.8571
MEdarkturquoise	ICM	-0.08175	0.2794	0.7185
MEgrey60	ICM	-0.02961	0.6956	0.8571
MEyellow	ICM	-0.03363	0.6568	0.8571
MEdarkred	ICM	-0.0297	0.6947	0.8571
MEgreenyellow	ICM	-0.1202	0.1111	0.5953

S6 Table (page 2 of 4)

Module–clinical-trait relationships in the HF network. Correlation of every HF module eigengene with ischaemic aetiology, sex, and age across the 177 failing ventricles. Disease status itself is not a trait here: the network was built on disease samples only, so all samples are failing myocardium and there is no within-network control group to correlate against. No module survives FDR correction against any of the three traits.

module	trait	r	p	fdr
MElightyellow	ICM	-0.01234	0.8705	0.8964
MEpurple	male	-0.06267	0.4073	0.8266
MElightgreen	male	-0.1085	0.1505	0.641
MEtan	male	0.0271	0.7203	0.8575
MEgreen	male	-0.02705	0.7208	0.8575
MEblack	male	-0.08714	0.2488	0.7152
MEmidnightblue	male	0.01433	0.8499	0.8896
MEroyalblue	male	-0.05149	0.4961	0.8571
MEbrown	male	-0.04969	0.5113	0.8571
MEmagenta	male	0.03608	0.6335	0.8571
MEdarkgreen	male	-0.04476	0.5542	0.8571
MEred	male	0.04017	0.5955	0.8571
MEsalmon	male	0.0002034	0.9979	0.9979
MEblue	male	0.02463	0.7449	0.8669
MElightcyan	male	0.04728	0.532	0.8571
MEpink	male	-0.01961	0.7956	0.8735
MEturquoise	male	0.01596	0.833	0.8896
MEcyan	male	0.02249	0.7664	0.8669
MEdarkturquoise	male	0.04498	0.5522	0.8571
MEgrey60	male	0.04235	0.5757	0.8571
MEyellow	male	0.05305	0.4831	0.8571
MEdarkred	male	0.03873	0.6088	0.8571

S6 Table (page 3 of 4)

Module–clinical-trait relationships in the HF network. Correlation of every HF module eigengene with ischaemic aetiology, sex, and age across the 177 failing ventricles. Disease status itself is not a trait here: the network was built on disease samples only, so all samples are failing myocardium and there is no within-network control group to correlate against. No module survives FDR correction against any of the three traits.

module	trait	r	p	fdr
MEgreenyellow	male	0.0874	0.2474	0.7152
MElightyellow	male	0.07802	0.302	0.7185
MEpurple	age	-0.07914	0.2951	0.7185
MElightgreen	age	0.09953	0.1875	0.641
MEtan	age	0.001922	0.9797	0.9941
MEgreen	age	-0.04006	0.5965	0.8571
MEblack	age	0.125	0.09733	0.5953
MEmidnightblue	age	0.08237	0.2757	0.7185
MEroyalblue	age	-0.09237	0.2214	0.6943
MEbrown	age	-0.03028	0.6891	0.8571
MEmagenta	age	-0.01423	0.8509	0.8896
MEdarkgreen	age	0.04931	0.5145	0.8571
MEred	age	0.1216	0.1068	0.5953
MEsalmon	age	0.1124	0.1364	0.641
MEblue	age	0.05918	0.434	0.8555
MElightcyan	age	0.05548	0.4633	0.8571
MEpink	age	0.1217	0.1067	0.5953
MEturquoise	age	0.03442	0.6492	0.8571
MEcyan	age	-0.1469	0.05097	0.5953
MEdarkturquoise	age	-0.1345	0.07433	0.5953
MEgrey60	age	-0.104	0.1682	0.641
MEyellow	age	-0.01942	0.7975	0.8735

S6 Table (page 4 of 4)

Module–clinical-trait relationships in the HF network. Correlation of every HF module eigengene with ischaemic aetiology, sex, and age across the 177 failing ventricles. Disease status itself is not a trait here: the network was built on disease samples only, so all samples are failing myocardium and there is no within-network control group to correlate against. No module survives FDR correction against any of the three traits.

module	trait	r	p	fdr
MEdarkred	age	0.1231	0.1027	0.5953
MEgreenyellow	age	0.07285	0.3353	0.7462
MElightyellow	age	0.08058	0.2863	0.7185

S7 Table (page 1 of 3)

Gene module membership. Module assignment for every gene in the universe in each network, with intramodular connectivity and module membership.

disease	module	gene	kME
HF	black	ATP6AP2	0.7941
HF	black	TXNDC12	0.7661
HF	black	SSR1	0.7545
HF	black	MOB1A	0.7531
HF	black	STX12	0.7488
HF	black	COPB1	0.7433
HF	black	SNX2	0.7417
HF	black	CD164	0.7417
HF	black	SELENOF	0.7386
HF	black	CCNC	0.7372
HF	black	SGK3	0.7363
HF	black	TMX1	0.7332
HF	black	MDM2	0.7265
HF	black	TBK1	0.7252
HF	black	TMEM87A	0.7127
HF	black	ERO1A	0.7101
HF	black	NIP7	0.7063
HF	black	BZW1	0.6987
HF	black	NIFK	0.6908
HF	black	RAB2B	0.6902
HF	black	OAT	0.688
HF	black	ARF4	0.6842

S7 Table (page 2 of 3)

Gene module membership. Module assignment for every gene in the universe in each network, with intramodular connectivity and module membership.

disease	module	gene	kME
HF	black	ATF6	0.6833
HF	black	HMGN4	0.6793
HF	black	ERP44	0.6772
HF	black	WDR75	0.6769
HF	black	HACD3	0.6763
HF	black	UTP18	0.6745
HF	black	ACTR3	0.6744
HF	black	GNAI3	0.671
HF	black	SRFBP1	0.6689
HF	black	MTDH	0.6644
HF	black	PGM3	0.6636
HF	black	LMAN2	0.6634
HF	black	GABPB1	0.6619
HF	black	DCK	0.655
HF	black	CYRIB	0.6542
HF	black	ARFGAP3	0.6493
HF	black	UBLCP1	0.6488
HF	black	VAMP4	0.647
HF	black	SNX6	0.6463
HF	black	RPL7L1	0.6449
HF	black	NOC3L	0.6412
HF	black	UFM1	0.6392

S7 Table (page 3 of 3)

Gene module membership. Module assignment for every gene in the universe in each network, with intramodular connectivity and module membership.

Showing the first 60 of 20,426 rows. The complete table is provided as the machine-readable file Table_S7_gene_module_membership.csv.

disease	module	gene	kME
HF	black	CHIC2	0.6388
HF	black	CREG1	0.637
HF	black	SLC25A32	0.6352
HF	black	MAP3K2	0.635
HF	black	NFE2L2	0.6323
HF	black	MORC3	0.6322
HF	black	FKBP14	0.63
HF	black	SPAST	0.6289
HF	black	OSBPL8	0.6287
HF	black	CISD2	0.6283
HF	black	SLC31A1	0.6267
HF	black	GNPNAT1	0.6259
HF	black	GNS	0.6248
HF	black	TCTN3	0.6234
HF	black	ACER3	0.6209
HF	black	RSRC1	0.6208

S8 Table (page 1 of 3)

Module x gene-set ssGSEA score matrix. The mean ssGSEA score of each a priori translation set within each module of each network, from which the translation modules were designated without manual input.

set	module	r	tier	disease	p.fwer	module.agg	p.module
KEGG_RIBOSOME	purple	0.9294	TierA	HF	0.000999	0.5034	0.000999
KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	purple	-0.2787	TierA	HF	0.03197	0.5034	0.000999
REACTOME_TRANSLATION	purple	0.2061	TierA	HF	0.4356	0.5034	0.000999
...	purple	0.9283	TierA	HF	0.000999	0.5034	0.000999
...	purple	0.9207	TierA	HF	0.000999	0.5034	0.000999
REACTOME_TRNA_AMINOACYLATION	purple	-0.2221	TierA	HF	0.2637	0.5034	0.000999
...	purple	-0.02213	TierA	HF	1	0.5034	0.000999
REACTOME_RRNA_PROCESSING	purple	0.729	TierA	HF	0.000999	0.5034	0.000999
...	purple	0.9465	TierA	HF	0.000999	0.5034	0.000999
...	purple	0.8207	TierA	HF	0.000999	0.5034	0.000999
...	purple	0.6753	TierA	HF	0.000999	0.5034	0.000999
REACTOME_MITOCHONDRIAL_TRANSLATION	purple	-0.1229	TierA	HF	0.995	0.5034	0.000999
...	purple	-0.08344	TierA	HF	1	0.5034	0.000999
...	purple	-0.3336	TierA	HF	0.003996	0.5034	0.000999
...	purple	-0.3322	TierA	HF	0.003996	0.5034	0.000999
KEGG_RIBOSOME	lightgreen	0.2565	TierA	HF	0.08192	0.3275	0.000999
KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	lightgreen	-0.5188	TierA	HF	0.000999	0.3275	0.000999
REACTOME_TRANSLATION	lightgreen	-0.3996	TierA	HF	0.000999	0.3275	0.000999
...	lightgreen	0.2081	TierA	HF	0.4036	0.3275	0.000999
...	lightgreen	0.2347	TierA	HF	0.1728	0.3275	0.000999
REACTOME_TRNA_AMINOACYLATION	lightgreen	-0.48	TierA	HF	0.000999	0.3275	0.000999
...	lightgreen	-0.2445	TierA	HF	0.1229	0.3275	0.000999

S8 Table (page 2 of 3)

Module x gene-set ssGSEA score matrix. The mean ssGSEA score of each a priori translation set within each module of each network, from which the translation modules were designated without manual input.

set	module	r	tier	disease	p.fwer	module.agg	p.module
REACTOME_RRNA_PROCESSING	lightgreen	0.02247	TierA	HF	1	0.3275	0.000999
...	lightgreen	0.2685	TierA	HF	0.04895	0.3275	0.000999
...	lightgreen	0.01078	TierA	HF	1	0.3275	0.000999
...	lightgreen	-0.08884	TierA	HF	1	0.3275	0.000999
REACTOME_MITOCHONDRIAL_TRANSLATION	lightgreen	-0.511	TierA	HF	0.000999	0.3275	0.000999
...	lightgreen	-0.4983	TierA	HF	0.000999	0.3275	0.000999
...	lightgreen	-0.5887	TierA	HF	0.000999	0.3275	0.000999
...	lightgreen	-0.581	TierA	HF	0.000999	0.3275	0.000999
KEGG_RIBOSOME	tan	0.1877	TierA	HF	0.6454	0.302	0.000999
KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	tan	-0.468	TierA	HF	0.000999	0.302	0.000999
REACTOME_TRANSLATION	tan	-0.4171	TierA	HF	0.000999	0.302	0.000999
...	tan	0.151	TierA	HF	0.9361	0.302	0.000999
...	tan	0.1665	TierA	HF	0.8432	0.302	0.000999
REACTOME_TRNA_AMINOACYLATION	tan	-0.465	TierA	HF	0.000999	0.302	0.000999
...	tan	-0.3454	TierA	HF	0.002997	0.302	0.000999
REACTOME_RRNA_PROCESSING	tan	-0.023	TierA	HF	1	0.302	0.000999
...	tan	0.154	TierA	HF	0.9271	0.302	0.000999
...	tan	-0.09133	TierA	HF	1	0.302	0.000999
...	tan	-0.1824	TierA	HF	0.7003	0.302	0.000999
REACTOME_MITOCHONDRIAL_TRANSLATION	tan	-0.4723	TierA	HF	0.000999	0.302	0.000999
...	tan	-0.4649	TierA	HF	0.000999	0.302	0.000999
...	tan	-0.4812	TierA	HF	0.000999	0.302	0.000999

S8 Table (page 3 of 3)

Module x gene-set ssGSEA score matrix. The mean ssGSEA score of each a priori translation set within each module of each network, from which the translation modules were designated without manual input.

Showing the first 60 of 2,967 rows. The complete table is provided as the machine-readable file Table_S8_set_module_correlation.csv.

set	module	r	tier	disease	p.fwer	module.agg	p.module
...	tan	-0.4607	TierA	HF	0.000999	0.302	0.000999
KEGG_RIBOSOME	green	-0.1519	TierA	HF	0.9321	0.2514	0.000999
KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	green	-0.1224	TierA	HF	0.995	0.2514	0.000999
REACTOME_TRANSLATION	green	-0.4351	TierA	HF	0.000999	0.2514	0.000999
...	green	-0.1591	TierA	HF	0.9011	0.2514	0.000999
...	green	-0.188	TierA	HF	0.6384	0.2514	0.000999
REACTOME_TRNA_AMINOACYLATION	green	-0.2081	TierA	HF	0.4036	0.2514	0.000999
...	green	-0.2144	TierA	HF	0.3437	0.2514	0.000999
REACTOME_RRNA_PROCESSING	green	-0.1226	TierA	HF	0.995	0.2514	0.000999
...	green	-0.1045	TierA	HF	0.999	0.2514	0.000999
...	green	-0.3076	TierA	HF	0.008991	0.2514	0.000999
...	green	-0.3895	TierA	HF	0.000999	0.2514	0.000999
REACTOME_MITOCHONDRIAL_TRANSLATION	green	-0.4906	TierA	HF	0.000999	0.2514	0.000999
...	green	-0.516	TierA	HF	0.000999	0.2514	0.000999
...	green	-0.152	TierA	HF	0.9321	0.2514	0.000999
...	green	-0.2087	TierA	HF	0.3976	0.2514	0.000999

S9 Table (page 1 of 3)

Translation over-representation by module. One-sided hypergeometric test of Tier A set membership within each module, with fold enrichment and BH-corrected P-values, for both networks.

module	n.module	n.overlap	expected	fold	p	fdr	disease
purple	153	73	6.8	10.69	2.367e-58	5.682e-57	HF
blue	1525	115	68.1	1.69	2.921e-09	3.505e-08	HF
pink	173	23	7.7	2.98	2.512e-06	2.01e-05	HF
turquoise	2132	119	95.2	1.25	0.003604	0.02162	HF
black	184	14	8.2	1.7	0.03598	0.1727	HF
midnightblue	93	7	4.2	1.69	0.1214	0.4856	HF
lightyellow	58	3	2.6	1.16	0.4828	1	HF
lightcyan	73	2	3.3	0.61	0.8438	1	HF
greenyellow	150	4	6.7	0.6	0.9078	1	HF
salmon	105	2	4.7	0.43	0.952	1	HF
cyan	103	1	4.6	0.22	0.9912	1	HF
yellow	496	7	22.1	0.32	1	1	HF
green	425	5	19	0.26	1	1	HF
grey	2210	65	98.7	0.66	1	1	HF
brown	1516	16	67.7	0.24	1	1	HF
red	189	0	8.4	0	1	1	HF
magenta	167	0	7.5	0	1	1	HF
tan	146	0	6.5	0	1	1	HF
grey60	70	0	3.1	0	1	1	HF
lightgreen	70	0	3.1	0	1	1	HF
royalblue	52	0	2.3	0	1	1	HF
darkred	43	0	1.9	0	1	1	HF

S9 Table (page 2 of 3)

Translation over-representation by module. One-sided hypergeometric test of Tier A set membership within each module, with fold enrichment and BH-corrected P-values, for both networks.

module	n.module	n.overlap	expected	fold	p	fdr	disease
darkgreen	41	0	1.8	0	1	1	HF
darkturquoise	39	0	1.7	0	1	1	HF
blue	1948	173	87	1.99	4.698e-22	9.866e-21	HCC
magenta	146	26	6.5	3.99	1.165e-09	1.223e-08	HCC
brown	389	37	17.4	2.13	1.053e-05	7.368e-05	HCC
lightgreen	68	8	3	2.63	0.01047	0.05496	HCC
grey60	78	8	3.5	2.3	0.02262	0.09502	HCC
cyan	99	8	4.4	1.81	0.07454	0.2609	HCC
salmon	110	6	4.9	1.22	0.368	1	HCC
lightyellow	57	3	2.5	1.18	0.4712	1	HCC
royalblue	43	2	1.9	1.04	0.5783	1	HCC
pink	254	8	11.3	0.71	0.8861	1	HCC
green	323	9	14.4	0.62	0.9558	1	HCC
purple	145	2	6.5	0.31	0.99	1	HCC
black	255	5	11.4	0.44	0.9903	1	HCC
grey	1445	44	64.5	0.68	0.9987	1	HCC
yellow	347	1	15.5	0.06	1	1	HCC
turquoise	3771	116	168.4	0.69	1	1	HCC
red	302	0	13.5	0	1	1	HCC
greenyellow	130	0	5.8	0	1	1	HCC
tan	117	0	5.2	0	1	1	HCC
midnightblue	96	0	4.3	0	1	1	HCC

S9 Table (page 3 of 3)

Translation over-representation by module. One-sided hypergeometric test of Tier A set membership within each module, with fold enrichment and BH-corrected P-values, for both networks.

module	n.module	n.overlap	expected	fold	p	fdr	disease
lightcyan	90	0	4	0	1	1	HCC

S10 Table (page 1 of 3)

Translation module gene lists. The genes of HF purple (153) and HCC magenta (146), with their connectivity and membership statistics in their own and the other network.

disease	module	gene
HF	purple	AHCY
HF	purple	AIP
HF	purple	ASL
HF	purple	ATOX1
HF	purple	BAD
HF	purple	BLOC1S1
HF	purple	BLOC1S3
HF	purple	BLVRA
HF	purple	BTF3
HF	purple	C12orf57
HF	purple	CCT2
HF	purple	CCT3
HF	purple	CNIH1
HF	purple	COA3
HF	purple	COMMD6
HF	purple	COPS9
HF	purple	CWC15
HF	purple	DDX49
HF	purple	DPP7
HF	purple	DUSP23
HF	purple	DYNLT1
HF	purple	EEF1B2

S10 Table (page 2 of 3)

Translation module gene lists. The genes of HF purple (153) and HCC magenta (146), with their connectivity and membership statistics in their own and the other network.

disease	module	gene
HF	purple	EEF1D
HF	purple	EIF3D
HF	purple	EIF3I
HF	purple	EIF4EBP1
HF	purple	ELOB
HF	purple	EVA1B
HF	purple	FAM216A
HF	purple	FKBP8
HF	purple	GET3
HF	purple	GPBP1
HF	purple	GRAMD1A
HF	purple	GUK1
HF	purple	HBB
HF	purple	HCFC1R1
HF	purple	HINT1
HF	purple	HSBP1
HF	purple	IMPDH2
HF	purple	JTB
HF	purple	LAMTOR1
HF	purple	LGALS1
HF	purple	LSM10
HF	purple	MAGEH1

S10 Table (page 3 of 3)

Translation module gene lists. The genes of HF purple (153) and HCC magenta (146), with their connectivity and membership statistics in their own and the other network.

Showing the first 60 of 299 rows. The complete table is provided as the machine-readable file Table_S10_translation_module_genes.csv.

disease	module	gene
HF	purple	MAZ
HF	purple	MCRIP1
HF	purple	MYL6B
HF	purple	NAT9
HF	purple	NBEAL1
HF	purple	NENF
HF	purple	NHP2
HF	purple	NME1
HF	purple	NME2
HF	purple	NOP10
HF	purple	NOSIP
HF	purple	NSMCE1
HF	purple	NT5C
HF	purple	PFDN5
HF	purple	PHF5A
HF	purple	PIH1D1

S11 Table

Cohort coverage. Every cohort with its sample count, group sizes, technology, and platform, including the platforms GSE14520 is split into.

cohort	disease	tech	n.genes	n.samples	n.disease	n.control	n.unlabeled	tierA.covered	tierA.total	tierA.pct
GSE57338	HF	array	12090	313	177	136	0	459	497	92.4
GSE141198	HCC	RNA-seq	13178	148	148	0	0	487	497	98
TCGA_LIHC	HCC	RNA-seq	17146	421	371	50	0	483	497	97.2
GSE116250	HF	RNA-seq	26261	64	50	14	0	439	497	88.3
GSE14520_GPL3921	HCC	array	7629	445	225	220	0	299	497	60.2
GSE14520_GPL571	HCC	array	7733	41	22	19	0	322	497	64.8
GSE76427	HCC	array	16907	167	115	52	0	417	497	83.9
GSE141910	HF	RNA-seq	16978	366	200	166	0	482	497	97

S12 Table (page 1 of 3)

Set coverage per cohort. The number of measured genes for each scored set in each cohort, so that a set's contribution to any cohort-level statistic is auditable.

cohort	set	n.genes.in cohort	scored
GSE57338	KEGG_RIBOSOME	81	True
GSE57338	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	40	True
GSE57338	REACTOME_TRANSLATION	342	True
GSE57338	...	112	True
GSE57338	...	87	True
GSE57338	REACTOME_TRNA_AMINOACYLATION	41	True
GSE57338	...	23	True
GSE57338	REACTOME_RRNA_PROCESSING	196	True
GSE57338	...	106	True
GSE57338	...	138	True
GSE57338	...	98	True
GSE57338	REACTOME_MITOCHONDRIAL_TRANSLATION	95	True
GSE57338	...	83	True
GSE57338	...	21	True
GSE57338	...	17	True
GSE57338	REACTOME_TRNA_PROCESSING	97	True
GSE57338	...	52	True
GSE57338	...	36	True
GSE57338	...	58	True
GSE57338	...	11	True
GSE57338	...	5	True
GSE57338	...	27	True

S12 Table (page 2 of 3)

Set coverage per cohort. The number of measured genes for each scored set in each cohort, so that a set's contribution to any cohort-level statistic is auditable.

cohort	set	n.genes.in cohort	scored
GSE57338	HALLMARK_MYC_TARGETS_V1	188	True
GSE57338	...	82	True
GSE57338	...	76	True
GSE57338	...	17	True
GSE57338	HALLMARK_MYC_TARGETS_V2	53	True
GSE57338	...	44	True
GSE57338	...	30	True
GSE57338	...	58	True
GSE57338	...	35	True
GSE57338	HALLMARK_MTORC1_SIGNALING	180	True
GSE57338	...	13	True
GSE57338	...	9	True
GSE57338	...	43	True
GSE57338	...	43	True
GSE57338	...	9	True
GSE57338	...	29	True
GSE57338	...	9	True
GSE57338	...	8	True
GSE57338	...	9	True
GSE141198	KEGG_RIBOSOME	85	True
GSE141198	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	41	True
GSE141198	REACTOME_TRANSLATION	364	True

S12 Table (page 3 of 3)

Set coverage per cohort. The number of measured genes for each scored set in each cohort, so that a set's contribution to any cohort-level statistic is auditable.

Showing the first 60 of 328 rows. The complete table is provided as the machine-readable file Table_S12_set_coverage_per_cohort.csv.

cohort	set	n.genes.in cohort	scored
GSE141198	...	117	True
GSE141198	...	90	True
GSE141198	REACTOME_TRNA_AMINOACYLATION	42	True
GSE141198	...	24	True
GSE141198	REACTOME_RRNA_PROCESSING	205	True
GSE141198	...	110	True
GSE141198	...	142	True
GSE141198	...	101	True
GSE141198	REACTOME_MITOCHONDRIAL_TRANSLATION	112	True
GSE141198	...	87	True
GSE141198	...	21	True
GSE141198	...	17	True
GSE141198	REACTOME_TRNA_PROCESSING	110	True
GSE141198	...	58	True
GSE141198	...	43	True
GSE141198	...	58	True

S13 Table

Scored set counts. The number of sets scored per cohort, broken down by tier, with any set dropped for insufficient measured genes.

cohort	n.sets.total	n.tierA	n.tierB	n.tierC
GSE57338	41	15	21	33
GSE141198	41	15	21	33
TCGA_LIHC	41	15	21	33
GSE116250	40	15	20	33
GSE14520_GPL3921	34	14	19	27
GSE14520_GPL571	36	15	20	29
GSE76427	35	14	19	28
GSE141910	41	15	21	33

S14 Table (columns 1 of 2) (page 1 of 3)

Within-disease effect sizes. Hedges g with n-weighted pooled SD, the small-sample correction J, bootstrap confidence intervals, and test statistics for every set in every cohort.

cohort	disease	tech	set	tier	n.disease	n.control	mean.disease	mean.control	sd.disease	sd.control	cohens.d equalweight
GSE57338	HF	array	KEGG_RIBOSOME	A	177	136	0.5646	0.5741	0.0186	0.02437	-0.4394
GSE57338	HF	array	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	177	136	0.2202	0.2253	0.02426	0.02403	-0.2086
GSE57338	HF	array	REACTOME_TRANSLATION	A	177	136	0.4068	0.4156	0.01644	0.01585	-0.5403
GSE57338	HF	array	...	A	177	136	0.5612	0.5725	0.01395	0.02099	-0.6342
GSE57338	HF	array	...	A	177	136	0.5774	0.5867	0.01697	0.02238	-0.4672
GSE57338	HF	array	REACTOME_TRNA_AMINOACYLATION	A	177	136	0.2053	0.2137	0.02271	0.0243	-0.3604
GSE57338	HF	array	...	A	177	136	0.3976	0.4241	0.01923	0.03777	-0.8835
GSE57338	HF	array	REACTOME_RRNA_PROCESSING	A	177	136	0.2403	0.2588	0.012	0.03178	-0.7702
GSE57338	HF	array	...	A	177	136	0.5349	0.5467	0.01535	0.02382	-0.5904
GSE57338	HF	array	...	A	177	136	0.5226	0.5318	0.01232	0.0183	-0.5843
GSE57338	HF	array	...	A	177	136	0.5239	0.5318	0.01239	0.01635	-0.5421
GSE57338	HF	array	REACTOME_MITOCHONDRIAL_TRANSLATION	A	177	136	0.2456	0.2489	0.0396	0.03456	-0.08902
GSE57338	HF	array	...	A	177	136	0.2579	0.2609	0.03892	0.03476	-0.08181
GSE57338	HF	array	...	A	177	136	0.03087	0.01304	0.03422	0.02874	0.5645
GSE57338	HF	array	...	A	177	136	-0.05406	-0.04939	0.05149	0.03886	-0.1023
GSE57338	HF	array	REACTOME_TRNA_PROCESSING	B_extra	177	136	-0.0367	-0.02758	0.01723	0.02446	-0.4312
GSE57338	HF	array	...	B_extra	177	136	0.08264	0.09369	0.01417	0.02392	-0.5622
GSE57338	HF	array	...	B_extra	177	136	-0.2042	-0.2005	0.02095	0.03245	-0.1379
GSE57338	HF	array	...	B_extra	177	136	-0.03546	-0.0005126	0.01828	0.05765	-0.8172
GSE57338	HF	array	...	B_extra	177	136	-0.1646	-0.1413	0.05361	0.0449	-0.4705
GSE57338	HF	array	...	B_extra	177	136	0.04544	0.06177	0.04657	0.04029	-0.3749
GSE57338	HF	array	...	C_v1	177	136	0.07477	0.09175	0.02215	0.0305	-0.6368

S14 Table (columns 1 of 2) (page 2 of 3)

Within-disease effect sizes. Hedges g with n-weighted pooled SD, the small-sample correction J, bootstrap confidence intervals, and test statistics for every set in every cohort.

cohort	disease	tech	set	tier	n.disease	n.control	mean.disease	mean.control	sd.disease	sd.control	cohens.d equalweight
GSE57338	HF	array	HALLMARK_MYC_TARGETS_V1	C_v1	177	136	0.3974	0.413	0.01082	0.02495	-0.8128
GSE57338	HF	array	...	C_v1	177	136	0.4703	0.4788	0.01321	0.01587	-0.5862
GSE57338	HF	array	...	C_v1	177	136	0.4415	0.4527	0.0103	0.01838	-0.7499
GSE57338	HF	array	...	C_v1	177	136	0.53	0.5492	0.01097	0.02334	-1.052
GSE57338	HF	array	HALLMARK_MYC_TARGETS_V2	C_v1	177	136	0.008979	0.0551	0.01753	0.05439	-1.141
GSE57338	HF	array	...	C_v1	177	136	0.4304	0.4457	0.02052	0.02858	-0.6168
GSE57338	HF	array	...	C_v1	177	136	0.4032	0.4212	0.01119	0.01902	-1.159
GSE57338	HF	array	...	C_v1	177	136	0.2601	0.2788	0.01217	0.0186	-1.188
GSE57338	HF	array	...	C_v1	177	136	0.587	0.5993	0.02061	0.02605	-0.5252
GSE57338	HF	array	HALLMARK_MTORC1_SIGNALING	C_v1	177	136	0.2644	0.2885	0.0187	0.02974	-0.9715
GSE57338	HF	array	...	C_v1	177	136	0.2797	0.279	0.01463	0.01578	0.04576
GSE57338	HF	array	...	C_v1	177	136	0.2104	0.2212	0.03877	0.03605	-0.289
GSE57338	HF	array	...	C_v1	177	136	-0.06274	-0.0471	0.03267	0.02557	-0.5332
GSE57338	HF	array	...	C_v1	177	136	0.328	0.3503	0.009551	0.01584	-1.704
GSE57338	HF	array	...	C_v1	177	136	0.2285	0.2164	0.03819	0.03963	0.3108
GSE57338	HF	array	...	C_v1	177	136	0.2067	0.2217	0.01335	0.01403	-1.093
GSE57338	HF	array	...	C_v1	177	136	0.3019	0.2969	0.02787	0.03689	0.153
GSE57338	HF	array	...	C_v1	177	136	0.2492	0.2416	0.03219	0.04233	0.2038
GSE57338	HF	array	...	C_v1	177	136	0.2038	0.2067	0.03154	0.04014	-0.08206
TCGA_LIHC	HCC	RNA-seq	KEGG_RIBOSOME	A	371	50	0.792	0.7747	0.02806	0.01869	0.7289
TCGA_LIHC	HCC	RNA-seq	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	371	50	0.4534	0.4175	0.03659	0.01923	1.229
TCGA_LIHC	HCC	RNA-seq	REACTOME_TRANSLATION	A	371	50	0.6296	0.6093	0.03076	0.01805	0.8056

S14 Table (columns 1 of 2) (page 3 of 3)

Within-disease effect sizes. Hedges g with n-weighted pooled SD, the small-sample correction J, bootstrap confidence intervals, and test statistics for every set in every cohort.

cohort	disease	tech	set	tier	n.disease	n.control	mean.disease	mean.control	sd.disease	sd.control	cohens.d equalweight
TCGA_LIHC	HCC	RNA-seq	...	A	371	50	0.7508	0.7332	0.02471	0.0159	0.8435
TCGA_LIHC	HCC	RNA-seq	...	A	371	50	0.7869	0.764	0.02679	0.01825	0.9973
TCGA_LIHC	HCC	RNA-seq	REACTOME_TRNA_AMINOACYLATION	A	371	50	0.4821	0.4387	0.04057	0.0207	1.35
TCGA_LIHC	HCC	RNA-seq	...	A	371	50	0.6209	0.5669	0.04774	0.02882	1.369
TCGA_LIHC	HCC	RNA-seq	REACTOME_RRNA_PROCESSING	A	371	50	0.5345	0.5035	0.03875	0.02174	0.9885
TCGA_LIHC	HCC	RNA-seq	...	A	371	50	0.7756	0.7557	0.02403	0.01581	0.9795
TCGA_LIHC	HCC	RNA-seq	...	A	371	50	0.7128	0.6916	0.02268	0.01479	1.108
TCGA_LIHC	HCC	RNA-seq	...	A	371	50	0.7007	0.6754	0.02376	0.01487	1.274
TCGA_LIHC	HCC	RNA-seq	REACTOME_MITOCHONDRIAL_TRANSLATION	A	371	50	0.515	0.508	0.05875	0.03596	0.1436
TCGA_LIHC	HCC	RNA-seq	...	A	371	50	0.4779	0.4658	0.07105	0.04409	0.2046
TCGA_LIHC	HCC	RNA-seq	...	A	371	50	0.3515	0.3206	0.04906	0.02866	0.7708
TCGA_LIHC	HCC	RNA-seq	...	A	371	50	0.3547	0.3693	0.05927	0.0277	-0.3153
TCGA_LIHC	HCC	RNA-seq	REACTOME_TRNA_PROCESSING	B_extra	371	50	0.2498	0.2012	0.04002	0.02053	1.528
TCGA_LIHC	HCC	RNA-seq	...	B_extra	371	50	0.3304	0.2701	0.04348	0.01624	1.835
TCGA_LIHC	HCC	RNA-seq	...	B_extra	371	50	0.118	0.07734	0.05377	0.03415	0.903
TCGA_LIHC	HCC	RNA-seq	...	B_extra	371	50	0.4245	0.3754	0.05238	0.03418	1.109

S14 Table (columns 2 of 2) (page 1 of 3)

Within-disease effect sizes. Hedges g with n-weighted pooled SD, the small-sample correction J, bootstrap confidence intervals, and test statistics for every set in every cohort.

cohort	pooled.sd nweighted	hedges.J	d.nweighted	hedges.g	ci.lo	ci.hi	p.ttest	ci.excludes.0	meets.min effect	mirror candidate	p.fdr
GSE57338	0.0213	0.9976	-0.4473	-0.4462	-0.6826	-0.2193	0.0001906	True	True	True	0.0002894
GSE57338	0.02416	0.9976	-0.2085	-0.208	-0.4249	0.02243	0.06822	False	True	False	0.08475
GSE57338	0.01619	0.9976	-0.539	-0.5377	-0.7463	-0.3254	3.18e-06	True	True	True	6.154e-06
GSE57338	0.01736	0.9976	-0.6511	-0.6495	-0.8878	-0.4249	1.499e-07	True	True	True	4.389e-07
GSE57338	0.0195	0.9976	-0.4757	-0.4746	-0.7172	-0.2489	7.56e-05	True	True	True	0.0001192
GSE57338	0.02342	0.9976	-0.3621	-0.3612	-0.5781	-0.138	0.001827	True	True	True	0.002496
GSE57338	0.02879	0.9976	-0.9199	-0.9177	-1.138	-0.7103	2.977e-12	True	True	True	1.526e-11
GSE57338	0.0228	0.9976	-0.8114	-0.8094	-1.007	-0.6242	1.223e-09	True	True	True	4.557e-09
GSE57338	0.01948	0.9976	-0.6071	-0.6057	-0.8485	-0.3791	9.641e-07	True	True	True	2.196e-06
GSE57338	0.01521	0.9976	-0.5993	-0.5979	-0.83	-0.3833	1.143e-06	True	True	True	2.467e-06
GSE57338	0.01424	0.9976	-0.5521	-0.5507	-0.7818	-0.3368	4.927e-06	True	True	True	8.783e-06
GSE57338	0.0375	0.9976	-0.08824	-0.08803	-0.2911	0.1401	0.4315	False	False	False	0.4655
GSE57338	0.03717	0.9976	-0.08121	-0.08101	-0.2864	0.1497	0.4704	False	False	False	0.4911
GSE57338	0.03196	0.9976	0.5582	0.5569	0.3469	0.7993	9.278e-07	True	True	True	2.196e-06
GSE57338	0.04643	0.9976	-0.1005	-0.1003	-0.3014	0.1268	0.3614	False	False	False	0.4005
GSE57338	0.02068	0.9976	-0.4411	-0.4401	-0.6467	-0.2139	0.0002682	True	True	True	0.0003927
GSE57338	0.01903	0.9976	-0.5808	-0.5794	-0.7853	-0.3714	3.302e-06	True	True	True	6.154e-06
GSE57338	0.02656	0.9976	-0.1418	-0.1414	-0.3563	0.1073	0.2402	False	False	False	0.2735
GSE57338	0.04039	0.9976	-0.8652	-0.8631	-1.05	-0.6899	1.977e-10	True	True	True	8.107e-10
GSE57338	0.05002	0.9976	-0.4652	-0.464	-0.6902	-0.2378	3.889e-05	True	True	True	6.378e-05
GSE57338	0.04396	0.9976	-0.3714	-0.3705	-0.591	-0.1429	0.00101	True	True	True	0.001428
GSE57338	0.02611	0.9976	-0.6502	-0.6486	-0.9084	-0.4232	1.117e-07	True	True	True	3.523e-07

S14 Table (columns 2 of 2) (page 2 of 3)

Within-disease effect sizes. Hedges g with n-weighted pooled SD, the small-sample correction J, bootstrap confidence intervals, and test statistics for every set in every cohort.

cohort	pooled.sd nweighted	hedges.J	d.nweighted	hedges.g	ci.lo	ci.hi	p.ttest	ci.excludes.0	meets.min effect	mirror candidate	p.fdr
GSE57338	0.01834	0.9976	-0.8521	-0.8501	-1.049	-0.66	1.372e-10	True	True	True	6.249e-10
GSE57338	0.01443	0.9976	-0.5934	-0.5919	-0.8014	-0.3855	7.167e-07	True	True	True	1.837e-06
GSE57338	0.01438	0.9976	-0.7772	-0.7753	-1.034	-0.5518	1.336e-09	True	True	True	4.565e-09
GSE57338	0.01745	0.9976	-1.099	-1.097	-1.334	-0.8822	7.292e-16	True	True	True	4.983e-15
GSE57338	0.03819	0.9976	-1.208	-1.205	-1.397	-1.04	3.172e-17	True	True	True	2.601e-16
GSE57338	0.02435	0.9976	-0.6302	-0.6287	-0.8657	-0.4055	2.674e-07	True	True	True	7.31e-07
GSE57338	0.0151	0.9976	-1.198	-1.195	-1.437	-0.9874	5.309e-19	True	True	True	6.3e-18
GSE57338	0.01529	0.9976	-1.22	-1.217	-1.495	-0.9778	4.163e-20	True	True	True	8.534e-19
GSE57338	0.02313	0.9976	-0.5334	-0.5321	-0.7744	-0.302	8.81e-06	True	True	True	1.505e-05
GSE57338	0.02412	0.9976	-1.001	-0.9981	-1.237	-0.7759	1.258e-14	True	True	True	7.37e-14
GSE57338	0.01514	0.9976	0.04598	0.04587	-0.179	0.2749	0.69	False	False	False	0.69
GSE57338	0.03761	0.9976	-0.2876	-0.2869	-0.5148	-0.07689	0.01137	True	True	True	0.01457
GSE57338	0.02979	0.9976	-0.5249	-0.5236	-0.7695	-0.2906	3.096e-06	True	True	True	6.154e-06
GSE57338	0.01267	0.9976	-1.759	-1.755	-2.064	-1.479	1.979e-33	True	True	True	8.113e-32
GSE57338	0.03882	0.9976	0.3116	0.3108	0.08858	0.5274	0.006941	True	True	True	0.009181
GSE57338	0.01365	0.9976	-1.097	-1.094	-1.402	-0.8238	6.146e-19	True	True	True	6.3e-18
GSE57338	0.0321	0.9976	0.1559	0.1555	-0.07886	0.3735	0.1885	False	False	False	0.2208
GSE57338	0.03694	0.9976	0.2075	0.207	-0.02192	0.419	0.08026	False	True	False	0.09678
GSE57338	0.03553	0.9976	-0.08337	-0.08317	-0.3287	0.1251	0.4791	False	False	False	0.4911
TCGA_LIHC	0.02714	0.9982	0.6404	0.6393	0.4269	0.8579	1.413e-07	True	True	True	2.07e-07
TCGA_LIHC	0.03501	0.9982	1.026	1.024	0.8293	1.24	8.292e-19	True	True	True	3.4e-18
TCGA_LIHC	0.02956	0.9982	0.6874	0.6862	0.4841	0.8868	1.268e-09	True	True	True	2.362e-09

S14 Table (columns 2 of 2) (page 3 of 3)

Within-disease effect sizes. Hedges g with n-weighted pooled SD, the small-sample correction J, bootstrap confidence intervals, and test statistics for every set in every cohort.

Showing the first 60 of 268 rows. The complete table is provided as the machine-readable file Table_S14_effect_sizes_within_disease.csv.

cohort	pooled.sd nweighted	hedges.J	d.nweighted	hedges.g	ci.lo	ci.hi	p.ttest	ci.excludes.0	meets.min effect	mirror candidate	p.fdr
TCGA_LIHC	0.02385	0.9982	0.7349	0.7336	0.5244	0.9482	1.576e-09	True	True	True	2.809e-09
TCGA_LIHC	0.02594	0.9982	0.8813	0.8797	0.6589	1.11	1.923e-11	True	True	True	4.637e-11
TCGA_LIHC	0.03878	0.9982	1.121	1.119	0.9216	1.334	8.724e-22	True	True	True	5.962e-21
TCGA_LIHC	0.04593	0.9982	1.175	1.173	0.9511	1.418	5.301e-19	True	True	True	3.105e-18
TCGA_LIHC	0.03716	0.9982	0.8357	0.8342	0.6499	1.039	2.81e-13	True	True	True	7.973e-13
TCGA_LIHC	0.02322	0.9982	0.858	0.8564	0.639	1.071	1.736e-11	True	True	True	4.45e-11
TCGA_LIHC	0.0219	0.9982	0.9686	0.9669	0.746	1.194	1.271e-13	True	True	True	4.008e-13
TCGA_LIHC	0.0229	0.9982	1.102	1.1	0.8834	1.312	7.429e-17	True	True	True	2.769e-16
TCGA_LIHC	0.05656	0.9982	0.1237	0.1234	-0.07905	0.3292	0.2414	False	False	False	0.2538
TCGA_LIHC	0.06845	0.9982	0.1767	0.1764	-0.03141	0.3826	0.09858	False	False	False	0.1123
TCGA_LIHC	0.04713	0.9982	0.657	0.6559	0.4395	0.8653	4.49e-09	True	True	True	7.67e-09
TCGA_LIHC	0.0565	0.9982	-0.2582	-0.2577	-0.4263	-0.07756	0.004072	True	True	True	0.004911
TCGA_LIHC	0.03826	0.9982	1.27	1.268	1.082	1.473	3.402e-25	True	True	True	2.79e-24
TCGA_LIHC	0.04124	0.9982	1.461	1.458	1.302	1.631	5.522e-43	True	True	True	2.264e-41
TCGA_LIHC	0.05186	0.9982	0.7843	0.7829	0.5684	0.9958	1.411e-10	True	True	True	2.994e-10
TCGA_LIHC	0.05059	0.9982	0.9697	0.968	0.7535	1.194	1.24e-13	True	True	True	4.008e-13

S15 Table (page 1 of 2)

Set redundancy clusters. Jaccard similarity hierarchical clustering of the scored sets at cut height 0.7, giving the redundancy clusters used for the cluster-level mirror test.

set	tier	cluster	is representative	n.genes
REACTOME_RRNA_PROCESSING	A	1	True	248
...	A	1	False	167
...	A	1	False	140
...	A	1	False	133
...	A	1	False	115
...	A	1	False	108
KEGG_RIBOSOME	A	1	False	88
REACTOME_TRNA_AMINOACYLATION	A	2	True	50
KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	2	False	41
...	A	2	False	30
...	A	2	False	23
REACTOME_TRANSLATION	A	3	True	414
REACTOME_MITOCHONDRIAL_TRANSLATION	A	4	True	124
...	A	4	False	99
...	A	5	True	20
...	B_extra	5	False	7
REACTOME_TRNA_PROCESSING	B_extra	6	True	136
...	B_extra	6	False	76
...	B_extra	7	True	49
...	B_extra	8	True	80
...	B_extra	9	True	11
...	C_v1	10	True	47

S15 Table (page 2 of 2)

Set redundancy clusters. Jaccard similarity hierarchical clustering of the scored sets at cut height 0.7, giving the redundancy clusters used for the cluster-level mirror test.

set	tier	cluster	is representative	n.genes
HALLMARK_MYC_TARGETS_V1	C_v1	11	True	200
...	C_v1	12	True	103
...	C_v1	13	True	95
...	C_v1	14	True	23
HALLMARK_MYC_TARGETS_V2	C_v1	15	True	58
...	C_v1	16	True	71
...	C_v1	16	False	54
...	C_v1	17	True	40
...	C_v1	18	True	64
...	C_v1	18	False	30
HALLMARK_MTORC1_SIGNALING	C_v1	19	True	200
...	C_v1	20	True	15
...	C_v1	21	True	15
...	C_v1	21	False	14
...	C_v1	21	False	10
...	C_v1	21	False	10
...	C_v1	21	False	9
...	C_v1	22	True	108
...	C_v1	23	True	67

S16 Table

Effective number of tests. Nyholt and Li & Ji effective test counts per cohort and tier, alongside the nominal set counts.

cohort	tierA.m	tierA.nyholt	tierA.li.ji	all.m	all.nyholt	all.li.ji
GSE57338	15	10.25	5	41	32.18	13
GSE141198	15	10.54	6	41	33.47	15
TCGA_LIHC	15	8.63	5	41	32.56	17
GSE116250	15	10.37	6	40	35.18	17
GSE14520_GPL3921	14	9.3	6	34	25.43	15
GSE14520_GPL571	15	8.53	6	36	25.42	13
GSE76427	14	7.99	5	35	26.77	16
GSE141910	15	8.05	4	41	28.63	13

S17 Table (page 1 of 8)

Mirror sets. Per-set effect sizes in both diseases, the mirror flag, and the redundancy cluster assignment, for each of the four analysed pairings and every tier. The four are TCGA-LIHC x GSE57338 (the primary v1-cohort pairing), GSE76427 x GSE116250 and GSE14520_GPL3921 x GSE141910 (HCC array against an independent HF RNA-seq cohort), and GSE14520_GPL571 x GSE57338 (HCC array against the HF array cohort). Fig 2d gives the effect-size correlation and Tier A mirror count for all twelve HCC x HF combinations; the permutation tests and effective counts in S18 are computed on these four.

set	g.HCC	g.HF	hcc.solid	hf.solid	mirror	same direction	hcc.cohort	hf.cohort	tier	cluster
KEGG_RIBOSOME	0.6393	-0.4462	True	True	True	False	TCGA_LIHC	GSE57338	A	1
KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	1.024	-0.208	True	False	False	False	TCGA_LIHC	GSE57338	A	2
REACTOME_TRANSLATION	0.6862	-0.5377	True	True	True	False	TCGA_LIHC	GSE57338	A	3
...	0.7336	-0.6495	True	True	True	False	TCGA_LIHC	GSE57338	A	1
...	0.8797	-0.4746	True	True	True	False	TCGA_LIHC	GSE57338	A	1
REACTOME_TRNA_AMINOACYLATION	1.119	-0.3612	True	True	True	False	TCGA_LIHC	GSE57338	A	2
...	1.173	-0.9177	True	True	True	False	TCGA_LIHC	GSE57338	A	2
REACTOME_RRNA_PROCESSING	0.8342	-0.8094	True	True	True	False	TCGA_LIHC	GSE57338	A	1
...	0.8564	-0.6057	True	True	True	False	TCGA_LIHC	GSE57338	A	1
...	0.9669	-0.5979	True	True	True	False	TCGA_LIHC	GSE57338	A	1
...	1.1	-0.5507	True	True	True	False	TCGA_LIHC	GSE57338	A	1
REACTOME_MITOCHONDRIAL_TRANSLATION	0.1234	-0.08803	False	False	False	False	TCGA_LIHC	GSE57338	A	4
...	0.1764	-0.08101	False	False	False	False	TCGA_LIHC	GSE57338	A	4
...	0.6559	0.5569	True	True	False	True	TCGA_LIHC	GSE57338	A	2
...	-0.2577	-0.1003	True	False	False	True	TCGA_LIHC	GSE57338	A	5
...	1.681	-0.6486	True	True	True	False	TCGA_LIHC	GSE57338	C	10
HALLMARK_MYC_TARGETS_V1	1.46	-0.8501	True	True	True	False	TCGA_LIHC	GSE57338	C	11
...	1.173	-0.9177	True	True	True	False	TCGA_LIHC	GSE57338	C	2
REACTOME_TRNA_AMINOACYLATION	1.119	-0.3612	True	True	True	False	TCGA_LIHC	GSE57338	C	2
...	1.1	-0.5507	True	True	True	False	TCGA_LIHC	GSE57338	C	1
...	0.9669	-0.5979	True	True	True	False	TCGA_LIHC	GSE57338	C	1
...	1.03	-0.5919	True	True	True	False	TCGA_LIHC	GSE57338	C	12

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Mirror sets. Per-set effect sizes in both diseases, the mirror flag, and the redundancy cluster assignment, for each of the four analysed pairings and every tier. The four are TCGA-LIHC x GSE57338 (the primary v1-cohort pairing), GSE76427 x GSE116250 and GSE14520_GPL3921 x GSE141910 (HCC array against an independent HF RNA-seq cohort), and GSE14520_GPL571 x GSE57338 (HCC array against the HF array cohort). Fig 2d gives the effect-size correlation and Tier A mirror count for all twelve HCC x HF combinations; the permutation tests and effective counts in S18 are computed on these four.

set	g.HCC	g.HF	hcc.solid	hf.solid	mirror	same direction	hcc.cohort	hf.cohort	tier	cluster
...	1.065	-0.7753	True	True	True	False	TCGA_LIHC	GSE57338	C	13
...	1.348	-1.097	True	True	True	False	TCGA_LIHC	GSE57338	C	14
HALLMARK_MYC_TARGETS_V2	0.9592	-1.205	True	True	True	False	TCGA_LIHC	GSE57338	C	15
...	0.8797	-0.4746	True	True	True	False	TCGA_LIHC	GSE57338	C	1
...	0.8564	-0.6057	True	True	True	False	TCGA_LIHC	GSE57338	C	1
REACTOME_RRNA_PROCESSING	0.8342	-0.8094	True	True	True	False	TCGA_LIHC	GSE57338	C	1
...	0.8201	-0.6287	True	True	True	False	TCGA_LIHC	GSE57338	C	16
...	0.6066	-1.195	True	True	True	False	TCGA_LIHC	GSE57338	C	17
...	0.7587	-1.217	True	True	True	False	TCGA_LIHC	GSE57338	C	18
...	0.7336	-0.6495	True	True	True	False	TCGA_LIHC	GSE57338	C	1
REACTOME_TRANSLATION	0.6862	-0.5377	True	True	True	False	TCGA_LIHC	GSE57338	C	3
...	0.5474	-0.5321	True	True	True	False	TCGA_LIHC	GSE57338	C	16
HALLMARK_MTORC1_SIGNALING	0.5742	-0.9981	True	True	True	False	TCGA_LIHC	GSE57338	C	19
...	0.6559	0.5569	True	True	False	True	TCGA_LIHC	GSE57338	C	2
...	-0.5065	0.04587	True	False	False	False	TCGA_LIHC	GSE57338	C	20
...	-0.5947	-0.2869	True	True	False	True	TCGA_LIHC	GSE57338	C	21
...	-0.2428	-0.5236	True	True	False	True	TCGA_LIHC	GSE57338	C	22
...	0.4754	-1.755	True	True	True	False	TCGA_LIHC	GSE57338	C	23
...	0.352	0.3108	True	True	False	True	TCGA_LIHC	GSE57338	C	21
...	0.1483	-1.094	False	True	False	False	TCGA_LIHC	GSE57338	C	18
...	-0.02133	0.1555	False	False	False	False	TCGA_LIHC	GSE57338	C	21
...	0.1764	-0.08101	False	False	False	False	TCGA_LIHC	GSE57338	C	4

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Mirror sets. Per-set effect sizes in both diseases, the mirror flag, and the redundancy cluster assignment, for each of the four analysed pairings and every tier. The four are TCGA-LIHC x GSE57338 (the primary v1-cohort pairing), GSE76427 x GSE116250 and GSE14520_GPL3921 x GSE141910 (HCC array against an independent HF RNA-seq cohort), and GSE14520_GPL571 x GSE57338 (HCC array against the HF array cohort). Fig 2d gives the effect-size correlation and Tier A mirror count for all twelve HCC x HF combinations; the permutation tests and effective counts in S18 are computed on these four.

set	g.HCC	g.HF	hcc.solid	hf.solid	mirror	same direction	hcc.cohort	hf.cohort	tier	cluster
...	-0.2577	-0.1003	True	False	False	True	TCGA_LIHC	GSE57338	C	5
...	0.1561	0.207	False	False	False	True	TCGA_LIHC	GSE57338	C	21
REACTOME_MITOCHONDRIAL_TRANSLATION	0.1234	-0.08803	False	False	False	False	TCGA_LIHC	GSE57338	C	4
...	-0.03293	-0.08317	False	False	False	True	TCGA_LIHC	GSE57338	C	21
KEGG_RIBOSOME	1.013	0.2236	True	False	False	True	GSE76427	GSE116250	A	1
KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	1.042	0.2177	True	False	False	True	GSE76427	GSE116250	A	2
REACTOME_TRANSLATION	1.603	-0.4416	True	False	False	False	GSE76427	GSE116250	A	3
...	1.206	-0.04907	True	False	False	False	GSE76427	GSE116250	A	1
...	1.189	0.4075	True	False	False	True	GSE76427	GSE116250	A	1
REACTOME_TRNA_AMINOACYLATION	1.377	0.1044	True	False	False	True	GSE76427	GSE116250	A	2
REACTOME_RRNA_PROCESSING	1.265	-0.02748	True	False	False	False	GSE76427	GSE116250	A	1
...	1.341	0.3627	True	False	False	True	GSE76427	GSE116250	A	1
...	1.412	0.3047	True	False	False	True	GSE76427	GSE116250	A	1
...	1.53	0.21	True	False	False	True	GSE76427	GSE116250	A	1
REACTOME_MITOCHONDRIAL_TRANSLATION	1.081	-0.8947	True	True	True	False	GSE76427	GSE116250	A	4
...	1.104	-0.7284	True	True	True	False	GSE76427	GSE116250	A	4
...	1.175	0.4217	True	False	False	True	GSE76427	GSE116250	A	2
...	-0.07637	-0.8351	False	True	False	True	GSE76427	GSE116250	A	5
...	0.8942	-1.149	True	True	True	False	GSE76427	GSE116250	C	10
HALLMARK_MYC_TARGETS_V1	1.905	-0.2655	True	False	False	False	GSE76427	GSE116250	C	11
REACTOME_TRNA_AMINOACYLATION	1.377	0.1044	True	False	False	True	GSE76427	GSE116250	C	2
...	1.53	0.21	True	False	False	True	GSE76427	GSE116250	C	1

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Mirror sets. Per-set effect sizes in both diseases, the mirror flag, and the redundancy cluster assignment, for each of the four analysed pairings and every tier. The four are TCGA-LIHC x GSE57338 (the primary v1-cohort pairing), GSE76427 x GSE116250 and GSE14520_GPL3921 x GSE141910 (HCC array against an independent HF RNA-seq cohort), and GSE14520_GPL571 x GSE57338 (HCC array against the HF array cohort). Fig 2d gives the effect-size correlation and Tier A mirror count for all twelve HCC x HF combinations; the permutation tests and effective counts in S18 are computed on these four.

set	g.HCC	g.HF	hcc.solid	hf.solid	mirror	same direction	hcc.cohort	hf.cohort	tier	cluster
...	1.412	0.3047	True	False	False	True	GSE76427	GSE116250	C	1
...	1.725	0.03991	True	False	False	True	GSE76427	GSE116250	C	12
...	1.661	-0.1027	True	False	False	False	GSE76427	GSE116250	C	13
...	1.397	-0.1673	True	False	False	False	GSE76427	GSE116250	C	14
HALLMARK_MYC_TARGETS_V2	0.5758	0.254	True	False	False	True	GSE76427	GSE116250	C	15
...	1.189	0.4075	True	False	False	True	GSE76427	GSE116250	C	1
...	1.341	0.3627	True	False	False	True	GSE76427	GSE116250	C	1
REACTOME_RRNA_PROCESSING	1.265	-0.02748	True	False	False	False	GSE76427	GSE116250	C	1
...	1.251	0.4944	True	False	False	True	GSE76427	GSE116250	C	16
...	0.3253	0.2745	True	False	False	True	GSE76427	GSE116250	C	17
...	1.232	-0.9052	True	True	True	False	GSE76427	GSE116250	C	18
...	1.206	-0.04907	True	False	False	False	GSE76427	GSE116250	C	1
REACTOME_TRANSLATION	1.603	-0.4416	True	False	False	False	GSE76427	GSE116250	C	3
...	0.8099	0.541	True	False	False	True	GSE76427	GSE116250	C	16
HALLMARK_MTORC1_SIGNALING	0.8698	0.07753	True	False	False	True	GSE76427	GSE116250	C	19
...	1.175	0.4217	True	False	False	True	GSE76427	GSE116250	C	2
...	0.9936	0.02631	True	False	False	True	GSE76427	GSE116250	C	20
...	-0.305	-0.4975	True	True	False	True	GSE76427	GSE116250	C	21
...	0.2824	-0.2943	False	False	False	False	GSE76427	GSE116250	C	22
...	0.4974	-0.8355	True	True	True	False	GSE76427	GSE116250	C	23
...	0.05087	-1.271	False	True	False	False	GSE76427	GSE116250	C	18
...	1.104	-0.7284	True	True	True	False	GSE76427	GSE116250	C	4

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Mirror sets. Per-set effect sizes in both diseases, the mirror flag, and the redundancy cluster assignment, for each of the four analysed pairings and every tier. The four are TCGA-LIHC x GSE57338 (the primary v1-cohort pairing), GSE76427 x GSE116250 and GSE14520_GPL3921 x GSE141910 (HCC array against an independent HF RNA-seq cohort), and GSE14520_GPL571 x GSE57338 (HCC array against the HF array cohort). Fig 2d gives the effect-size correlation and Tier A mirror count for all twelve HCC x HF combinations; the permutation tests and effective counts in S18 are computed on these four.

set	g.HCC	g.HF	hcc.solid	hf.solid	mirror	same direction	hcc.cohort	hf.cohort	tier	cluster
...	-0.07637	-0.8351	False	True	False	True	GSE76427	GSE116250	C	5
REACTOME_MITOCHONDRIAL_TRANSLATION	1.081	-0.8947	True	True	True	False	GSE76427	GSE116250	C	4
KEGG_RIBOSOME	1.348	0.3286	True	True	False	True	GSE14520_GPL3921	GSE141910	A	1
KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	0.5985	0.1062	True	False	False	True	GSE14520_GPL3921	GSE141910	A	2
REACTOME_TRANSLATION	2.34	0.06818	True	False	False	True	GSE14520_GPL3921	GSE141910	A	3
...	1.599	0.2034	True	False	False	True	GSE14520_GPL3921	GSE141910	A	1
...	1.07	0.3332	True	True	False	True	GSE14520_GPL3921	GSE141910	A	1
REACTOME_TRNA_AMINOACYLATION	1.392	0.0232	True	False	False	True	GSE14520_GPL3921	GSE141910	A	2
REACTOME_RRNA_PROCESSING	1.71	-0.01461	True	False	False	False	GSE14520_GPL3921	GSE141910	A	1
...	1.885	0.2271	True	True	False	True	GSE14520_GPL3921	GSE141910	A	1
...	2.355	0.184	True	False	False	True	GSE14520_GPL3921	GSE141910	A	1
...	2.499	0.08908	True	False	False	True	GSE14520_GPL3921	GSE141910	A	1
REACTOME_MITOCHONDRIAL_TRANSLATION	1.371	0.0652	True	False	False	True	GSE14520_GPL3921	GSE141910	A	4
...	1.355	0.1153	True	False	False	True	GSE14520_GPL3921	GSE141910	A	4
...	0.5985	0.5104	True	True	False	True	GSE14520_GPL3921	GSE141910	A	2
...	-0.5951	-0.2196	True	True	False	True	GSE14520_GPL3921	GSE141910	A	5
...	1.194	-1.125	True	True	True	False	GSE14520_GPL3921	GSE141910	C	10
HALLMARK_MYC_TARGETS_V1	2.819	-0.2095	True	True	True	False	GSE14520_GPL3921	GSE141910	C	11
REACTOME_TRNA_AMINOACYLATION	1.392	0.0232	True	False	False	True	GSE14520_GPL3921	GSE141910	C	2
...	2.499	0.08908	True	False	False	True	GSE14520_GPL3921	GSE141910	C	1
...	2.355	0.184	True	False	False	True	GSE14520_GPL3921	GSE141910	C	1
...	1.957	-0.2196	True	True	True	False	GSE14520_GPL3921	GSE141910	C	12

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Mirror sets. Per-set effect sizes in both diseases, the mirror flag, and the redundancy cluster assignment, for each of the four analysed pairings and every tier. The four are TCGA-LIHC x GSE57338 (the primary v1-cohort pairing), GSE76427 x GSE116250 and GSE14520_GPL3921 x GSE141910 (HCC array against an independent HF RNA-seq cohort), and GSE14520_GPL571 x GSE57338 (HCC array against the HF array cohort). Fig 2d gives the effect-size correlation and Tier A mirror count for all twelve HCC x HF combinations; the permutation tests and effective counts in S18 are computed on these four.

set	g.HCC	g.HF	hcc.solid	hf.solid	mirror	same direction	hcc.cohort	hf.cohort	tier	cluster
...	2.084	-0.2245	True	True	True	False	GSE14520_GPL3921	GSE141910	C	13
...	2.103	-0.4166	True	True	True	False	GSE14520_GPL3921	GSE141910	C	14
HALLMARK_MYC_TARGETS_V2	1.452	-0.3006	True	True	True	False	GSE14520_GPL3921	GSE141910	C	15
...	1.07	0.3332	True	True	False	True	GSE14520_GPL3921	GSE141910	C	1
...	1.885	0.2271	True	True	False	True	GSE14520_GPL3921	GSE141910	C	1
REACTOME_RRNA_PROCESSING	1.71	-0.01461	True	False	False	False	GSE14520_GPL3921	GSE141910	C	1
...	2.323	0.2908	True	True	False	True	GSE14520_GPL3921	GSE141910	C	16
...	0.3113	-0.02188	True	False	False	False	GSE14520_GPL3921	GSE141910	C	17
...	1.084	-0.5194	True	True	True	False	GSE14520_GPL3921	GSE141910	C	18
...	1.599	0.2034	True	False	False	True	GSE14520_GPL3921	GSE141910	C	1
REACTOME_TRANSLATION	2.34	0.06818	True	False	False	True	GSE14520_GPL3921	GSE141910	C	3
...	0.7	0.4147	True	True	False	True	GSE14520_GPL3921	GSE141910	C	16
HALLMARK_MTORC1_SIGNALING	1.977	-0.9319	True	True	True	False	GSE14520_GPL3921	GSE141910	C	19
...	0.5985	0.5104	True	True	False	True	GSE14520_GPL3921	GSE141910	C	2
...	0.411	-0.6531	True	True	True	False	GSE14520_GPL3921	GSE141910	C	20
...	0.03561	-0.2948	False	True	False	False	GSE14520_GPL3921	GSE141910	C	22
...	0.86	-1.079	True	True	True	False	GSE14520_GPL3921	GSE141910	C	23
...	-0.3197	0.32	True	True	True	False	GSE14520_GPL3921	GSE141910	C	18
...	1.355	0.1153	True	False	False	True	GSE14520_GPL3921	GSE141910	C	4
...	-0.5951	-0.2196	True	True	False	True	GSE14520_GPL3921	GSE141910	C	5
REACTOME_MITOCHONDRIAL_TRANSLATION	1.371	0.0652	True	False	False	True	GSE14520_GPL3921	GSE141910	C	4
KEGG_RIBOSOME	0.6263	-0.4462	True	True	True	False	GSE14520_GPL571	GSE57338	A	1

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Mirror sets. Per-set effect sizes in both diseases, the mirror flag, and the redundancy cluster assignment, for each of the four analysed pairings and every tier. The four are TCGA-LIHC x GSE57338 (the primary v1-cohort pairing), GSE76427 x GSE116250 and GSE14520_GPL3921 x GSE141910 (HCC array against an independent HF RNA-seq cohort), and GSE14520_GPL571 x GSE57338 (HCC array against the HF array cohort). Fig 2d gives the effect-size correlation and Tier A mirror count for all twelve HCC x HF combinations; the permutation tests and effective counts in S18 are computed on these four.

set	g.HCC	g.HF	hcc.solid	hf.solid	mirror	same direction	hcc.cohort	hf.cohort	tier	cluster
KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	0.6662	-0.208	True	False	False	False	GSE14520_GPL571	GSE57338	A	2
REACTOME_TRANSLATION	1.203	-0.5377	True	True	True	False	GSE14520_GPL571	GSE57338	A	3
...	0.9839	-0.6495	True	True	True	False	GSE14520_GPL571	GSE57338	A	1
...	0.444	-0.4746	False	True	False	False	GSE14520_GPL571	GSE57338	A	1
REACTOME_TRNA_AMINOACYLATION	0.6957	-0.3612	True	True	True	False	GSE14520_GPL571	GSE57338	A	2
...	1.29	-0.9177	True	True	True	False	GSE14520_GPL571	GSE57338	A	2
REACTOME_RRNA_PROCESSING	1.244	-0.8094	True	True	True	False	GSE14520_GPL571	GSE57338	A	1
...	0.9826	-0.6057	True	True	True	False	GSE14520_GPL571	GSE57338	A	1
...	1.105	-0.5979	True	True	True	False	GSE14520_GPL571	GSE57338	A	1
...	1.247	-0.5507	True	True	True	False	GSE14520_GPL571	GSE57338	A	1
REACTOME_MITOCHONDRIAL_TRANSLATION	0.8371	-0.08803	True	False	False	False	GSE14520_GPL571	GSE57338	A	4
...	0.8308	-0.08101	True	False	False	False	GSE14520_GPL571	GSE57338	A	4
...	0.05438	0.5569	False	True	False	True	GSE14520_GPL571	GSE57338	A	2
...	0.3713	-0.1003	False	False	False	False	GSE14520_GPL571	GSE57338	A	5
...	1.989	-0.6486	True	True	True	False	GSE14520_GPL571	GSE57338	C	10
HALLMARK_MYC_TARGETS_V1	2.082	-0.8501	True	True	True	False	GSE14520_GPL571	GSE57338	C	11
...	1.29	-0.9177	True	True	True	False	GSE14520_GPL571	GSE57338	C	2
REACTOME_TRNA_AMINOACYLATION	0.6957	-0.3612	True	True	True	False	GSE14520_GPL571	GSE57338	C	2
...	1.247	-0.5507	True	True	True	False	GSE14520_GPL571	GSE57338	C	1
...	1.105	-0.5979	True	True	True	False	GSE14520_GPL571	GSE57338	C	1
...	0.9742	-0.5919	True	True	True	False	GSE14520_GPL571	GSE57338	C	12
...	1.794	-0.7753	True	True	True	False	GSE14520_GPL571	GSE57338	C	13

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Mirror sets. Per-set effect sizes in both diseases, the mirror flag, and the redundancy cluster assignment, for each of the four analysed pairings and every tier. The four are TCGA-LIHC x GSE57338 (the primary v1-cohort pairing), GSE76427 x GSE116250 and GSE14520_GPL3921 x GSE141910 (HCC array against an independent HF RNA-seq cohort), and GSE14520_GPL571 x GSE57338 (HCC array against the HF array cohort). Fig 2d gives the effect-size correlation and Tier A mirror count for all twelve HCC x HF combinations; the permutation tests and effective counts in S18 are computed on these four.

set	g.HCC	g.HF	hcc.solid	hf.solid	mirror	same direction	hcc.cohort	hf.cohort	tier	cluster
...	1.963	-1.097	True	True	True	False	GSE14520_GPL571	GSE57338	C	14
HALLMARK_MYC_TARGETS_V2	1.739	-1.205	True	True	True	False	GSE14520_GPL571	GSE57338	C	15
...	0.444	-0.4746	False	True	False	False	GSE14520_GPL571	GSE57338	C	1
...	0.9826	-0.6057	True	True	True	False	GSE14520_GPL571	GSE57338	C	1
REACTOME_RRNA_PROCESSING	1.244	-0.8094	True	True	True	False	GSE14520_GPL571	GSE57338	C	1
...	1.392	-0.6287	True	True	True	False	GSE14520_GPL571	GSE57338	C	16
...	0.7816	-1.195	True	True	True	False	GSE14520_GPL571	GSE57338	C	17
...	1.762	-1.217	True	True	True	False	GSE14520_GPL571	GSE57338	C	18
...	0.9839	-0.6495	True	True	True	False	GSE14520_GPL571	GSE57338	C	1
REACTOME_TRANSLATION	1.203	-0.5377	True	True	True	False	GSE14520_GPL571	GSE57338	C	3
...	-0.0319	-0.5321	False	True	False	True	GSE14520_GPL571	GSE57338	C	16
HALLMARK_MTORC1_SIGNALING	1.047	-0.9981	True	True	True	False	GSE14520_GPL571	GSE57338	C	19
...	0.05438	0.5569	False	True	False	True	GSE14520_GPL571	GSE57338	C	2
...	0.6134	0.04587	True	False	False	True	GSE14520_GPL571	GSE57338	C	20
...	-0.3837	-0.2869	False	True	False	True	GSE14520_GPL571	GSE57338	C	21
...	-0.3434	-0.5236	False	True	False	True	GSE14520_GPL571	GSE57338	C	22
...	0.9781	-1.755	True	True	True	False	GSE14520_GPL571	GSE57338	C	23
...	0.1993	-1.094	False	True	False	False	GSE14520_GPL571	GSE57338	C	18
...	0.8308	-0.08101	True	False	False	False	GSE14520_GPL571	GSE57338	C	4
...	0.3713	-0.1003	False	False	False	False	GSE14520_GPL571	GSE57338	C	5
REACTOME_MITOCHONDRIAL_TRANSLATION	0.8371	-0.08803	True	False	False	False	GSE14520_GPL571	GSE57338	C	4

S18 Table (columns 1 of 2)

Mirror summary. For each of the four pairings named in S17: number of sets, clusters, mirror sets and mirror clusters, the sample-level permutation P-values at both levels, the effective test counts, and — reported for transparency only — the binomial P-value v1-style, which is not used for inference. The effective counts are computed per pairing on the measured sets of both cohorts, so they differ from the per-cohort values in S16 wherever set membership differs. The pairing labels in the first column are abbreviated in the CSV and read as: primary (v1 cohorts) = TCGA-LIHC x GSE57338; HCC array vs HF RNA-seq = GSE76427 x GSE116250; HCC array vs HF RNA-seq 2 = GSE14520_GPL3921 x GSE141910; HCC array vs HF array-adjacent = GSE14520_GPL571 x GSE57338.

pair	tier	n.sets	n.clusters	n.mirror.sets	null.sets mean	p.perm.sets	n.mirror clusters	null.clusters mean	p.perm clusters	p.binom.v1 style	n.eff.nyholt
primary (v1 cohorts)	A	15	5	10	0.11	9.999e-05	3	0.06	0.0031	1	9.44
primary (v1 cohorts)	C	33	19	21	0.23	9.999e-05	14	0.18	9.999e-05	0.06357	32.37
HCC array vs HF RNA-seq	A	14	5	2	0.79	0.1744	1	0.39	0.2705	0.375	9.18
HCC array vs HF RNA-seq	C	28	19	5	1.61	0.09339	4	1.26	0.08679	0.01921	30.98
HCC array vs HF RNA-seq 2	A	14	5	0	0.01	1	0	0.01	1	0.0625	8.68
HCC array vs HF RNA-seq 2	C	27	18	11	0.03	0.0002	10	0.02	9.999e-05	0.8145	27.03
HCC array vs HF array-adjacent	A	15	5	9	0.31	0.0014	3	0.18	0.009099	1	9.39
HCC array vs HF array-adjacent	C	29	19	19	0.61	0.0002	14	0.5	9.999e-05	0.06357	28.8

S18 Table (columns 2 of 2)

Mirror summary. For each of the four pairings named in S17: number of sets, clusters, mirror sets and mirror clusters, the sample-level permutation P-values at both levels, the effective test counts, and — reported for transparency only — the binomial P-value v1-style, which is not used for inference. The effective counts are computed per pairing on the measured sets of both cohorts, so they differ from the per-cohort values in S16 wherever set membership differs. The pairing labels in the first column are abbreviated in the CSV and read as: primary (v1 cohorts) = TCGA-LIHC x GSE57338; HCC array vs HF RNA-seq = GSE76427 x GSE116250; HCC array vs HF RNA-seq 2 = GSE14520_GPL3921 x GSE141910; HCC array vs HF array-adjacent = GSE14520_GPL571 x GSE57338.

pair	n.eff.li.ji
primary (v1 cohorts)	5
primary (v1 cohorts)	15
HCC array vs HF RNA-seq	5.5
HCC array vs HF RNA-seq	16.5
HCC array vs HF RNA-seq 2	5
HCC array vs HF RNA-seq 2	14
HCC array vs HF array-adjacent	5.5
HCC array vs HF array-adjacent	13

S19 Table (columns 1 of 4) (page 1 of 3)

Module preservation, all modules. Zsummary and median rank for every named module in both directions, 200 permutations. The legend states that modulePreservation subsamples modules larger than 1,000 genes, so the three largest modules per direction are computed on a random 1,000-gene subsample. Unassigned (grey) modules appear as rows in the table for completeness and are reported where relevant (the HF → HCC grey module reaches Zsummary = 10.76), but they are excluded from the named-module counts quoted in the main text.

reference	test	module	moduleSize	Zsummary.pres	Zdensity.pres	Zconnectivity pres	Z propVarExplained pres	Z meanSignAwareKME pres	Z meanSignAwareCorDat pres	Z.meanAdj pres	Z.meanMAR pres
HCC	HF	red	302	15.01	17.57	12.45	12.93	9.313	22.21	39.45	15.15
HCC	HF	greenyellow	130	12.31	16.96	7.661	12.7	8.239	21.21	47.54	16.5
HCC	HF	yellow	347	8.89	8.963	8.817	5.612	6.359	11.57	17.27	11.25
HCC	HF	lightcyan	90	8.752	11.34	6.164	8.302	6.191	14.38	27.75	10.33
HCC	HF	turquoise	1000	7.961	8.269	7.654	0.03611	8.883	23.87	7.655	-1.197
HCC	HF	magenta	146	7.094	10.35	3.842	5.852	6.455	14.74	14.23	3.643
HCC	HF	brown	389	5.221	7.563	2.878	3.259	6.803	13.41	8.323	1.742
HCC	HF	green	323	4.812	6.306	3.319	0.2649	9.167	20.14	3.445	-1.529
HCC	HF	midnightblue	96	4.722	3.245	6.2	-0.6513	4.574	7.146	1.916	0.3767
HCC	HF	black	255	3.784	5.183	2.385	3.579	1.945	6.797	6.787	3.539
HCC	HF	purple	145	2.913	4.864	0.9621	5.202	2.783	4.525	9.273	6.8
HCC	HF	royalblue	43	2.807	3.129	2.485	0.6888	4.026	6.393	2.233	-0.3897
HCC	HF	cyan	99	2.457	3.057	1.857	0.3205	4.13	7.253	1.983	-0.4936
HCC	HF	pink	254	2.347	3.415	1.279	-0.1436	4.696	8.043	2.133	-0.8684
HCC	HF	blue	1000	1.048	-1.443	3.539	-4.225	0.8927	5.296	-3.778	-4.67
HCC	HF	grey	1000	0.839	-2.318	3.996	-3.459	-1.178	24.9	-4.231	-1.787
HCC	HF	lightyellow	57	0.7656	2.13	-0.5987	0.8985	2.423	3.162	1.837	0.5584
HCC	HF	lightgreen	68	0.7004	0.8951	0.5057	-1.814	2.675	3.509	-0.8852	-0.981
HCC	HF	salmon	110	0.6828	1.664	-0.298	-1.061	3.147	3.44	0.1802	-1.3
HCC	HF	grey60	78	-0.08104	-1.075	0.9129	-3.757	-0.06678	0.7671	-2.083	-1.876
HCC	HF	tan	117	-0.8069	-0.91	-0.7038	-1.114	-1.1	0.3069	-0.7202	-0.1261
HF	HCC	green	425	11.51	18.94	4.074	8.644	14.05	30.8	23.84	7.363

S19 Table (columns 1 of 4) (page 2 of 3)

Module preservation, all modules. Zsummary and median rank for every named module in both directions, 200 permutations. The legend states that modulePreservation subsamples modules larger than 1,000 genes, so the three largest modules per direction are computed on a random 1,000-gene subsample. Unassigned (grey) modules appear as rows in the table for completeness and are reported where relevant (the HF → HCC grey module reaches Zsummary = 10.76), but they are excluded from the named-module counts quoted in the main text.

reference	test	module	moduleSize	Zsummary.pres	Zdensity.pres	Zconnectivity pres	Z propVarExplained pres	Z meanSignAwareKME pres	Z meanSignAwareCorDat pres	Z.meanAdj pres	Z.meanMAR pres
HF	HCC	purple	153	11.03	17.78	4.27	10.85	10.22	24.71	44	20.65
HF	HCC	grey	1000	10.76	3.636	17.88	-2.021	9.292	23.88	-2.444	-1.198
HF	HCC	pink	173	10.59	17.04	4.14	8.103	9.971	24.41	24.11	8.271
HF	HCC	turquoise	1000	9.294	16.55	2.033	9.392	13.19	24.49	19.92	7.33
HF	HCC	black	184	8.377	13.96	2.796	6.417	9.742	22.57	18.18	6.239
HF	HCC	midnightblue	93	5.585	11.06	0.1058	5.411	7.39	16.96	14.74	4.439
HF	HCC	royalblue	52	4.769	6.913	2.626	2.018	6.026	13.1	7.799	4.146
HF	HCC	blue	1000	2.863	-1.078	6.804	-0.4464	-6.039	-1.58	-0.5756	0.9857
HF	HCC	salmon	105	2.478	3.291	1.666	0.8381	3.687	5.965	2.895	0.1667
HF	HCC	lightyellow	58	2.386	4.666	0.1049	1.864	3.413	7.059	5.92	3.386
HF	HCC	darkturquoise	39	2.222	2.053	2.392	-1.484	1.534	4.741	2.572	2.387
HF	HCC	lightgreen	70	2.022	2.122	1.923	-0.1563	3.108	6.583	1.136	0.03779
HF	HCC	yellow	496	1.72	0.1031	3.336	-1.952	0.5816	4.829	-0.3754	-2.388
HF	HCC	brown	1000	1.103	-2.779	4.985	-7.386	-1.926	6.178	-3.632	-2.083
HF	HCC	grey60	70	0.9942	0.7998	1.189	-3.429	1.768	4.663	-0.1686	-0.08374
HF	HCC	cyan	103	0.9133	-0.7346	2.561	-3.454	0.3587	2.544	-1.828	-1.862
HF	HCC	darkred	43	0.8175	1.856	-0.221	0.9283	1.862	3.897	1.85	0.4778
HF	HCC	darkgreen	41	0.6741	-0.4533	1.801	-3.569	1.068	1.282	-1.974	-2.977
HF	HCC	tan	146	0.2566	0.4895	0.02373	-2.355	1.596	3.833	-0.6174	-0.1636
HF	HCC	greenyellow	150	0.192	-1.127	1.511	-1.225	-2.019	0.7304	-1.029	-1.277
HF	HCC	red	189	0.1829	-0.3931	0.7589	-3.342	1.608	2.208	-2.394	-3.355
HF	HCC	magenta	167	-1.068	-1.984	-0.1513	-4.995	-0.7298	0.3908	-3.239	-3.583

S19 Table (columns 1 of 4) (page 3 of 3)

Module preservation, all modules. Zsummary and median rank for every named module in both directions, 200 permutations. The legend states that modulePreservation subsamples modules larger than 1,000 genes, so the three largest modules per direction are computed on a random 1,000-gene subsample. Unassigned (grey) modules appear as rows in the table for completeness and are reported where relevant (the HF → HCC grey module reaches Zsummary = 10.76), but they are excluded from the named-module counts quoted in the main text.

reference	test	module	moduleSize	Zsummary.pres	Zdensity.pres	Zconnectivity pres	Z propVarExplained pres	Z meanSignAwareKME pres	Z meanSignAwareCorDat pres	Z.meanAdj pres	Z.meanMAR pres
HF	HCC	lightcyan	73	-1.128	-1.124	-1.131	-2.204	-0.6234	0.3726	-1.625	-2.05

S19 Table (columns 2 of 4) (page 1 of 3)

Module preservation, all modules. Zsummary and median rank for every named module in both directions, 200 permutations. The legend states that modulePreservation subsamples modules larger than 1,000 genes, so the three largest modules per direction are computed on a random 1,000-gene subsample. Unassigned (grey) modules appear as rows in the table for completeness and are reported where relevant (the HF → HCC grey module reaches Zsummary = 10.76), but they are excluded from the named-module counts quoted in the main text.

reference	Z.cor.kIM	Z.cor.kME	Z.cor.kMEall	Z.cor.cor	Z.cor.MAR	moduleSize obs	medianRank pres	medianRankDensity pres	medianRankConnectivity pres	propVarExplained pres	meanSignAwareKME pres
HCC	10.44	12.45	37.94	26.29	9.015	302	3	3.5	2	0.2957	0.3638
HCC	7.438	7.661	60.6	15.77	7.774	130	1	1	1	0.3623	0.4462
HCC	7.152	8.817	24.41	20.28	8.141	347	6	7.5	5	0.2193	0.2602
HCC	4.586	6.164	44.51	12.11	4.399	90	2	2	3	0.3382	0.4207
HCC	6.004	7.654	24.31	22.74	1.89	1000	9	13.5	8	0.1678	0.2243
HCC	3.135	3.842	12.39	9.956	4.241	146	5	4	7	0.2505	0.3558
HCC	2.878	1.277	17.41	10.29	2.982	389	10	9.5	12	0.1997	0.2543
HCC	3.035	3.319	37.31	10.28	0.2575	323	11	8.5	11	0.1733	0.3257
HCC	4.839	6.2	38.47	11.18	1.682	96	7	10.5	4	0.1642	0.3082
HCC	2.385	0.8749	10.39	6.579	1.023	255	13	11	13	0.2046	0.1466
HCC	0.8117	0.9621	10.36	4.736	2.984	145	16	11	16	0.2447	0.1952
HCC	1.621	2.485	33.51	4.21	0.4036	43	6	6	6	0.2012	0.3743
HCC	1.857	-0.3446	26.07	5.841	2.034	99	10	9.5	10	0.1811	0.28
HCC	1.279	-0.3124	13.8	3.563	0.5989	254	15	12.5	20	0.1686	0.2452
HCC	3.539	2.059	0.3436	12.29	3.477	1000	18	19	14	0.147	0.1002
HCC	0.1457	3.996	28.25	36.48	27.04	1000	19	21.5	19	0.1504	0.01058
HCC	-0.5987	-0.6415	0.286	4.01	-0.3869	57	14	12	17	0.1995	0.2341
HCC	0.5057	-1.147	11.25	4.28	-0.2897	68	17	16	17	0.1423	0.2508
HCC	-0.5158	-0.298	23.24	2.382	0.3089	110	16	16	20	0.1569	0.2348
HCC	0.9129	-0.9963	7.528	5.11	0.5632	78	19	20.5	14	0.109	0.09574
HCC	-1.884	-0.7038	11.78	0.468	-1.336	117	20	19	22	0.1571	0.04449
HF	4.074	3.677	53.54	9.993	2.707	425	6	6	13	0.4191	0.5034

S19 Table (columns 2 of 4) (page 2 of 3)

Module preservation, all modules. Zsummary and median rank for every named module in both directions, 200 permutations. The legend states that modulePreservation subsamples modules larger than 1,000 genes, so the three largest modules per direction are computed on a random 1,000-gene subsample. Unassigned (grey) modules appear as rows in the table for completeness and are reported where relevant (the HF → HCC grey module reaches Zsummary = 10.76), but they are excluded from the named-module counts quoted in the main text.

reference	Z.cor.kIM	Z.cor.kME	Z.cor.kMEall	Z.cor.cor	Z.cor.MAR	moduleSize obs	medianRank pres	medianRankDensity pres	medianRankConnectivity pres	propVarExplained pres	meanSignAwareKME pres
HF	4.27	3.692	28.21	9.84	3.757	153	3	2	4	0.5479	0.557
HF	3.528	17.88	66.31	36.24	7.873	1000	14	17.5	12	0.3026	0.1962
HF	3.102	4.14	34.65	9.171	3.102	173	4	3	5	0.4717	0.5567
HF	1.943	2.033	33.22	9.661	1.042	1000	10	8	18	0.3854	0.3733
HF	2.796	2.509	39.38	7.52	2.483	184	5	4.5	7	0.4346	0.5323
HF	0.1058	-1.202	59.9	1.607	-0.795	93	3	2	24	0.4543	0.5793
HF	2.626	1.664	32.04	6.075	2.948	52	2	3.5	2	0.3998	0.5768
HF	1.958	6.804	-0.6577	13.52	0.1893	1000	15	20	13	0.3134	0.05141
HF	-0.04777	1.666	22.91	6.455	-0.7855	105	10	10.5	9	0.3394	0.3292
HF	0.1049	-1.087	20.76	0.5359	0.07774	58	7	7	21	0.3873	0.3834
HF	2.392	0.4216	6.62	6.628	1.392	39	9	10.5	2	0.2702	0.2721
HF	1.551	1.923	8.245	5.058	1.088	70	8	10.5	6	0.3198	0.3624
HF	3.336	3.278	6.371	7.538	2.676	496	16	16.5	12	0.2941	0.1612
HF	4.985	-0.4313	4.254	19.47	7.842	1000	19	19.5	11	0.2606	0.1171
HF	-0.04517	1.189	22.87	2.139	0.6891	70	16	14.5	21	0.2228	0.2688
HF	2.561	0.6487	7.536	5.623	1.722	103	16	20	11	0.2313	0.1664
HF	-0.221	-1.2	12.97	1.263	-0.01629	43	11	10.5	24	0.3611	0.3043
HF	1.255	1.801	11.26	3.121	0.9212	41	14	20	5	0.1931	0.2513
HF	0.02373	-1.02	3.608	5.087	3.928	146	17	16	20	0.2648	0.2251
HF	1.511	0.4219	4.125	4.237	1.691	150	18	20	16	0.2984	0.07249
HF	0.2441	0.7589	3.031	4.259	0.1807	189	19	19.5	19	0.2529	0.2185
HF	-0.1513	-0.8802	-0.728	1.918	0.6857	167	23	23	21	0.2246	0.1232

S19 Table (columns 2 of 4) (page 3 of 3)

Module preservation, all modules. Zsummary and median rank for every named module in both directions, 200 permutations. The legend states that modulePreservation subsamples modules larger than 1,000 genes, so the three largest modules per direction are computed on a random 1,000-gene subsample. Unassigned (grey) modules appear as rows in the table for completeness and are reported where relevant (the HF → HCC grey module reaches Zsummary = 10.76), but they are excluded from the named-module counts quoted in the main text.

reference	Z.cor.kIM	Z.cor.kME	Z.cor.kMEall	Z.cor.cor	Z.cor.MAR	moduleSize obs	medianRank pres	medianRankDensity pres	medianRankConnectivity pres	propVarExplained pres	meanSignAwareKME pres
HF	-1.131	-1.133	19.1	2.765	-1.237	73	22	21.5	23	0.2559	0.1216

S19 Table (columns 3 of 4) (page 1 of 3)

Module preservation, all modules. Zsummary and median rank for every named module in both directions, 200 permutations. The legend states that modulePreservation subsamples modules larger than 1,000 genes, so the three largest modules per direction are computed on a random 1,000-gene subsample. Unassigned (grey) modules appear as rows in the table for completeness and are reported where relevant (the HF → HCC grey module reaches Zsummary = 10.76), but they are excluded from the named-module counts quoted in the main text.

reference	meanSignAwareCorDat pres	meanAdj.pres	meanMAR.pres	cor.kIM	cor.kME	cor.kMEall	cor.cor	cor.MAR	log.p Bonfsummary pres	log.p Bonfdensity pres	log.p Bonfconnectivity pres
HCC	0.1386	0.01434	0.05754	0.5788	0.4877	0.2714	0.3661	0.5155	-52.91	-71.99	-33.84
HCC	0.201	0.02575	0.07915	0.6229	0.4776	0.4205	0.3814	0.6275	-39.66	-66.64	-12.69
HCC	0.08849	0.007374	0.04923	0.387	0.3386	0.1745	0.2523	0.4376	-17.9	-18.91	-16.89
HCC	0.1832	0.02059	0.07268	0.5011	0.4896	0.3196	0.3579	0.4698	-19.07	-30.03	-8.107
HCC	0.08926	0.00398	0.02049	0.2053	0.1765	0.1746	0.18	0.06062	-13.79	-14.91	-12.67
HCC	0.1396	0.008446	0.03157	0.2681	0.2421	0.0997	0.1938	0.3534	-14.72	-26.57	-2.872
HCC	0.08915	0.00513	0.02529	0.1411	0.08507	0.1214	0.1214	0.141	-6.922	-12.49	-1.356
HCC	0.118	0.00402	0.01737	0.1617	0.1489	0.2778	0.1327	0.0132	-6.12	-10.24	-2.002
HCC	0.1014	0.003961	0.02178	0.4861	0.5002	0.2573	0.3223	0.178	-5.226	-2.246	-8.206
HCC	0.06927	0.005032	0.0306	0.1617	0.07815	0.07821	0.09578	0.07948	-3.443	-6.16	-0.7257
HCC	0.06547	0.00678	0.04251	0.05759	0.1137	0.07365	0.08835	0.2343	-2.46	-4.921	0
HCC	0.1394	0.004572	0.01532	0.2314	0.3727	0.2357	0.2013	0.0401	-1.362	-1.878	-0.8459
HCC	0.0934	0.004006	0.01815	0.186	0.05863	0.1798	0.1469	0.2131	-0.9991	-1.841	-0.1568
HCC	0.07837	0.003552	0.01912	0.08298	0.03615	0.1013	0.05343	0.03929	-1.244	-2.488	0
HCC	0.04475	0.002356	0.01597	0.1134	0.06625	0.01092	0.09947	0.1131	-1.178	0	-2.355
HCC	0.01305	0.00226	0.01955	0.006	0.1014	0.2103	0.07756	0.8716	-1.575	0	-3.15
HCC	0.06875	0.004274	0.02183	-0.08185	0.05836	0.01068	0.1486	-0.04749	-0.2271	-0.4543	0
HCC	0.07199	0.002371	0.01443	0.05541	0.012	0.08444	0.1417	-0.04205	-0.2714	-0.5428	0
HCC	0.06552	0.002967	0.01476	-0.03737	0.06059	0.1668	0.06763	0.03973	-0.4354	-0.8707	0
HCC	0.04217	0.001656	0.01032	0.1061	0.0175	0.06005	0.1418	0.05825	0	0	0
HCC	0.03597	0.002558	0.01971	-0.1818	0.03655	0.09065	0.01532	-0.1329	0	0	0
HF	0.256	0.03649	0.08519	0.185	0.1358	0.3661	0.1132	0.1462	-43.31	-83.39	-3.239

S19 Table (columns 3 of 4) (page 2 of 3)

Module preservation, all modules. Zsummary and median rank for every named module in both directions, 200 permutations. The legend states that modulePreservation subsamples modules larger than 1,000 genes, so the three largest modules per direction are computed on a random 1,000-gene subsample. Unassigned (grey) modules appear as rows in the table for completeness and are reported where relevant (the HF → HCC grey module reaches Zsummary = 10.76), but they are excluded from the named-module counts quoted in the main text.

reference	meanSignAwareCorDat pres	meanAdj.pres	meanMAR.pres	cor.kIM	cor.kME	cor.kMEall	cor.cor	cor.MAR	log.p Bonfsummary pres	log.p Bonfdensity pres	log.p Bonfconnectivity pres
HF	0.3213	0.1026	0.2107	0.3538	0.2433	0.2037	0.1977	0.3228	-41.46	-79.3	-3.613
HF	0.03126	0.009098	0.05392	0.1004	0.3931	0.4714	0.1202	0.2428	-39.52	-9.362	-69.68
HF	0.3158	0.05321	0.1084	0.2365	0.2482	0.2867	0.1752	0.2149	-38.72	-74.08	-3.362
HF	0.1415	0.02464	0.07466	0.06348	0.06298	0.271	0.07389	0.03819	-31.23	-62.18	-0.2795
HF	0.2849	0.04085	0.09364	0.2167	0.1878	0.2821	0.1508	0.1897	-23.74	-46.3	-1.189
HF	0.3365	0.04505	0.09053	0.0267	0.008329	0.4224	0.04494	-0.0741	-14.77	-29.54	0
HF	0.3339	0.03915	0.1043	0.4039	0.2472	0.2256	0.2489	0.4589	-5.68	-10.39	-0.9666
HF	0.01566	0.01028	0.05933	0.0615	0.1593	0.003908	0.1187	0.007814	-4.947	0	-9.894
HF	0.1108	0.01669	0.0559	-0.00674	0.1814	0.1666	0.1521	-0.07147	-0.9678	-1.936	0
HF	0.1727	0.02882	0.09081	0.02857	0.01779	0.1503	0.02407	0.02038	-2.372	-4.745	0
HF	0.1456	0.02115	0.08778	0.3803	0.1833	0.05709	0.356	0.2142	-0.5639	-0.4491	-0.6786
HF	0.1641	0.01438	0.05412	0.1894	0.2114	0.07186	0.1977	0.1262	-0.4904	-0.8143	-0.1665
HF	0.0548	0.01029	0.04731	0.1634	0.1464	0.0567	0.09365	0.1238	-0.9872	0	-1.974
HF	0.0523	0.00806	0.05096	0.1564	0.01779	0.04083	0.1532	0.2549	-2.556	0	-5.112
HF	0.1165	0.009969	0.05144	-0.02239	0.1787	0.1827	0.06287	0.07058	-0.004122	-0.008245	0
HF	0.06203	0.006051	0.03711	0.2643	0.1043	0.06506	0.1344	0.1906	-0.4424	0	-0.8847
HF	0.13	0.0171	0.0581	-0.02598	0.01081	0.103	0.05781	0.006053	-0.05027	-0.1005	0
HF	0.06581	0.003164	0.01353	0.1933	0.3155	0.07861	0.1918	0.1246	-0.02399	0	-0.04798
HF	0.07762	0.009414	0.0539	0.004223	0.01438	0.0377	0.1027	0.3206	0	0	0
HF	0.03498	0.008938	0.04729	0.1254	0.08158	0.03847	0.08342	0.1411	0	0	0
HF	0.05239	0.006352	0.03316	0.01347	0.09228	0.03159	0.08316	0.008723	0	0	0
HF	0.03138	0.005073	0.03346	-0.005852	0.01901	0.003571	0.04755	0.05581	0	0	0

S19 Table (columns 3 of 4) (page 3 of 3)

Module preservation, all modules. Zsummary and median rank for every named module in both directions, 200 permutations. The legend states that modulePreservation subsamples modules larger than 1,000 genes, so the three largest modules per direction are computed on a random 1,000-gene subsample. Unassigned (grey) modules appear as rows in the table for completeness and are reported where relevant (the HF → HCC grey module reaches Zsummary = 10.76), but they are excluded from the named-module counts quoted in the main text.

reference	meanSignAwareCorDat pres	meanAdj.pres	meanMAR.pres	cor.kIM	cor.kME	cor.kMEall	cor.cor	cor.MAR	log.p Bonfsummary pres	log.p Bonfdensity pres	log.p Bonfconnectivity pres
HF	0.03314	0.005796	0.03174	-0.1241	0.01192	0.1407	0.09396	-0.1424	0	0	0

S19 Table (columns 4 of 4) (page 1 of 3)

Module preservation, all modules. Zsummary and median rank for every named module in both directions, 200 permutations. The legend states that modulePreservation subsamples modules larger than 1,000 genes, so the three largest modules per direction are computed on a random 1,000-gene subsample. Unassigned (grey) modules appear as rows in the table for completeness and are reported where relevant (the HF → HCC grey module reaches Zsummary = 10.76), but they are excluded from the named-module counts quoted in the main text.

reference	p.summary	p.summary.bh	preservation class
HCC	1.219e-53	2.56e-52	strong (>10)
HCC	2.168e-40	2.276e-39	strong (>10)
HCC	1.255e-18	6.588e-18	weak-moderate (2-10)
HCC	8.538e-20	5.976e-19	weak-moderate (2-10)
HCC	1.628e-14	5.699e-14	weak-moderate (2-10)
HCC	1.904e-15	7.995e-15	weak-moderate (2-10)
HCC	1.197e-07	3.592e-07	weak-moderate (2-10)
HCC	7.593e-07	1.993e-06	weak-moderate (2-10)
HCC	5.942e-06	1.386e-05	weak-moderate (2-10)
HCC	0.0003607	0.0007575	weak-moderate (2-10)
HCC	0.003464	0.006613	weak-moderate (2-10)
HCC	0.04347	0.07022	weak-moderate (2-10)
HCC	0.1002	0.1315	weak-moderate (2-10)
HCC	0.05702	0.08552	weak-moderate (2-10)
HCC	0.06645	0.09303	none (<2)
HCC	0.02661	0.04656	none (<2)
HCC	0.5927	0.6551	none (<2)
HCC	0.5353	0.6245	none (<2)
HCC	0.367	0.4533	none (<2)
HCC	1	1	none (<2)
HCC	1	1	none (<2)
HF	4.855e-44	1.165e-42	strong (>10)

S19 Table (columns 4 of 4) (page 2 of 3)

Module preservation, all modules. Zsummary and median rank for every named module in both directions, 200 permutations. The legend states that modulePreservation subsamples modules larger than 1,000 genes, so the three largest modules per direction are computed on a random 1,000-gene subsample. Unassigned (grey) modules appear as rows in the table for completeness and are reported where relevant (the HF → HCC grey module reaches Zsummary = 10.76), but they are excluded from the named-module counts quoted in the main text.

reference	p.summary	p.summary.bh	preservation class
HF	3.494e-42	4.193e-41	strong (>10)
HF	3.027e-40	2.422e-39	strong (>10)
HF	1.903e-39	1.142e-38	strong (>10)
HF	5.909e-32	2.836e-31	weak-moderate (2-10)
HF	1.8e-24	7.199e-24	weak-moderate (2-10)
HF	1.701e-15	5.831e-15	weak-moderate (2-10)
HF	2.089e-06	6.266e-06	weak-moderate (2-10)
HF	1.129e-05	3.011e-05	weak-moderate (2-10)
HF	0.1077	0.1988	weak-moderate (2-10)
HF	0.004244	0.009259	weak-moderate (2-10)
HF	0.273	0.468	weak-moderate (2-10)
HF	0.3233	0.5173	weak-moderate (2-10)
HF	0.103	0.1988	none (<2)
HF	0.002781	0.006674	none (<2)
HF	0.9906	1	none (<2)
HF	0.3611	0.5417	none (<2)
HF	0.8907	1	none (<2)
HF	0.9463	1	none (<2)
HF	1	1	none (<2)
HF	1	1	none (<2)
HF	1	1	none (<2)
HF	1	1	none (<2)

S19 Table (columns 4 of 4) (page 3 of 3)

Module preservation, all modules. Zsummary and median rank for every named module in both directions, 200 permutations. The legend states that modulePreservation subsamples modules larger than 1,000 genes, so the three largest modules per direction are computed on a random 1,000-gene subsample. Unassigned (grey) modules appear as rows in the table for completeness and are reported where relevant (the HF → HCC grey module reaches Zsummary = 10.76), but they are excluded from the named-module counts quoted in the main text.

reference	p.summary	p.summary.bh	preservation class
HF	1	1	none (<2)

S20 Table (columns 1 of 4)

Module preservation, designated modules. Zsummary, median rank, BH FDR, rank among named modules, and preservation class for HF purple tested in HCC and HCC magenta tested in HF.

reference	module	test	moduleSize	Zsummary.pres	Zdensity.pres	Zconnectivity pres	Z propVarExplained pres	Z meanSignAwareKME pres	Z meanSignAwareCorDat pres	Z.meanAdj pres	Z.meanMAR pres
HCC	magenta	HF	146	7.094	10.35	3.842	5.852	6.455	14.74	14.23	3.643
HF	purple	HCC	153	11.03	17.78	4.27	10.85	10.22	24.71	44	20.65

S20 Table (columns 2 of 4)

Module preservation, designated modules. Zsummary, median rank, BH FDR, rank among named modules, and preservation class for HF purple tested in HCC and HCC magenta tested in HF.

reference	Z.cor.kIM	Z.cor.kME	Z.cor.kMEall	Z.cor.cor	Z.cor.MAR	moduleSize obs	medianRank pres	medianRankDensity pres	medianRankConnectivity pres	propVarExplained pres	meanSignAwareKME pres
HCC	3.135	3.842	12.39	9.956	4.241	146	5	4	7	0.2505	0.3558
HF	4.27	3.692	28.21	9.84	3.757	153	3	2	4	0.5479	0.557

S20 Table (columns 3 of 4)

Module preservation, designated modules. Zsummary, median rank, BH FDR, rank among named modules, and preservation class for HF purple tested in HCC and HCC magenta tested in HF.

reference	meanSignAwareCorDat pres	meanAdj.pres	meanMAR.pres	cor.kIM	cor.kME	cor.kMEall	cor.cor	cor.MAR	log.p Bonfsummary pres	log.p Bonfdensity pres	log.p Bonfconnectivity pres
HCC	0.1396	0.008446	0.03157	0.2681	0.2421	0.0997	0.1938	0.3534	-14.72	-26.57	-2.872
HF	0.3213	0.1026	0.2107	0.3538	0.2433	0.2037	0.1977	0.3228	-41.46	-79.3	-3.613

S20 Table (columns 4 of 4)

Module preservation, designated modules. Zsummary, median rank, BH FDR, rank among named modules, and preservation class for HF purple tested in HCC and HCC magenta tested in HF.

reference	p.summary	p.summary.bh	preservation class
HCC	1.904e-15	7.995e-15	weak-moderate (2-10)
HF	3.494e-42	4.193e-41	strong (>10)

S20b Table

Designated module preservation verdicts. The two-row summary of S20 in the form used in the Results text.

reference	module	test	Zsummary	verdict
HCC	magenta	HF	7.094	weakly preserved
HF	purple	HCC	11.03	strongly preserved

S21 Table (page 1 of 4)

Between-cohort heterogeneity. Cochran's Q, pQ, I^2, and pooled effect size with standard error for every set within each disease.

disease	set	tier	n.cohorts	Q	p.Q	I2.pct	g.pooled	se.pooled	cohorts
HCC	...	B_extra	4	67.68	1.338e-14	95.6	2.255	0.4344	TCGA_LIHC; GSE14520_GPL3921;...
HCC	REACTOME_TRNA_PROCESSING	B_extra	4	59.91	6.132e-13	95	2.099	0.4104	TCGA_LIHC; GSE14520_GPL3921;...
HCC	HALLMARK_MYC_TARGETS_V1	C_v1	4	71.97	1.618e-15	95.8	2.065	0.4062	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	C_v1	4	15.87	0.001207	81.1	1.662	0.2013	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	C_v1	4	25	1.543e-05	88	1.627	0.2879	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	A	4	51.83	3.255e-11	94.2	1.618	0.3937	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	A	4	49.8	8.794e-11	94	1.485	0.3862	TCGA_LIHC; GSE14520_GPL3921;...
HCC	REACTOME_TRANSLATION	A	4	103.2	3.251e-22	97.1	1.472	0.5056	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	C_v1	4	31.94	5.398e-07	90.6	1.463	0.2888	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	C_v1	4	78.04	8.072e-17	96.2	1.451	0.4348	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	C_v1	4	21.28	9.224e-05	85.9	1.377	0.2171	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	A	4	29.02	2.213e-06	89.7	1.298	0.2946	TCGA_LIHC; GSE14520_GPL3921;...
HCC	REACTOME_RRNA_PROCESSING	A	4	36.61	5.566e-08	91.8	1.264	0.2481	TCGA_LIHC; GSE14520_GPL3921;...
HCC	REACTOME_TRNA_AMINOACYLATION	A	4	5.957	0.1137	49.6	1.247	0.1064	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	B_extra	4	20.71	0.0001207	85.5	1.213	0.2154	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	A	2	0.05818	0.8094	0	1.18	0.1155	TCGA_LIHC; GSE14520_GPL571
HCC	...	A	4	30.94	8.767e-07	90.3	1.15	0.2631	TCGA_LIHC; GSE14520_GPL3921;...
HCC	HALLMARK_MYC_TARGETS_V2	C_v1	4	28.78	2.488e-06	89.6	1.136	0.227	TCGA_LIHC; GSE14520_GPL3921;...
HCC	HALLMARK_MTORC1_SIGNALING	C_v1	4	72.66	1.15e-15	95.9	1.119	0.3813	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	C_v1	4	11.81	0.008056	74.6	1.098	0.1521	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	B_extra	4	27.56	4.502e-06	89.1	1.092	0.2161	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	A	4	4.487	0.2135	33.1	0.9885	0.09697	TCGA_LIHC; GSE14520_GPL3921;...

S21 Table (page 2 of 4)

Between-cohort heterogeneity. Cochran's Q, pQ, I^2, and pooled effect size with standard error for every set within each disease.

disease	set	tier	n.cohorts	Q	p.Q	I2.pct	g.pooled	se.pooled	cohorts
HCC	KEGG_RIBOSOME	A	4	20.42	0.000139	85.3	0.9477	0.215	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	A	4	60.86	3.847e-13	95.1	0.8647	0.3536	TCGA_LIHC; GSE14520_GPL3921;...
HCC	REACTOME_MITOCHONDRIAL_TRANSLATION	A	4	68.77	7.823e-15	95.6	0.8507	0.373	TCGA_LIHC; GSE14520_GPL3921;...
HCC	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	4	11.17	0.01084	73.1	0.8458	0.1395	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	A	4	13.76	0.003257	78.2	0.6806	0.155	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	C_v1	4	9.17	0.02711	67.3	0.6775	0.1263	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	C_v1	4	6.559	0.08735	54.3	0.6008	0.1143	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	B_extra	1				-0.5117	0.08026	TCGA_LIHC
HCC	...	B_extra	4	33.41	2.634e-07	91	0.5025	0.228	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	C_v1	3	2.459	0.2925	18.7	-0.4679	0.1019	TCGA_LIHC; GSE14520_GPL571;...
HCC	...	C_v1	4	6.216	0.1016	51.7	0.4498	0.09852	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	C_v1	4	87.55	7.361e-19	96.6	0.3573	0.359	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	C_v1	1				0.352	0.1097	TCGA_LIHC
HCC	...	A	4	16.78	0.0007854	82.1	-0.2118	0.1556	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	C_v1	1				0.1561	0.1195	TCGA_LIHC
HCC	...	C_v1	4	10.87	0.01247	72.4	-0.04296	0.1256	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	C_v1	1				-0.03293	0.1165	TCGA_LIHC
HCC	...	C_v1	1				-0.02133	0.1186	TCGA_LIHC
HCC	...	C_v1	4	12.95	0.004753	76.8	-0.01429	0.145	TCGA_LIHC; GSE14520_GPL3921;...
HF	...	C_v1	3	15.79	0.0003728	87.3	-1.261	0.2737	GSE57338; GSE116250; GSE141910
HF	...	C_v1	3	8.03	0.01804	75.1	-0.9447	0.1878	GSE57338; GSE116250; GSE141910
HF	...	C_v1	3	17.59	0.0001512	88.6	-0.8735	0.2644	GSE57338; GSE116250; GSE141910

S21 Table (page 3 of 4)

Between-cohort heterogeneity. Cochran's Q, pQ, I^2, and pooled effect size with standard error for every set within each disease.

disease	set	tier	n.cohorts	Q	p.Q	I2.pct	g.pooled	se.pooled	cohorts
HF	HALLMARK_MTORC1_SIGNALING	C_v1	3	8.608	0.01351	76.8	-0.788	0.1777	GSE57338; GSE116250; GSE141910
HF	...	A	3	11.3	0.003524	82.3	-0.7436	0.2037	GSE57338; GSE116250; GSE141910
HF	...	B_extra	3	6.884	0.03201	70.9	-0.7142	0.1649	GSE57338; GSE116250; GSE141910
HF	...	C_v1	3	77.32	1.619e-17	97.4	-0.6666	0.5688	GSE57338; GSE116250; GSE141910
HF	...	C_v1	3	21.76	1.887e-05	90.8	-0.6042	0.2853	GSE57338; GSE116250; GSE141910
HF	REACTOME_TRNA_PROCESSING	B_extra	3	2.342	0.31	14.6	-0.5561	0.08326	GSE57338; GSE116250; GSE141910
HF	...	A	3	0.2268	0.8928	0	0.5241	0.07572	GSE57338; GSE116250; GSE141910
HF	HALLMARK_MYC_TARGETS_V1	C_v1	3	21.06	2.671e-05	90.5	-0.4596	0.2519	GSE57338; GSE116250; GSE141910
HF	HALLMARK_MYC_TARGETS_V2	C_v1	3	55.1	1.085e-12	96.4	-0.4542	0.396	GSE57338; GSE116250; GSE141910
HF	...	B_extra	3	5.04	0.08045	60.3	-0.429	0.1327	GSE57338; GSE116250; GSE141910
HF	...	B_extra	3	8.048	0.01788	75.2	-0.415	0.1725	GSE57338; GSE116250; GSE141910
HF	...	B_extra	3	23.31	8.669e-06	91.4	-0.4108	0.2641	GSE57338; GSE116250; GSE141910
HF	...	C_v1	3	13.19	0.001369	84.8	-0.3975	0.2212	GSE57338; GSE116250; GSE141910
HF	...	C_v1	3	2.185	0.3354	8.5	-0.3816	0.07956	GSE57338; GSE116250; GSE141910
HF	...	C_v1	3	62.88	2.222e-14	96.8	-0.3475	0.4796	GSE57338; GSE116250; GSE141910
HF	...	B_extra	2	0.329	0.5662	0	-0.3219	0.07671	GSE57338; GSE141910
HF	...	C_v1	3	8.559	0.01385	76.6	-0.3151	0.169	GSE57338; GSE116250; GSE141910
HF	REACTOME_RRNA_PROCESSING	A	3	31.84	1.22e-07	93.7	-0.3074	0.3201	GSE57338; GSE116250; GSE141910
HF	...	C_v1	3	1.141	0.5651	0	-0.2913	0.07213	GSE57338; GSE116250; GSE141910
HF	...	C_v1	3	8.216	0.01644	75.7	-0.288	0.1685	GSE57338; GSE116250; GSE141910
HF	...	A	3	6.473	0.03931	69.1	-0.2856	0.1432	GSE57338; GSE116250; GSE141910
HF	REACTOME_TRANSLATION	A	3	16.98	0.0002056	88.2	-0.2826	0.2445	GSE57338; GSE116250; GSE141910

S21 Table (page 4 of 4)

Between-cohort heterogeneity. Cochran's Q, pQ, I^2, and pooled effect size with standard error for every set within each disease.

disease	set	tier	n.cohorts	Q	p.Q	I2.pct	g.pooled	se.pooled	cohorts
HF	...	C_v1	3	18.29	0.0001065	89.1	-0.2668	0.245	GSE57338; GSE116250; GSE141910
HF	...	C_v1	3	20.32	3.875e-05	90.2	-0.2356	0.2584	GSE57338; GSE116250; GSE141910
HF	...	C_v1	3	21.73	1.911e-05	90.8	-0.2122	0.2722	GSE57338; GSE116250; GSE141910
HF	REACTOME_MITOCHONDRIAL_TRANSLATION	A	3	9.682	0.0079	79.3	-0.1982	0.1838	GSE57338; GSE116250; GSE141910
HF	...	A	3	28.95	5.162e-07	93.1	-0.1739	0.3325	GSE57338; GSE116250; GSE141910
HF	...	A	3	8.333	0.0155	76	-0.1355	0.1697	GSE57338; GSE116250; GSE141910
HF	...	A	3	18.18	0.0001129	89	-0.1277	0.2668	GSE57338; GSE116250; GSE141910
HF	REACTOME_TRNA_AMINOACYLATION	A	3	7.223	0.02701	72.3	-0.1125	0.1565	GSE57338; GSE116250; GSE141910
HF	...	C_v1	3	36.73	1.056e-08	94.6	0.1105	0.3827	GSE57338; GSE116250; GSE141910
HF	...	A	3	26.95	1.403e-06	92.6	-0.07609	0.3231	GSE57338; GSE116250; GSE141910
HF	...	C_v1	3	14.19	0.0008305	85.9	-0.05933	0.2222	GSE57338; GSE116250; GSE141910
HF	...	A	3	26.62	1.658e-06	92.5	0.0533	0.3303	GSE57338; GSE116250; GSE141910
HF	...	A	3	28.83	5.494e-07	93.1	-0.04594	0.3434	GSE57338; GSE116250; GSE141910
HF	KEGG_RIBOSOME	A	3	24.03	6.065e-06	91.7	0.01552	0.309	GSE57338; GSE116250; GSE141910
HF	...	C_v1	3	36.42	1.232e-08	94.5	0.01364	0.3782	GSE57338; GSE116250; GSE141910
HF	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	3	4.976	0.08306	59.8	-0.002771	0.1301	GSE57338; GSE116250; GSE141910

S22 Table (page 1 of 2)

HF aetiology stratification (summary). Each HF subgroup against TCGA-LIHC: subgroup sizes, cross-disease Spearman rho, mirror set counts before and after the effect-size gate, and mean g.

set	g.DCM.vs.NF	g.ICM.vs.NF	tier	same direction
KEGG_RIBOSOME	-0.2746	-0.5563	A	True
KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	-0.207	-0.2082	A	True
REACTOME_TRANSLATION	-0.4619	-0.6029	A	True
...	-0.4714	-0.7269	A	True
...	-0.2922	-0.5914	A	True
REACTOME_TRNA_AMINOACYLATION	-0.3564	-0.3557	A	True
...	-0.8103	-0.8461	A	True
REACTOME_RRNA_PROCESSING	-0.6626	-0.7588	A	True
...	-0.4564	-0.6578	A	True
...	-0.3782	-0.7275	A	True
...	-0.3541	-0.6801	A	True
REACTOME_MITOCHONDRIAL_TRANSLATION	-0.05858	-0.1123	A	True
...	-0.05923	-0.0992	A	True
...	0.5985	0.5506	A	True
...	-0.1222	-0.09263	A	True
REACTOME_TRNA_PROCESSING	-0.4041	-0.4249	B_extra	True
...	-0.4822	-0.5813	B_extra	True
...	-0.1847	-0.08593	B_extra	True
...	-0.7435	-0.76	B_extra	True
...	-0.4767	-0.4773	B_extra	True
...	-0.3308	-0.4195	B_extra	True
...	-0.5631	-0.6587	C_v1	True

S22 Table (page 2 of 2)

HF aetiology stratification (summary). Each HF subgroup against TCGA-LIHC: subgroup sizes, cross-disease Spearman rho, mirror set counts before and after the effect-size gate, and mean g.

set	g.DCM.vs.NF	g.ICM.vs.NF	tier	same direction
HALLMARK_MYC_TARGETS_V1	-0.7464	-0.7666	C_v1	True
...	-0.5091	-0.6278	C_v1	True
...	-0.6841	-0.7331	C_v1	True
...	-0.9324	-1.031	C_v1	True
HALLMARK_MYC_TARGETS_V2	-1.098	-1.01	C_v1	True
...	-0.4667	-0.7082	C_v1	True
...	-1.013	-1.179	C_v1	True
...	-1.126	-1.142	C_v1	True
...	-0.3616	-0.6411	C_v1	True
HALLMARK_MTORC1_SIGNALING	-1.066	-0.8098	C_v1	True
...	-0.06269	0.1354	C_v1	False
...	-0.2431	-0.3304	C_v1	True
...	-0.8361	-0.3439	C_v1	True
...	-1.548	-1.683	C_v1	True
...	0.4368	0.1951	C_v1	True
...	-1.133	-1.041	C_v1	True
...	0.2685	0.03703	C_v1	True
...	0.3304	0.07489	C_v1	True
...	0.06916	-0.217	C_v1	False

S22b Table (page 1 of 4)

HF aetiology effect sizes. Per-set Hedges g for each aetiology subgroup, with a flag for whether the DCM and ICM arms of GSE57338 agree in direction.

cohort	aetiology	n.aetiology	n.control	set	tier	hedges.g	direction
GSE57338	DCM	82	136	KEGG_RIBOSOME	A	-0.2746	down in disease
GSE57338	DCM	82	136	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	-0.207	down in disease
GSE57338	DCM	82	136	REACTOME_TRANSLATION	A	-0.4619	down in disease
GSE57338	DCM	82	136	...	A	-0.4714	down in disease
GSE57338	DCM	82	136	...	A	-0.2922	down in disease
GSE57338	DCM	82	136	REACTOME_TRNA_AMINOACYLATION	A	-0.3564	down in disease
GSE57338	DCM	82	136	...	A	-0.8103	down in disease
GSE57338	DCM	82	136	REACTOME_RRNA_PROCESSING	A	-0.6626	down in disease
GSE57338	DCM	82	136	...	A	-0.4564	down in disease
GSE57338	DCM	82	136	...	A	-0.3782	down in disease
GSE57338	DCM	82	136	...	A	-0.3541	down in disease
GSE57338	DCM	82	136	REACTOME_MITOCHONDRIAL_TRANSLATION	A	-0.05858	down in disease
GSE57338	DCM	82	136	...	A	-0.05923	down in disease
GSE57338	DCM	82	136	...	A	0.5985	up in disease
GSE57338	DCM	82	136	...	A	-0.1222	down in disease
GSE57338	ICM	95	136	KEGG_RIBOSOME	A	-0.5563	down in disease
GSE57338	ICM	95	136	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	-0.2082	down in disease
GSE57338	ICM	95	136	REACTOME_TRANSLATION	A	-0.6029	down in disease
GSE57338	ICM	95	136	...	A	-0.7269	down in disease
GSE57338	ICM	95	136	...	A	-0.5914	down in disease
GSE57338	ICM	95	136	REACTOME_TRNA_AMINOACYLATION	A	-0.3557	down in disease
GSE57338	ICM	95	136	...	A	-0.8461	down in disease

S22b Table (page 2 of 4)

HF aetiology effect sizes. Per-set Hedges g for each aetiology subgroup, with a flag for whether the DCM and ICM arms of GSE57338 agree in direction.

cohort	aetiology	n.aetiology	n.control	set	tier	hedges.g	direction
GSE57338	ICM	95	136	REACTOME_RRNA_PROCESSING	A	-0.7588	down in disease
GSE57338	ICM	95	136	...	A	-0.6578	down in disease
GSE57338	ICM	95	136	...	A	-0.7275	down in disease
GSE57338	ICM	95	136	...	A	-0.6801	down in disease
GSE57338	ICM	95	136	REACTOME_MITOCHONDRIAL_TRANSLATION	A	-0.1123	down in disease
GSE57338	ICM	95	136	...	A	-0.0992	down in disease
GSE57338	ICM	95	136	...	A	0.5506	up in disease
GSE57338	ICM	95	136	...	A	-0.09263	down in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	KEGG_RIBOSOME	A	0.3306	up in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	0.1561	up in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	REACTOME_TRANSLATION	A	0.09562	up in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	...	A	0.2051	up in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	...	A	0.3325	up in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	REACTOME_TRNA_AMINOACYLATION	A	0.07557	up in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	...	A	-0.4163	down in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	REACTOME_RRNA_PROCESSING	A	-0.005967	down in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	...	A	0.2235	up in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	...	A	0.191	up in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	...	A	0.09461	up in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	REACTOME_MITOCHONDRIAL_TRANSLATION	A	0.1033	up in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	...	A	0.1514	up in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	...	A	0.5812	up in disease

S22b Table (page 3 of 4)

HF aetiology effect sizes. Per-set Hedges g for each aetiology subgroup, with a flag for whether the DCM and ICM arms of GSE57338 agree in direction.

cohort	aetiology	n.aetiology	n.control	set	tier	hedges.g	direction
GSE141910	Dilated cardiomyopathy (DCM)	166	166	...	A	-0.1394	down in disease
GSE141910	Hypertrophic cardiomyopathy (HCM)	28	166	KEGG_RIBOSOME	A	0.3192	up in disease
GSE141910	Hypertrophic cardiomyopathy (HCM)	28	166	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	-0.1521	down in disease
GSE141910	Hypertrophic cardiomyopathy (HCM)	28	166	REACTOME_TRANSLATION	A	-0.03284	down in disease
GSE141910	Hypertrophic cardiomyopathy (HCM)	28	166	...	A	0.1966	up in disease
GSE141910	Hypertrophic cardiomyopathy (HCM)	28	166	...	A	0.3358	up in disease
GSE141910	Hypertrophic cardiomyopathy (HCM)	28	166	REACTOME_TRNA_AMINOACYLATION	A	-0.2584	down in disease
GSE141910	Hypertrophic cardiomyopathy (HCM)	28	166	...	A	-0.55	down in disease
GSE141910	Hypertrophic cardiomyopathy (HCM)	28	166	REACTOME_RRNA_PROCESSING	A	-0.06591	down in disease
GSE141910	Hypertrophic cardiomyopathy (HCM)	28	166	...	A	0.2378	up in disease
GSE141910	Hypertrophic cardiomyopathy (HCM)	28	166	...	A	0.1696	up in disease
GSE141910	Hypertrophic cardiomyopathy (HCM)	28	166	...	A	0.0668	up in disease
GSE141910	Hypertrophic cardiomyopathy (HCM)	28	166	REACTOME_MITOCHONDRIAL_TRANSLATION	A	-0.07351	down in disease
GSE141910	Hypertrophic cardiomyopathy (HCM)	28	166	...	A	-0.005165	down in disease
GSE141910	Hypertrophic cardiomyopathy (HCM)	28	166	...	A	0.2338	up in disease
GSE141910	Hypertrophic cardiomyopathy (HCM)	28	166	...	A	-0.7135	down in disease
GSE141910	Peripartum cardiomyopathy (PPCM)	6	166	KEGG_RIBOSOME	A	0.1651	up in disease
GSE141910	Peripartum cardiomyopathy (PPCM)	6	166	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	-0.1461	down in disease
GSE141910	Peripartum cardiomyopathy (PPCM)	6	166	REACTOME_TRANSLATION	A	-0.2133	down in disease
GSE141910	Peripartum cardiomyopathy (PPCM)	6	166	...	A	0.1093	up in disease
GSE141910	Peripartum cardiomyopathy (PPCM)	6	166	...	A	0.1788	up in disease
GSE141910	Peripartum cardiomyopathy (PPCM)	6	166	REACTOME_TRNA_AMINOACYLATION	A	-0.2554	down in disease

S22b Table (page 4 of 4)

HF aetiology effect sizes. Per-set Hedges g for each aetiology subgroup, with a flag for whether the DCM and ICM arms of GSE57338 agree in direction.

cohort	aetiology	n.aetiology	n.control	set	tier	hedges.g	direction
GSE141910	Peripartum cardiomyopathy (PPCM)	6	166	...	A	-0.4534	down in disease
GSE141910	Peripartum cardiomyopathy (PPCM)	6	166	REACTOME_RRNA_PROCESSING	A	-0.005683	down in disease
GSE141910	Peripartum cardiomyopathy (PPCM)	6	166	...	A	0.1444	up in disease
GSE141910	Peripartum cardiomyopathy (PPCM)	6	166	...	A	0.0198	up in disease
GSE141910	Peripartum cardiomyopathy (PPCM)	6	166	...	A	0.03514	up in disease
GSE141910	Peripartum cardiomyopathy (PPCM)	6	166	REACTOME_MITOCHONDRIAL_TRANSLATION	A	-0.3693	down in disease
GSE141910	Peripartum cardiomyopathy (PPCM)	6	166	...	A	-0.2895	down in disease
GSE141910	Peripartum cardiomyopathy (PPCM)	6	166	...	A	0.1204	up in disease
GSE141910	Peripartum cardiomyopathy (PPCM)	6	166	...	A	-0.9343	down in disease

S22c Table

HF aetiology versus HCC. Cross-disease correlations computed separately within each aetiology arm, so that aetiology composition can be assessed as an explanation for cohort discordance.

cohort	aetiology	n.aetiology	n.control	n.sets	spearman.rho vs.HCC	mirror.sets abs.g.ge.020	mirror.sets gated	n.gated	mean.g
GSE141910	Dilated cardiomyopathy (DCM)	166	166	15	-0.1929	1	1	13	0.1319
GSE141910	Peripartum cardiomyopathy (PPCM)	6	166	15	0.1143	3	3	13	-0.1263
GSE141910	Hypertrophic cardiomyopathy (HCM)	28	166	15	-0.1	2	2	13	-0.01946
GSE57338	ICM	95	136	15	-0.5964	11	11	13	-0.431
GSE57338	DCM	82	136	15	-0.5536	11	11	13	-0.2911

S22d Table

DCM across cohorts. The same aetiology (dilated cardiomyopathy) sampled in GSE57338 and GSE141910 and compared per set, showing the same-aetiology, cross-cohort discordance.

set	g.GSE57338 DCM	g.GSE141910 DCM
KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	-0.207	0.1561
KEGG_RIBOSOME	-0.2746	0.3306
...	-0.8103	-0.4163
...	-0.2922	0.3325
...	-0.4714	0.2051
REACTOME_MITOCHONDRIAL_TRANSLATION	-0.05858	0.1033
...	-0.05923	0.1514
...	0.5985	0.5812
...	-0.3782	0.191
...	-0.3541	0.09461
REACTOME_RRNA_PROCESSING	-0.6626	-0.005967
...	-0.1222	-0.1394
...	-0.4564	0.2235
REACTOME_TRANSLATION	-0.4619	0.09562
REACTOME_TRNA_AMINOACYLATION	-0.3564	0.07557

S23 Table (page 1 of 3)

Composition marker coverage. The marker genes available in each cohort for each composition panel, with the panels lost entirely on the GSE57338 array made explicit — 10 of 24 pan-immune markers are retained, and the lymphoid and proliferation panels are lost outright, so that cohort's "immune-adjusted" model is myeloid-only.

cohort	panel	n.used	n.total	frac.used	mean.rank	mean.z	missing
GSE57338	immune_pan	10	24	0.417	4699	1.836e-17	...
GSE57338	myeloid	7	9	0.778	6445	-2.781e-17	FCGR3A,ITGAX
GSE57338	fibroblast	12	14	0.857	8480	6.298e-16	FAP,THY1
GSE57338	endothelial	8	8	1	8504	-1.035e-16	
GSE57338	hepatocyte	3	14	0.214	3871	-1.382e-16	...
GSE57338	cardiomyocyte	14	14	1	1.121e+04	2.348e-16	
GSE141198	immune_pan	19	24	0.792	5036	1.223e-17	CD3G,CD19,MS4A1,CD68,IL2RA
GSE141198	myeloid	8	9	0.889	7613	-1.144e-16	CD68
GSE141198	lymphocyte	8	10	0.8	3771	1.07e-16	CD19,MS4A1
GSE141198	fibroblast	11	14	0.786	7927	5.157e-17	LUM,FAP,POSTN
GSE141198	endothelial	8	8	1	7502	3.277e-17	
GSE141198	hepatocyte	14	14	1	1.271e+04	-3.124e-17	
GSE141198	cardiomyocyte	3	14	0.214	5105	-1.594e-16	...
GSE141198	proliferation	11	11	1	5475	-1.427e-16	
TCGA_LIHC	immune_pan	23	24	0.958	7512	-7.409e-18	CD68
TCGA_LIHC	myeloid	8	9	0.889	1.124e+04	-8.18e-17	CD68
TCGA_LIHC	lymphocyte	10	10	1	5991	7.239e-17	
TCGA_LIHC	fibroblast	14	14	1	1.2e+04	1.316e-16	
TCGA_LIHC	endothelial	8	8	1	1.202e+04	9.696e-18	
TCGA_LIHC	hepatocyte	14	14	1	1.637e+04	-8.279e-17	
TCGA_LIHC	cardiomyocyte	4	14	0.286	6527	-2.027e-16	...
TCGA_LIHC	proliferation	11	11	1	9213	1.261e-16	

S23 Table (page 2 of 3)

Composition marker coverage. The marker genes available in each cohort for each composition panel, with the panels lost entirely on the GSE57338 array made explicit — 10 of 24 pan-immune markers are retained, and the lymphoid and proliferation panels are lost outright, so that cohort's "immune-adjusted" model is myeloid-only.

cohort	panel	n.used	n.total	frac.used	mean.rank	mean.z	missing
GSE116250	immune_pan	24	24	1	1.215e+04	7.571e-18	
GSE116250	myeloid	9	9	1	1.819e+04	1.301e-17	
GSE116250	lymphocyte	10	10	1	9689	-3.753e-18	
GSE116250	fibroblast	14	14	1	2.316e+04	-2.038e-17	
GSE116250	endothelial	7	8	0.875	2.167e+04	-1.493e-17	PECAM1
GSE116250	hepatocyte	9	14	0.643	1.123e+04	-4.857e-17	CYP3A4,FGA,FGB,FGG,APOC3
GSE116250	cardiomyocyte	14	14	1	2.594e+04	2.835e-17	
GSE116250	proliferation	11	11	1	1.058e+04	3.523e-18	
GSE14520_GPL3921	immune_pan	11	24	0.458	3502	1.67e-16	...
GSE14520_GPL3921	myeloid	5	9	0.556	3817	2.984e-16	CD68,FCGR3A,LYZ,ITGAX
GSE14520_GPL3921	lymphocyte	4	10	0.4	2172	6.282e-17	CD3E,CD19,MS4A1,NKG7,GNLY,KLRD1
GSE14520_GPL3921	fibroblast	13	14	0.929	4271	-5.479e-17	FAP
GSE14520_GPL3921	endothelial	7	8	0.875	2139	-1.149e-17	TEK
GSE14520_GPL3921	hepatocyte	14	14	1	7379	-1.547e-16	
GSE14520_GPL3921	proliferation	4	11	0.364	3759	-2.159e-17	...
GSE14520_GPL571	immune_pan	12	24	0.5	3480	-4.649e-17	...
GSE14520_GPL571	myeloid	5	9	0.556	4757	1.232e-18	CD68,FCGR3A,ITGAM,ITGAX
GSE14520_GPL571	lymphocyte	5	10	0.5	1871	-2.372e-17	CD19,MS4A1,NKG7,GNLY,KLRD1
GSE14520_GPL571	fibroblast	13	14	0.929	4435	-5.566e-17	FAP
GSE14520_GPL571	endothelial	7	8	0.875	2590	6.362e-17	TEK
GSE14520_GPL571	hepatocyte	14	14	1	7403	-9.841e-17	
GSE14520_GPL571	proliferation	5	11	0.455	3510	-9.904e-17	MKI67,CCNB1,CDK1,BUB1,PLK1,CCNA2

S23 Table (page 3 of 3)

Composition marker coverage. The marker genes available in each cohort for each composition panel, with the panels lost entirely on the GSE57338 array made explicit — 10 of 24 pan-immune markers are retained, and the lymphoid and proliferation panels are lost outright, so that cohort's "immune-adjusted" model is myeloid-only.

cohort	panel	n.used	n.total	frac.used	mean.rank	mean.z	missing
GSE76427	immune_pan	20	24	0.833	8051	-5.602e-17	PTPRC,CD3E,CD19,KLRD1
GSE76427	myeloid	9	9	1	1.11e+04	-5.703e-18	
GSE76427	lymphocyte	7	10	0.7	6696	-5.195e-17	CD3E,CD19,KLRD1
GSE76427	fibroblast	14	14	1	9007	-6.833e-17	
GSE76427	endothelial	8	8	1	9292	1.401e-17	
GSE76427	hepatocyte	14	14	1	1.644e+04	-5.303e-18	
GSE76427	cardiomyocyte	7	14	0.5	4566	3.056e-17	...
GSE76427	proliferation	8	11	0.727	8704	-2.806e-17	MKI67,CDK1,RRM2
GSE141910	immune_pan	21	24	0.875	5716	1.999e-16	CD19,CD68,FOXP3
GSE141910	myeloid	8	9	0.889	9415	5.042e-16	CD68
GSE141910	lymphocyte	9	10	0.9	3961	5.46e-17	CD19
GSE141910	fibroblast	14	14	1	1.432e+04	-3.238e-16	
GSE141910	endothelial	7	8	0.875	1.378e+04	-2.856e-16	PECAM1
GSE141910	hepatocyte	8	14	0.571	6099	-1.768e-16	TTR,CYP3A4,FGA,FGB,FGG,APOC3
GSE141910	cardiomyocyte	14	14	1	1.658e+04	7.583e-16	
GSE141910	proliferation	11	11	1	5087	1.822e-17	

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Composition shift. Marker-based composition scores by group in each cohort, showing the direction and magnitude of the composition difference between disease and control arms.

cohort	disease	panel	n.disease	n.control	mean.disease	mean.control	hedges.g	p.ttest	p.fdr
GSE57338	HF	immune_pan	177	136	4527	4923	-0.3597	0.001153	0.001383
GSE57338	HF	myeloid	177	136	6199	6765	-0.4904	1.184e-05	7.105e-05
GSE57338	HF	fibroblast	177	136	8645	8266	0.445	0.0001025	0.0002511
GSE57338	HF	endothelial	177	136	8448	8575	-0.3728	0.0009568	0.001383
GSE57338	HF	hepatocyte	177	136	3706	4086	-0.4323	0.0001256	0.0002511
GSE57338	HF	cardiomyocyte	177	136	1.129e+04	1.11e+04	0.3223	0.003833	0.003833
TCGA_LIHC	HCC	immune_pan	371	50	7419	8204	-0.4506	0.0002319	0.0003092
TCGA_LIHC	HCC	myeloid	371	50	1.113e+04	1.208e+04	-0.5322	2.787e-05	4.459e-05
TCGA_LIHC	HCC	lymphocyte	371	50	5855	7005	-0.5972	4.992e-08	9.984e-08
TCGA_LIHC	HCC	fibroblast	371	50	1.198e+04	1.217e+04	-0.08356	0.2665	0.2665
TCGA_LIHC	HCC	endothelial	371	50	1.195e+04	1.248e+04	-0.324	0.001166	0.001333
TCGA_LIHC	HCC	hepatocyte	371	50	1.627e+04	1.712e+04	-0.5822	5.993e-23	2.397e-22
TCGA_LIHC	HCC	cardiomyocyte	371	50	6603	5966	0.6018	8.247e-14	2.199e-13
TCGA_LIHC	HCC	proliferation	371	50	9791	4922	2.227	3.36e-47	2.688e-46
GSE116250	HF	immune_pan	50	14	1.234e+04	1.149e+04	0.5469	0.03318	0.04424
GSE116250	HF	myeloid	50	14	1.811e+04	1.844e+04	-0.2004	0.4697	0.4697
GSE116250	HF	lymphocyte	50	14	1.002e+04	8493	0.9507	0.0001904	0.0007617
GSE116250	HF	fibroblast	50	14	2.352e+04	2.189e+04	1.717	0.0004642	0.001238
GSE116250	HF	endothelial	50	14	2.183e+04	2.108e+04	1.341	5.045e-07	4.036e-06
GSE116250	HF	hepatocyte	50	14	1.117e+04	1.141e+04	-0.2595	0.4187	0.4697
GSE116250	HF	cardiomyocyte	50	14	2.604e+04	2.561e+04	1.674	0.002228	0.004456
GSE116250	HF	proliferation	50	14	1.027e+04	1.166e+04	-1.266	0.01355	0.02168

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Composition shift. Marker-based composition scores by group in each cohort, showing the direction and magnitude of the composition difference between disease and control arms.

cohort	disease	panel	n.disease	n.control	mean.disease	mean.control	hedges.g	p.ttest	p.fdr
GSE14520_GPL3921	HCC	immune_pan	225	220	3135	3877	-0.8265	5.764e-17	8.069e-17
GSE14520_GPL3921	HCC	myeloid	225	220	3478	4164	-0.9374	5.975e-21	1.046e-20
GSE14520_GPL3921	HCC	lymphocyte	225	220	1711	2644	-0.9854	9.099e-23	3.185e-22
GSE14520_GPL3921	HCC	fibroblast	225	220	4059	4487	-0.4215	1.014e-05	1.183e-05
GSE14520_GPL3921	HCC	endothelial	225	220	2080	2200	-0.181	0.05597	0.05597
GSE14520_GPL3921	HCC	hepatocyte	225	220	7198	7564	-0.9779	1.149e-21	2.68e-21
GSE14520_GPL3921	HCC	proliferation	225	220	5130	2357	3.017	7.528e-111	5.269e-110
GSE14520_GPL571	HCC	immune_pan	22	19	3171	3837	-0.8078	0.009723	0.01702
GSE14520_GPL571	HCC	myeloid	22	19	4390	5183	-1.026	0.001421	0.003316
GSE14520_GPL571	HCC	lymphocyte	22	19	1615	2166	-0.7622	0.01396	0.01955
GSE14520_GPL571	HCC	fibroblast	22	19	4332	4555	-0.1795	0.5398	0.6005
GSE14520_GPL571	HCC	endothelial	22	19	2642	2528	0.1546	0.6005	0.6005
GSE14520_GPL571	HCC	hepatocyte	22	19	7170	7671	-1.147	0.0006099	0.002135
GSE14520_GPL571	HCC	proliferation	22	19	4978	1811	3.116	1.15e-12	8.05e-12
GSE76427	HCC	immune_pan	115	52	7718	8788	-0.6595	1.921e-05	3.074e-05
GSE76427	HCC	myeloid	115	52	1.082e+04	1.173e+04	-0.6432	1.531e-05	3.062e-05
GSE76427	HCC	lymphocyte	115	52	6086	8046	-0.8249	1.111e-07	2.964e-07
GSE76427	HCC	fibroblast	115	52	8705	9675	-0.4353	0.002453	0.002804
GSE76427	HCC	endothelial	115	52	9027	9877	-0.5002	0.001072	0.001429
GSE76427	HCC	hepatocyte	115	52	1.628e+04	1.681e+04	-0.8143	3.714e-10	1.486e-09
GSE76427	HCC	cardiomyocyte	115	52	4497	4719	-0.2888	0.04875	0.04875
GSE76427	HCC	proliferation	115	52	9717	6465	1.693	1.525e-22	1.22e-21

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Composition shift. Marker-based composition scores by group in each cohort, showing the direction and magnitude of the composition difference between disease and control arms.

cohort	disease	panel	n.disease	n.control	mean.disease	mean.control	hedges.g	p.ttest	p.fdr
GSE141910	HF	immune_pan	200	166	5979	5398	0.6198	3.872e-09	6.195e-09
GSE141910	HF	myeloid	200	166	9318	9533	-0.1431	0.1725	0.2048
GSE141910	HF	lymphocyte	200	166	4510	3298	1.429	6.069e-36	4.855e-35
GSE141910	HF	fibroblast	200	166	1.475e+04	1.38e+04	1.318	2.555e-29	1.022e-28
GSE141910	HF	endothelial	200	166	1.383e+04	1.373e+04	0.1405	0.1792	0.2048
GSE141910	HF	hepatocyte	200	166	6110	6086	0.03083	0.7714	0.7714
GSE141910	HF	cardiomyocyte	200	166	1.671e+04	1.641e+04	0.812	3.391e-13	9.042e-13
GSE141910	HF	proliferation	200	166	4813	5418	-0.7642	8.716e-12	1.743e-11

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Composition adjustment. The unadjusted and adjusted correlation, the attenuation, whether the direction survives adjustment, and the maximum variance inflation factor, for every set–cohort pair across all three tiers (268 rows: 103 Tier A, 38 Tier B-extra, 127 Tier C-v1, over seven cohorts). The summary quoted in the main text — 55 of 103 pairs survive, median attenuation 34.1%, maximum VIF 1.94 — is computed on the Tier A subset alone; the Tier B and C rows are provided for completeness and are not part of that summary.

cohort	disease	set	tier	n	g.unadjusted	p.unadjusted	g.adjusted	p.adjusted	n.covariates	max.vif	g.one.at.a time.worst
GSE57338	HF	KEGG_RIBOSOME	A	313	-0.4473	0.000108	-0.4774	9.783e-05	3	1.388	-0.2937
GSE57338	HF	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	313	-0.2085	0.06849	-0.2253	0.06345	3	1.388	-0.4726
GSE57338	HF	REACTOME_TRANSLATION	A	313	-0.539	3.462e-06	-0.4611	0.0001663	3	1.388	-0.4142
GSE57338	HF	...	A	313	-0.6511	2.644e-08	-0.6786	4.49e-08	3	1.388	-0.539
GSE57338	HF	...	A	313	-0.4757	3.924e-05	-0.5152	2.724e-05	3	1.388	-0.3303
GSE57338	HF	REACTOME_TRNA_AMINOACYLATION	A	313	-0.3621	0.001648	-0.3482	0.004265	3	1.388	-0.2201
GSE57338	HF	...	A	313	-0.9199	1.575e-14	-0.8983	1.095e-12	3	1.388	-0.9995
GSE57338	HF	REACTOME_RRNA_PROCESSING	A	313	-0.8114	7.726e-12	-0.7515	1.681e-09	3	1.388	-0.7821
GSE57338	HF	...	A	313	-0.6071	1.944e-07	-0.6397	2.332e-07	3	1.388	-0.481
GSE57338	HF	...	A	313	-0.5993	2.741e-07	-0.561	5.199e-06	3	1.388	-0.5369
GSE57338	HF	...	A	313	-0.5521	2.034e-06	-0.4676	0.0001347	3	1.388	-0.654
GSE57338	HF	REACTOME_MITOCHONDRIAL_TRANSLATION	A	313	-0.08824	0.4396	0.0285	0.8138	3	1.388	0.2223
GSE57338	HF	...	A	313	-0.08121	0.4769	0.03919	0.7462	3	1.388	0.2132
GSE57338	HF	...	A	313	0.5582	1.578e-06	0.7671	8.05e-10	3	1.388	0.3901
GSE57338	HF	...	A	313	-0.1005	0.3786	-0.04998	0.6797	3	1.388	0.2097
GSE57338	HF	REACTOME_TRNA_PROCESSING	B_extra	313	-0.4411	0.0001333	-0.4269	0.0004795	3	1.388	-0.6091
GSE57338	HF	...	B_extra	313	-0.5808	6.096e-07	-0.5856	2.041e-06	3	1.388	-0.6828
GSE57338	HF	...	B_extra	313	-0.1418	0.2147	-0.1195	0.3239	3	1.388	-0.2624
GSE57338	HF	...	B_extra	313	-0.8652	3.821e-13	-0.8064	1.201e-10	3	1.388	-0.9262
GSE57338	HF	...	B_extra	313	-0.4652	5.745e-05	-0.4201	0.0005874	3	1.388	-0.2729
GSE57338	HF	...	B_extra	313	-0.3714	0.001249	-0.3574	0.003366	3	1.388	-0.23
GSE57338	HF	...	C_v1	313	-0.6502	2.754e-08	-0.9312	1.879e-13	3	1.388	-0.9509

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Composition adjustment. The unadjusted and adjusted correlation, the attenuation, whether the direction survives adjustment, and the maximum variance inflation factor, for every set–cohort pair across all three tiers (268 rows: 103 Tier A, 38 Tier B-extra, 127 Tier C-v1, over seven cohorts). The summary quoted in the main text — 55 of 103 pairs survive, median attenuation 34.1%, maximum VIF 1.94 — is computed on the Tier A subset alone; the Tier B and C rows are provided for completeness and are not part of that summary.

cohort	disease	set	tier	n	g.unadjusted	p.unadjusted	g.adjusted	p.adjusted	n.covariates	max.vif	g.one.at.a time.worst
GSE57338	HF	HALLMARK_MYC_TARGETS_V1	C_v1	313	-0.8521	8.017e-13	-0.7738	5.842e-10	3	1.388	-0.9342
GSE57338	HF	...	C_v1	313	-0.5934	3.556e-07	-0.5416	1.062e-05	3	1.388	-0.5039
GSE57338	HF	...	C_v1	313	-0.7772	4.879e-11	-0.77	7.007e-10	3	1.388	-0.9262
GSE57338	HF	...	C_v1	313	-1.099	2.049e-19	-1.124	2.745e-18	3	1.388	-1.247
GSE57338	HF	HALLMARK_MYC_TARGETS_V2	C_v1	313	-1.208	1.38e-22	-1.188	5.599e-20	3	1.388	-1.287
GSE57338	HF	...	C_v1	313	-0.6302	6.9e-08	-0.6209	5.049e-07	3	1.388	-0.5266
GSE57338	HF	...	C_v1	313	-1.198	2.731e-22	-1.265	4.258e-22	3	1.388	-1.169
GSE57338	HF	...	C_v1	313	-1.22	5.855e-23	-1.313	1.821e-23	3	1.388	-1.172
GSE57338	HF	...	C_v1	313	-0.5334	4.339e-06	-0.5861	2.001e-06	3	1.388	-0.3908
GSE57338	HF	HALLMARK_MTORC1_SIGNALING	C_v1	313	-1.001	1.159e-16	-0.9993	4.283e-15	3	1.388	-0.9129
GSE57338	HF	...	C_v1	313	0.04598	0.687	0.06093	0.6148	3	1.388	-0.08425
GSE57338	HF	...	C_v1	313	-0.2876	0.01215	-0.3857	0.001573	3	1.388	-0.4426
GSE57338	HF	...	C_v1	313	-0.5249	6.066e-06	-0.4117	0.0007521	3	1.388	-0.2451
GSE57338	HF	...	C_v1	313	-1.759	2.86e-40	-1.83	4.557e-39	3	1.388	-1.693
GSE57338	HF	...	C_v1	313	0.3116	0.006643	0.003495	0.977	3	1.388	0.1009
GSE57338	HF	...	C_v1	313	-1.097	2.408e-19	-1.102	1.072e-17	3	1.388	-1.001
GSE57338	HF	...	C_v1	313	0.1559	0.1727	0.006693	0.9559	3	1.388	0.2755
GSE57338	HF	...	C_v1	313	0.2075	0.06983	0.05158	0.67	3	1.388	0.3377
GSE57338	HF	...	C_v1	313	-0.08337	0.4652	-0.2357	0.05226	3	1.388	-0.1567
TCGA_LIHC	HCC	KEGG_RIBOSOME	A	421	0.6404	2.624e-05	0.5654	0.002906	3	1.086	0.4832
TCGA_LIHC	HCC	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	421	1.026	3.369e-11	0.7066	0.000207	3	1.086	0.7754
TCGA_LIHC	HCC	REACTOME_TRANSLATION	A	421	0.6874	6.635e-06	0.603	0.001506	3	1.086	0.7711

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Composition adjustment. The unadjusted and adjusted correlation, the attenuation, whether the direction survives adjustment, and the maximum variance inflation factor, for every set–cohort pair across all three tiers (268 rows: 103 Tier A, 38 Tier B-extra, 127 Tier C-v1, over seven cohorts). The summary quoted in the main text — 55 of 103 pairs survive, median attenuation 34.1%, maximum VIF 1.94 — is computed on the Tier A subset alone; the Tier B and C rows are provided for completeness and are not part of that summary.

cohort	disease	set	tier	n	g.unadjusted	p.unadjusted	g.adjusted	p.adjusted	n.covariates	max.vif	g.one.at.a time.worst
TCGA_LIHC	HCC	...	A	421	0.7349	1.521e-06	0.4542	0.01656	3	1.086	0.3968
TCGA_LIHC	HCC	...	A	421	0.8813	9.914e-09	0.8346	1.252e-05	3	1.086	1.008
TCGA_LIHC	HCC	REACTOME_TRNA_AMINOACYLATION	A	421	1.121	5.662e-13	0.766	5.909e-05	3	1.086	0.8062
TCGA_LIHC	HCC	...	A	421	1.175	4.975e-14	0.748	8.719e-05	3	1.086	0.6828
TCGA_LIHC	HCC	REACTOME_RRNA_PROCESSING	A	421	0.8357	5.147e-08	0.4899	0.009774	3	1.086	0.5048
TCGA_LIHC	HCC	...	A	421	0.858	2.324e-08	0.668	0.0004473	3	1.086	0.5796
TCGA_LIHC	HCC	...	A	421	0.9686	3.488e-10	0.7555	7.421e-05	3	1.086	0.6689
TCGA_LIHC	HCC	...	A	421	1.102	1.308e-12	0.8261	1.527e-05	3	1.086	0.7529
TCGA_LIHC	HCC	REACTOME_MITOCHONDRIAL_TRANSLATION	A	421	0.1237	0.4122	0.297	0.1164	3	1.086	0.3475
TCGA_LIHC	HCC	...	A	421	0.1767	0.2415	0.293	0.1214	3	1.086	0.3305
TCGA_LIHC	HCC	...	A	421	0.657	1.629e-05	0.5939	0.001771	3	1.086	0.5408
TCGA_LIHC	HCC	...	A	421	-0.2582	0.08733	0.1775	0.3475	3	1.086	0.2453
TCGA_LIHC	HCC	REACTOME_TRNA_PROCESSING	B_extra	421	1.27	5.547e-16	0.2028	0.2833	3	1.086	0.3587
TCGA_LIHC	HCC	...	B_extra	421	1.461	3.471e-20	-0.222	0.2403	3	1.086	-0.0818
TCGA_LIHC	HCC	...	B_extra	421	0.7843	3.024e-07	0.5499	0.003771	3	1.086	0.6727
TCGA_LIHC	HCC	...	B_extra	421	0.9697	3.343e-10	0.462	0.01479	3	1.086	0.5135

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Composition adjustment. The unadjusted and adjusted correlation, the attenuation, whether the direction survives adjustment, and the maximum variance inflation factor, for every set–cohort pair across all three tiers (268 rows: 103 Tier A, 38 Tier B-extra, 127 Tier C-v1, over seven cohorts). The summary quoted in the main text — 55 of 103 pairs survive, median attenuation 34.1%, maximum VIF 1.94 — is computed on the Tier A subset alone; the Tier B and C rows are provided for completeness and are not part of that summary.

cohort	worst covariate	p.one.at.a time.worst	covariates	attenuation	sign.kept	survives	p.fdr.unadj	p.fdr.adj
GSE57338	myeloid	0.01892	...	-0.06748	True	True	0.000164	0.0001744
GSE57338	myeloid	0.0001641	...	-0.08072	True	False	0.0842	0.07883
GSE57338	fibroblast	0.0007884	...	0.1446	True	True	6.452e-06	0.0002727
GSE57338	myeloid	1.738e-05	...	-0.04225	True	True	8.067e-08	1.227e-07
GSE57338	myeloid	0.008532	...	-0.08291	True	True	6.435e-05	5.077e-05
GSE57338	fibroblast	0.0702	...	0.03821	True	True	0.002252	0.00564
GSE57338	cardiomyocyte	5.515e-16	...	0.02353	True	True	8.069e-14	4.99e-12
GSE57338	myeloid	4.616e-10	...	0.07386	True	True	2.88e-11	4.922e-09
GSE57338	myeloid	0.0001545	...	-0.05363	True	True	4.981e-07	5.976e-07
GSE57338	myeloid	1.908e-05	...	0.06401	True	True	6.611e-07	1.066e-05
GSE57338	cardiomyocyte	1.042e-07	...	0.153	True	True	3.971e-06	0.0002301
GSE57338	fibroblast	0.07569	...	1.323	False	False	0.4743	0.8556
GSE57338	fibroblast	0.08543	...	1.483	False	False	0.4888	0.8051
GSE57338	myeloid	0.003575	...	-0.3741	True	True	3.235e-06	2.539e-09
GSE57338	fibroblast	0.1046	...	0.5029	True	False	0.4196	0.7532
GSE57338	myeloid	6.471e-07	...	0.03232	True	True	0.0001952	0.0007562
GSE57338	myeloid	1.54e-08	...	-0.008124	True	True	1.316e-06	4.404e-06
GSE57338	myeloid	0.02832	...	0.1573	True	False	0.2445	0.3906
GSE57338	cardiomyocyte	5.114e-14	...	0.0679	True	True	1.741e-12	4.922e-10
GSE57338	fibroblast	0.02468	...	0.0969	True	True	9.059e-05	0.000892
GSE57338	fibroblast	0.05578	...	0.03778	True	True	0.001765	0.004601
GSE57338	fibroblast	3.665e-13	...	-0.4321	True	True	8.067e-08	9.63e-13

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Composition adjustment. The unadjusted and adjusted correlation, the attenuation, whether the direction survives adjustment, and the maximum variance inflation factor, for every set–cohort pair across all three tiers (268 rows: 103 Tier A, 38 Tier B-extra, 127 Tier C-v1, over seven cohorts). The summary quoted in the main text — 55 of 103 pairs survive, median attenuation 34.1%, maximum VIF 1.94 — is computed on the Tier A subset alone; the Tier B and C rows are provided for completeness and are not part of that summary.

cohort	worst covariate	p.one.at.a time.worst	covariates	attenuation	sign.kept	survives	p.fdr.unadj	p.fdr.adj
GSE57338	cardiomyocyte	6.234e-14	...	0.09191	True	True	3.287e-12	2.177e-09
GSE57338	fibroblast	5.39e-05	...	0.08724	True	True	8.099e-07	2.073e-05
GSE57338	cardiomyocyte	9.376e-14	...	0.009246	True	True	1.667e-10	2.394e-09
GSE57338	cardiomyocyte	2.79e-22	...	-0.02283	True	True	1.646e-18	2.251e-17
GSE57338	cardiomyocyte	4.487e-24	...	0.01666	True	True	1.886e-21	5.739e-19
GSE57338	myeloid	2.712e-05	...	0.0149	True	True	1.886e-07	1.218e-06
GSE57338	myeloid	2.528e-19	...	-0.0557	True	True	2.799e-21	5.82e-21
GSE57338	myeloid	6.027e-19	...	-0.07614	True	True	1.2e-21	3.734e-22
GSE57338	myeloid	0.00191	...	-0.09883	True	True	7.736e-06	4.404e-06
GSE57338	myeloid	7.852e-13	...	0.001226	True	True	6.791e-16	2.508e-14
GSE57338	myeloid	0.4812	...	-0.3251	True	False	0.687	0.7201
GSE57338	fibroblast	0.0002275	...	-0.341	True	True	0.01557	0.002224
GSE57338	myeloid	0.03646	...	0.2157	True	True	1.036e-05	0.001101
GSE57338	endothelial	7.523e-38	...	-0.04043	True	True	1.173e-38	1.868e-37
GSE57338	fibroblast	0.3982	...	0.9888	True	False	0.008786	0.977
GSE57338	myeloid	2.341e-15	...	-0.004543	True	True	1.646e-18	7.328e-17
GSE57338	endothelial	0.0192	...	0.9571	True	False	0.2022	0.977
GSE57338	endothelial	0.004025	...	0.7514	True	False	0.0842	0.7532
GSE57338	myeloid	0.1899	...	-1.827	True	False	0.4888	0.06696
TCGA_LIHC	proliferation	0.01015	...	0.1172	True	True	4.483e-05	0.009166
TCGA_LIHC	proliferation	4.302e-05	...	0.3114	True	True	1.256e-10	0.0011
TCGA_LIHC	cardiomyocyte	6.578e-07	...	0.1227	True	True	1.237e-05	0.005612

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Composition adjustment. The unadjusted and adjusted correlation, the attenuation, whether the direction survives adjustment, and the maximum variance inflation factor, for every set–cohort pair across all three tiers (268 rows: 103 Tier A, 38 Tier B-extra, 127 Tier C-v1, over seven cohorts). The summary quoted in the main text — 55 of 103 pairs survive, median attenuation 34.1%, maximum VIF 1.94 — is computed on the Tier A subset alone; the Tier B and C rows are provided for completeness and are not part of that summary.

Showing the first 60 of 268 rows. The complete table is provided as the machine-readable file Table_S25_composition_adjustment.csv.

cohort	worst covariate	p.one.at.a time.worst	covariates	attenuation	sign.kept	survives	p.fdr.unadj	p.fdr.adj
TCGA_LIHC	proliferation	0.03399	...	0.382	True	True	2.97e-06	0.03232
TCGA_LIHC	lymphocyte	7.713e-11	...	0.05302	True	True	2.71e-08	0.0002087
TCGA_LIHC	proliferation	1.813e-05	...	0.3168	True	True	3.316e-12	0.0005958
TCGA_LIHC	proliferation	0.000317	...	0.3635	True	True	3.4e-13	0.0005958
TCGA_LIHC	proliferation	0.006784	...	0.4137	True	True	1.241e-07	0.02341
TCGA_LIHC	proliferation	0.002084	...	0.2215	True	True	5.955e-08	0.001834
TCGA_LIHC	proliferation	0.0003913	...	0.2201	True	True	1.1e-09	0.0005958
TCGA_LIHC	proliferation	6.504e-05	...	0.2505	True	True	6.701e-12	0.0002087
TCGA_LIHC	proliferation	0.06666	...	-1.402	True	False	0.4334	0.1767
TCGA_LIHC	proliferation	0.07962	...	-0.658	True	False	0.275	0.1777
TCGA_LIHC	lymphocyte	0.0005525	...	0.0961	True	True	2.904e-05	0.006052
TCGA_LIHC	proliferation	0.1955	...	1.688	False	False	0.1053	0.3957
TCGA_LIHC	proliferation	0.06048	...	0.8404	True	False	4.549e-15	0.3465
TCGA_LIHC	proliferation	0.6682	...	1.152	False	False	4.744e-19	0.3178
TCGA_LIHC	proliferation	0.0004155	...	0.2989	True	True	6.526e-07	0.01031
TCGA_LIHC	proliferation	0.006054	...	0.5236	True	True	1.1e-09	0.03032

S26 Table (page 1 of 3)

Composition–set correlation. The correlation between each composition score and each translation set score, so that a set whose apparent disease association tracks composition rather than translation can be identified.

cohort	disease	set	panel	r	p	p.fdr
GSE57338	HF	KEGG_RIBOSOME	immune_pan	0.43	1.607e-15	1.206e-14
GSE57338	HF	KEGG_RIBOSOME	myeloid	0.4216	6.385e-15	4.105e-14
GSE57338	HF	KEGG_RIBOSOME	fibroblast	0.2595	3.278e-06	1.054e-05
GSE57338	HF	KEGG_RIBOSOME	endothelial	-0.01923	0.7348	0.8064
GSE57338	HF	KEGG_RIBOSOME	hepatocyte	0.1182	0.03658	0.06097
GSE57338	HF	KEGG_RIBOSOME	cardiomyocyte	0.1642	0.003569	0.007833
GSE57338	HF	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	immune_pan	-0.3859	1.488e-12	5.579e-12
GSE57338	HF	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	myeloid	-0.3705	1.277e-11	4.598e-11
GSE57338	HF	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	fibroblast	-0.4028	1.229e-13	5.821e-13
GSE57338	HF	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	endothelial	-0.03548	0.5317	0.6296
GSE57338	HF	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	hepatocyte	-0.0607	0.2843	0.3604
GSE57338	HF	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	cardiomyocyte	0.05599	0.3234	0.4043
GSE57338	HF	REACTOME_TRANSLATION	immune_pan	-0.1542	0.006262	0.01299
GSE57338	HF	REACTOME_TRANSLATION	myeloid	-0.1385	0.01421	0.02611
GSE57338	HF	REACTOME_TRANSLATION	fibroblast	-0.4231	5.023e-15	3.478e-14
GSE57338	HF	REACTOME_TRANSLATION	endothelial	-0.06726	0.2354	0.3116
GSE57338	HF	REACTOME_TRANSLATION	hepatocyte	0.09589	0.09036	0.1479
GSE57338	HF	REACTOME_TRANSLATION	cardiomyocyte	0.08024	0.1567	0.2312
GSE57338	HF	...	immune_pan	0.3952	3.822e-13	1.72e-12
GSE57338	HF	...	myeloid	0.3915	6.633e-13	2.616e-12
GSE57338	HF	...	fibroblast	0.1778	0.001588	0.003971
GSE57338	HF	...	endothelial	0.00808	0.8868	0.9163

S26 Table (page 2 of 3)

Composition–set correlation. The correlation between each composition score and each translation set score, so that a set whose apparent disease association tracks composition rather than translation can be identified.

cohort	disease	set	panel	r	p	p.fdr
GSE57338	HF	...	hepatocyte	0.1363	0.01579	0.02841
GSE57338	HF	...	cardiomyocyte	0.1456	0.009874	0.01932
GSE57338	HF	...	immune_pan	0.4211	6.936e-15	4.161e-14
GSE57338	HF	...	myeloid	0.4166	1.439e-14	8.092e-14
GSE57338	HF	...	fibroblast	0.2538	5.453e-06	1.583e-05
GSE57338	HF	...	endothelial	-0.000457	0.9936	0.9936
GSE57338	HF	...	hepatocyte	0.127	0.02463	0.04241
GSE57338	HF	...	cardiomyocyte	0.1759	0.001783	0.004338
GSE57338	HF	REACTOME_TRNA_AMINOACYLATION	immune_pan	-0.2848	2.985e-07	1.033e-06
GSE57338	HF	REACTOME_TRNA_AMINOACYLATION	myeloid	-0.2709	1.144e-06	3.813e-06
GSE57338	HF	REACTOME_TRNA_AMINOACYLATION	fibroblast	-0.3927	5.567e-13	2.386e-12
GSE57338	HF	REACTOME_TRNA_AMINOACYLATION	endothelial	-0.009898	0.8615	0.9016
GSE57338	HF	REACTOME_TRNA_AMINOACYLATION	hepatocyte	-0.0325	0.5667	0.6624
GSE57338	HF	REACTOME_TRNA_AMINOACYLATION	cardiomyocyte	0.04009	0.4797	0.5757
GSE57338	HF	...	immune_pan	0.06571	0.2464	0.3214
GSE57338	HF	...	myeloid	0.07354	0.1944	0.2707
GSE57338	HF	...	fibroblast	-0.154	0.00635	0.01299
GSE57338	HF	...	endothelial	0.0408	0.472	0.574
GSE57338	HF	...	hepatocyte	0.08521	0.1325	0.2021
GSE57338	HF	...	cardiomyocyte	0.08734	0.1231	0.191
GSE57338	HF	REACTOME_RRNA_PROCESSING	immune_pan	0.2109	0.0001705	0.000465
GSE57338	HF	REACTOME_RRNA_PROCESSING	myeloid	0.1992	0.0003929	0.00104

S26 Table (page 3 of 3)

Composition–set correlation. The correlation between each composition score and each translation set score, so that a set whose apparent disease association tracks composition rather than translation can be identified.

Showing the first 60 of 765 rows. The complete table is provided as the machine-readable file Table_S26_composition_set_correlation.csv.

cohort	disease	set	panel	r	p	p.fdr
GSE57338	HF	REACTOME_RRNA_PROCESSING	fibroblast	-0.07287	0.1985	0.2707
GSE57338	HF	REACTOME_RRNA_PROCESSING	endothelial	-0.01097	0.8467	0.8965
GSE57338	HF	REACTOME_RRNA_PROCESSING	hepatocyte	0.08826	0.1192	0.1882
GSE57338	HF	REACTOME_RRNA_PROCESSING	cardiomyocyte	0.09117	0.1074	0.1726
GSE57338	HF	...	immune_pan	0.4429	1.815e-16	1.815e-15
GSE57338	HF	...	myeloid	0.4334	9.168e-16	7.501e-15
GSE57338	HF	...	fibroblast	0.211	0.0001692	0.000465
GSE57338	HF	...	endothelial	-0.02222	0.6954	0.7823
GSE57338	HF	...	hepatocyte	0.1293	0.02209	0.03898
GSE57338	HF	...	cardiomyocyte	0.1667	0.003089	0.006951
GSE57338	HF	...	immune_pan	0.2581	3.717e-06	1.153e-05
GSE57338	HF	...	myeloid	0.2558	4.571e-06	1.371e-05
GSE57338	HF	...	fibroblast	0.01737	0.7595	0.8236
GSE57338	HF	...	endothelial	-0.0477	0.4004	0.4936
GSE57338	HF	...	hepatocyte	0.1632	0.003793	0.008128
GSE57338	HF	...	cardiomyocyte	0.1754	0.001843	0.004365

S27 Table (columns 1 of 2)

Translation score gene accounting. For every cohort: the number of Tier A sets measured, the number of genes available for each module-derived score, and the fraction of each module present, replacing v1's single "227 genes" figure, which did not hold across cohorts.

cohort	disease	tech	n.samples	n.tierA.sets	n.tierA expected	genes.HF module	genes.HF present	frac.HF present	genes.HCC module	genes.HCC present	frac.HCC present
GSE57338	HF	array	313	15	15	153	153	1	146	146	1
GSE141198	HCC	RNA-seq	148	15	15	153	153	1	146	146	1
TCGA_LIHC	HCC	RNA-seq	421	15	15	153	152	0.993	146	142	0.973
GSE116250	HF	RNA-seq	64	15	15	153	145	0.948	146	131	0.897
GSE14520_GPL3921	HCC	array	445	14	15	153	103	0.673	146	78	0.534
GSE14520_GPL571	HCC	array	41	15	15	153	97	0.634	146	91	0.623
GSE76427	HCC	array	167	14	15	153	135	0.882	146	116	0.795
GSE141910	HF	RNA-seq	366	15	15	153	152	0.993	146	143	0.979

S27 Table (columns 2 of 2)

Translation score gene accounting. For every cohort: the number of Tier A sets measured, the number of genes available for each module-derived score, and the fraction of each module present, replacing v1's single "227 genes" figure, which did not hold across cohorts.

cohort	HF.in.tierA	HCC.in.tierA	shared.HF.HCC
GSE57338	73	26	5
GSE141198	73	26	5
TCGA_LIHC	73	25	5
GSE116250	72	22	5
GSE14520_GPL3921	62	13	2
GSE14520_GPL571	53	18	3
GSE76427	71	20	5
GSE141910	72	25	5

S28 Table (page 1 of 2)

Translation score concordance. Pairwise correlations among the canonical translation score, the HF module eigengene score, and the HCC module eigengene score, in each cohort, showing whether the choice of module determines the result.

cohort	score.a	score.b	r
GSE57338	TCS	ME_HF_weighted	0.6358
GSE57338	TCS	ME_HCC_weighted	0.7157
GSE57338	ME_HF_weighted	ME_HCC_weighted	0.3084
GSE141198	TCS	ME_HF_weighted	0.4629
GSE141198	TCS	ME_HCC_weighted	0.549
GSE141198	ME_HF_weighted	ME_HCC_weighted	0.7729
TCGA_LIHC	TCS	ME_HF_weighted	0.6019
TCGA_LIHC	TCS	ME_HCC_weighted	0.4723
TCGA_LIHC	ME_HF_weighted	ME_HCC_weighted	0.9115
GSE116250	TCS	ME_HF_weighted	0.2654
GSE116250	TCS	ME_HCC_weighted	0.4233
GSE116250	ME_HF_weighted	ME_HCC_weighted	0.8728
GSE14520_GPL3921	TCS	ME_HF_weighted	0.7852
GSE14520_GPL3921	TCS	ME_HCC_weighted	0.7032
GSE14520_GPL3921	ME_HF_weighted	ME_HCC_weighted	0.7274
GSE14520_GPL571	TCS	ME_HF_weighted	0.9041
GSE14520_GPL571	TCS	ME_HCC_weighted	0.8523
GSE14520_GPL571	ME_HF_weighted	ME_HCC_weighted	0.8041
GSE76427	TCS	ME_HF_weighted	0.8034
GSE76427	TCS	ME_HCC_weighted	0.6864
GSE76427	ME_HF_weighted	ME_HCC_weighted	0.6881
GSE141910	TCS	ME_HF_weighted	0.874

S28 Table (page 2 of 2)

Translation score concordance. Pairwise correlations among the canonical translation score, the HF module eigengene score, and the HCC module eigengene score, in each cohort, showing whether the choice of module determines the result.

cohort	score.a	score.b	r
GSE141910	TCS	ME_HCC_weighted	0.9171
GSE141910	ME_HF_weighted	ME_HCC_weighted	0.9506

S29 Table

Translation score effect sizes. Disease-versus-control effect size, P-value, and BH FDR for each score in each cohort.

cohort	disease	score	n.disease	n.control	mean.disease	mean.control	hedges.g	p	p.fdr
GSE57338	HF	TCS	177	136	-0.1628	0.2119	-0.5692	1.99e-06	2.984e-06
GSE57338	HF	ME_HF_weighted	177	136	-0.1482	0.1929	-0.5526	3.905e-06	3.905e-06
GSE57338	HF	ME_HCC_weighted	177	136	-0.1088	0.1416	-0.6611	1.684e-08	5.053e-08
TCGA_LIHC	HCC	TCS	371	50	0.08138	-0.6039	0.8716	7.331e-15	7.331e-15
TCGA_LIHC	HCC	ME_HF_weighted	371	50	0.1042	-0.7732	1.356	1.003e-16	1.504e-16
TCGA_LIHC	HCC	ME_HCC_weighted	371	50	0.1102	-0.8174	1.433	2.134e-17	6.401e-17
GSE116250	HF	TCS	50	14	-0.023	0.08214	-0.1605	0.6378	0.9707
GSE116250	HF	ME_HF_weighted	50	14	0.02091	-0.07468	0.1541	0.7059	0.9707
GSE116250	HF	ME_HCC_weighted	50	14	-0.001622	0.005792	-0.01573	0.9707	0.9707
GSE14520_GPL3921	HCC	TCS	225	220	0.5148	-0.5265	2.043	8.106e-69	1.216e-68
GSE14520_GPL3921	HCC	ME_HF_weighted	225	220	0.3837	-0.3924	1.604	4.988e-50	4.988e-50
GSE14520_GPL3921	HCC	ME_HCC_weighted	225	220	0.3996	-0.4087	2.044	3.73e-71	1.119e-70
GSE14520_GPL571	HCC	TCS	22	19	0.3556	-0.4118	1.101	0.0008482	0.0008482
GSE14520_GPL571	HCC	ME_HF_weighted	22	19	0.3193	-0.3697	1.363	7.423e-05	0.0001113
GSE14520_GPL571	HCC	ME_HCC_weighted	22	19	0.3032	-0.3511	1.521	1.324e-05	3.972e-05
GSE76427	HCC	TCS	115	52	0.3121	-0.6902	1.512	1.888e-13	1.888e-13
GSE76427	HCC	ME_HF_weighted	115	52	0.1922	-0.425	1.765	1.209e-20	1.814e-20
GSE76427	HCC	ME_HCC_weighted	115	52	0.1628	-0.36	1.886	1.82e-23	5.459e-23
GSE141910	HF	TCS	200	166	0.04731	-0.057	0.1278	0.2178	0.6535
GSE141910	HF	ME_HF_weighted	200	166	0.02843	-0.03426	0.07868	0.4529	0.6793
GSE141910	HF	ME_HCC_weighted	200	166	-7.632e-05	9.196e-05	-0.000208	0.9984	0.9984

S30 Table (columns 1 of 2) (page 1 of 3)

Transcription factor association. For every candidate transcription factor, in every cohort and for every score: the unadjusted Spearman correlation, the partial correlation adjusting for composition, both P-values and BH FDRs, the covariates used, whether the direction survived adjustment, and a flag marking factors that are themselves members of translation gene sets. The 582 rows are 14 factors x 3 scores (TCS and both module eigengene projections) across the arms available in each cohort — 312 rows on the all-samples arm and 270 on the disease-samples-only arm, the latter absent for GSE141198, which has no control arm. Because the table exceeds the preview threshold, the printed supplement shows its first 60 rows; the machine-readable CSV carries all 582.

cohort	disease	arm	score	tf	tf.in translation sets	n	r.unadjusted	p.unadjusted	r.adjusted	p.adjusted	covariates
GSE57338	HF	all	TCS	ATF4	False	313	0.2842	3.148e-07	0.3065	3.136e-08	...
GSE57338	HF	all	TCS	MYC	False	313	0.1836	0.001105	0.2673	1.606e-06	...
GSE57338	HF	all	TCS	XBP1	False	313	0.3127	1.579e-08	0.3283	2.662e-09	...
GSE57338	HF	all	TCS	ATF6	False	313	0.1873	0.0008708	0.1918	0.0006455	...
GSE57338	HF	all	TCS	EIF4EBP1	True	313	0.3549	1.007e-10	0.3806	3.173e-12	...
GSE57338	HF	all	TCS	EIF2AK3	False	313	0.196	0.0004862	0.2081	0.0002092	...
GSE57338	HF	all	TCS	ERN1	False	313	-0.0123	0.8284	0.03411	0.5477	...
GSE57338	HF	all	TCS	DDIT3	False	313	0.2763	6.857e-07	0.2708	1.151e-06	...
GSE57338	HF	all	TCS	EIF4G1	True	313	0.274	8.55e-07	0.2821	3.899e-07	...
GSE57338	HF	all	TCS	MTOR	False	313	0.07435	0.1895	0.06145	0.2785	...
GSE57338	HF	all	TCS	RPTOR	False	313	-0.1278	0.02374	-0.05237	0.3557	...
GSE57338	HF	all	TCS	TSC1	False	313	-0.3244	4.213e-09	-0.3895	8.774e-13	...
GSE57338	HF	all	TCS	TSC2	False	313	-0.2821	3.886e-07	-0.281	4.33e-07	...
GSE57338	HF	all	ME_HF_weighted	ATF4	False	313	0.4307	1.448e-15	0.3732	8.822e-12	...
GSE57338	HF	all	ME_HF_weighted	MYC	False	313	0.5671	4.91e-28	0.4294	1.779e-15	...
GSE57338	HF	all	ME_HF_weighted	XBP1	False	313	0.3878	1.126e-12	0.2748	7.87e-07	...
GSE57338	HF	all	ME_HF_weighted	ATF6	False	313	0.382	2.579e-12	0.2101	0.0001807	...
GSE57338	HF	all	ME_HF_weighted	EIF4EBP1	True	313	0.4468	9.103e-17	0.4797	2.042e-19	...
GSE57338	HF	all	ME_HF_weighted	EIF2AK3	False	313	0.2839	3.252e-07	0.2212	7.927e-05	...
GSE57338	HF	all	ME_HF_weighted	ERN1	False	313	0.1199	0.03401	0.06297	0.2667	...
GSE57338	HF	all	ME_HF_weighted	DDIT3	False	313	0.1446	0.01041	0.1665	0.003127	...
GSE57338	HF	all	ME_HF_weighted	EIF4G1	True	313	0.08847	0.1183	0.1434	0.0111	...

S30 Table (columns 1 of 2) (page 2 of 3)

Transcription factor association. For every candidate transcription factor, in every cohort and for every score: the unadjusted Spearman correlation, the partial correlation adjusting for composition, both P-values and BH FDRs, the covariates used, whether the direction survived adjustment, and a flag marking factors that are themselves members of translation gene sets. The 582 rows are 14 factors x 3 scores (TCS and both module eigengene projections) across the arms available in each cohort — 312 rows on the all-samples arm and 270 on the disease-samples-only arm, the latter absent for GSE141198, which has no control arm. Because the table exceeds the preview threshold, the printed supplement shows its first 60 rows; the machine-readable CSV carries all 582.

cohort	disease	arm	score	tf	tf.in translation sets	n	r.unadjusted	p.unadjusted	r.adjusted	p.adjusted	covariates
GSE57338	HF	all	ME_HF_weighted	MTOR	False	313	-0.2457	1.099e-05	-0.06865	0.2259	...
GSE57338	HF	all	ME_HF_weighted	RPTOR	False	313	-0.1479	0.008775	-0.06692	0.2378	...
GSE57338	HF	all	ME_HF_weighted	TSC1	False	313	-0.5256	1.253e-23	-0.3805	3.181e-12	...
GSE57338	HF	all	ME_HF_weighted	TSC2	False	313	-0.3474	2.625e-10	-0.2235	6.627e-05	...
GSE57338	HF	all	ME_HCC_weighted	ATF4	False	313	0.1594	0.004697	0.238	2.085e-05	...
GSE57338	HF	all	ME_HCC_weighted	MYC	False	313	-0.1739	0.002017	-0.03474	0.5404	...
GSE57338	HF	all	ME_HCC_weighted	XBP1	False	313	0.02138	0.7064	0.08619	0.1281	...
GSE57338	HF	all	ME_HCC_weighted	ATF6	False	313	-0.1229	0.02967	-0.06995	0.2172	...
GSE57338	HF	all	ME_HCC_weighted	EIF4EBP1	True	313	0.3541	1.12e-10	0.4179	1.168e-14	...
GSE57338	HF	all	ME_HCC_weighted	EIF2AK3	False	313	-0.09802	0.08339	-0.06744	0.2342	...
GSE57338	HF	all	ME_HCC_weighted	ERN1	False	313	-0.229	4.33e-05	-0.1714	0.002341	...
GSE57338	HF	all	ME_HCC_weighted	DDIT3	False	313	0.1667	0.003093	0.1607	0.004372	...
GSE57338	HF	all	ME_HCC_weighted	EIF4G1	True	313	0.2856	2.736e-07	0.2882	2.117e-07	...
GSE57338	HF	all	ME_HCC_weighted	MTOR	False	313	0.3386	7.814e-10	0.2898	1.8e-07	...
GSE57338	HF	all	ME_HCC_weighted	RPTOR	False	313	-0.03838	0.4987	0.01867	0.7422	...
GSE57338	HF	all	ME_HCC_weighted	TSC1	False	313	0.02638	0.642	-0.0917	0.1054	...
GSE57338	HF	all	ME_HCC_weighted	TSC2	False	313	-0.04115	0.4682	-0.08997	0.1121	...
GSE57338	HF	disease_only	TCS	ATF4	False	177	-0.1381	0.06671	0.1879	0.01228	...
GSE57338	HF	disease_only	TCS	MYC	False	177	-0.3475	2.153e-06	-0.09669	0.2004	...
GSE57338	HF	disease_only	TCS	XBP1	False	177	-0.03649	0.6297	0.1067	0.1576	...
GSE57338	HF	disease_only	TCS	ATF6	False	177	-0.3121	2.352e-05	-0.2138	0.004266	...
GSE57338	HF	disease_only	TCS	EIF4EBP1	True	177	0.1441	0.05562	0.2894	9.364e-05	...

S30 Table (columns 1 of 2) (page 3 of 3)

Transcription factor association. For every candidate transcription factor, in every cohort and for every score: the unadjusted Spearman correlation, the partial correlation adjusting for composition, both P-values and BH FDRs, the covariates used, whether the direction survived adjustment, and a flag marking factors that are themselves members of translation gene sets. The 582 rows are 14 factors x 3 scores (TCS and both module eigengene projections) across the arms available in each cohort — 312 rows on the all-samples arm and 270 on the disease-samples-only arm, the latter absent for GSE141198, which has no control arm. Because the table exceeds the preview threshold, the printed supplement shows its first 60 rows; the machine-readable CSV carries all 582.

cohort	disease	arm	score	tf	tf.in translation sets	n	r.unadjusted	p.unadjusted	r.adjusted	p.adjusted	covariates
GSE57338	HF	disease_only	TCS	EIF2AK3	False	177	-0.1008	0.1817	-0.04962	0.5119	...
GSE57338	HF	disease_only	TCS	ERN1	False	177	-0.2211	0.003103	-0.105	0.1643	...
GSE57338	HF	disease_only	TCS	DDIT3	False	177	0.2861	0.0001134	0.2868	0.0001087	...
GSE57338	HF	disease_only	TCS	EIF4G1	True	177	0.01592	0.8334	-0.09077	0.2295	...
GSE57338	HF	disease_only	TCS	MTOR	False	177	0.1841	0.01416	-0.05144	0.4965	...
GSE57338	HF	disease_only	TCS	RPTOR	False	177	-0.2329	0.001807	-0.2727	0.00024	...
GSE57338	HF	disease_only	TCS	TSC1	False	177	-0.007399	0.9221	-0.2926	7.756e-05	...
GSE57338	HF	disease_only	TCS	TSC2	False	177	-0.1082	0.1517	-0.2571	0.0005517	...
GSE57338	HF	disease_only	ME_HF_weighted	ATF4	False	177	0.444	6.037e-10	0.3659	5.506e-07	...
GSE57338	HF	disease_only	ME_HF_weighted	MYC	False	177	0.4614	1.018e-10	0.3076	3.113e-05	...
GSE57338	HF	disease_only	ME_HF_weighted	XBP1	False	177	0.2434	0.001099	0.124	0.1002	...
GSE57338	HF	disease_only	ME_HF_weighted	ATF6	False	177	0.1638	0.02938	-0.03457	0.6478	...
GSE57338	HF	disease_only	ME_HF_weighted	EIF4EBP1	True	177	0.4416	7.628e-10	0.4573	1.563e-10	...
GSE57338	HF	disease_only	ME_HF_weighted	EIF2AK3	False	177	0.113	0.1341	0.06036	0.4248	...
GSE57338	HF	disease_only	ME_HF_weighted	ERN1	False	177	0.09883	0.1906	0.03192	0.6732	...
GSE57338	HF	disease_only	ME_HF_weighted	DDIT3	False	177	0.1502	0.04601	0.1855	0.01342	...

S30 Table (columns 2 of 2) (page 1 of 3)

Transcription factor association. For every candidate transcription factor, in every cohort and for every score: the unadjusted Spearman correlation, the partial correlation adjusting for composition, both P-values and BH FDRs, the covariates used, whether the direction survived adjustment, and a flag marking factors that are themselves members of translation gene sets. The 582 rows are 14 factors x 3 scores (TCS and both module eigengene projections) across the arms available in each cohort — 312 rows on the all-samples arm and 270 on the disease-samples-only arm, the latter absent for GSE141198, which has no control arm. Because the table exceeds the preview threshold, the printed supplement shows its first 60 rows; the machine-readable CSV carries all 582.

cohort	p.fdr unadjusted	p.fdr adjusted	direction kept
GSE57338	1.01e-06	1.019e-07	True
GSE57338	0.001437	2.609e-06	True
GSE57338	6.842e-08	1.153e-08	True
GSE57338	0.001258	0.0008392	True
GSE57338	1.31e-09	2.062e-11	True
GSE57338	0.0007901	0.0003022	True
GSE57338	0.8284	0.5477	False
GSE57338	1.486e-06	2.138e-06	True
GSE57338	1.588e-06	9.382e-07	True
GSE57338	0.2053	0.3291	True
GSE57338	0.02806	0.3854	True
GSE57338	2.739e-08	1.141e-11	True
GSE57338	1.01e-06	9.382e-07	True
GSE57338	4.707e-15	2.867e-11	True
GSE57338	6.383e-27	1.157e-14	True
GSE57338	2.928e-12	2.046e-06	True
GSE57338	5.587e-12	0.0002936	True
GSE57338	3.945e-16	2.655e-18	True
GSE57338	5.285e-07	0.0001472	True
GSE57338	0.03685	0.2667	True
GSE57338	0.01231	0.004517	True
GSE57338	0.1183	0.01443	True

S30 Table (columns 2 of 2) (page 2 of 3)

Transcription factor association. For every candidate transcription factor, in every cohort and for every score: the unadjusted Spearman correlation, the partial correlation adjusting for composition, both P-values and BH FDRs, the covariates used, whether the direction survived adjustment, and a flag marking factors that are themselves members of translation gene sets. The 582 rows are 14 factors x 3 scores (TCS and both module eigengene projections) across the arms available in each cohort — 312 rows on the all-samples arm and 270 on the disease-samples-only arm, the latter absent for GSE141198, which has no control arm. Because the table exceeds the preview threshold, the printed supplement shows its first 60 rows; the machine-readable CSV carries all 582.

cohort	p.fdr unadjusted	p.fdr adjusted	direction kept
GSE57338	1.587e-05	0.2576	True
GSE57338	0.01141	0.2576	True
GSE57338	8.146e-23	1.379e-11	True
GSE57338	4.875e-10	0.0001436	True
GSE57338	0.008723	6.776e-05	True
GSE57338	0.005244	0.5854	True
GSE57338	0.7064	0.185	True
GSE57338	0.04821	0.2768	True
GSE57338	1.455e-09	1.518e-13	True
GSE57338	0.1205	0.2768	True
GSE57338	0.0001407	0.006087	True
GSE57338	0.006701	0.009472	True
GSE57338	1.186e-06	9.174e-07	True
GSE57338	5.079e-09	9.174e-07	True
GSE57338	0.5893	0.7422	False
GSE57338	0.6955	0.1822	False
GSE57338	0.5893	0.1822	True
GSE57338	0.1084	0.0228	False
GSE57338	2.799e-05	0.2606	True
GSE57338	0.7442	0.2373	False
GSE57338	0.0001529	0.009243	True
GSE57338	0.1033	0.000471	True

S30 Table (columns 2 of 2) (page 3 of 3)

Transcription factor association. For every candidate transcription factor, in every cohort and for every score: the unadjusted Spearman correlation, the partial correlation adjusting for composition, both P-values and BH FDRs, the covariates used, whether the direction survived adjustment, and a flag marking factors that are themselves members of translation gene sets. The 582 rows are 14 factors x 3 scores (TCS and both module eigengene projections) across the arms available in each cohort — 312 rows on the all-samples arm and 270 on the disease-samples-only arm, the latter absent for GSE141198, which has no control arm. Because the table exceeds the preview threshold, the printed supplement shows its first 60 rows; the machine-readable CSV carries all 582.

Showing the first 60 of 582 rows. The complete table is provided as the machine-readable file Table_S30_TF_translation_association.csv.

cohort	p.fdr unadjusted	p.fdr adjusted	direction kept
GSE57338	0.2362	0.5119	True
GSE57338	0.008068	0.2373	True
GSE57338	0.0004915	0.000471	True
GSE57338	0.9029	0.2713	False
GSE57338	0.03069	0.5119	False
GSE57338	0.005873	0.0007801	True
GSE57338	0.9221	0.000471	True
GSE57338	0.2191	0.001434	True
GSE57338	3.306e-09	3.579e-06	True
GSE57338	1.324e-09	0.0001349	True
GSE57338	0.00204	0.1335	True
GSE57338	0.0382	0.6732	False
GSE57338	3.306e-09	2.031e-09	True
GSE57338	0.1453	0.502	True
GSE57338	0.1906	0.6732	True
GSE57338	0.05438	0.02493	True

S31 Table

Gene significance threshold comparison. Hub gene counts under the v1 threshold (|GS| > 0.20), a stricter |GS| > 0.50, and the FDR < 0.05 criterion used in v2, in both networks. The three criteria are nested on this data: every gene passing |GS| > 0.50 also passes FDR < 0.05. |GS| > 0.50 is the strictest of the three and FDR < 0.05 the most permissive, so FDR < 0.05 was adopted for its calibration to sample size rather than because it is the most conservative.

cohort	module	n.module genes	n.present	n.absGS.gt 020	n.absGS.gt 050	n.absGS.gt 050.and.FDR05	n.FDR05.alone	median.abs.GS
GSE57338	purple	153	153	76	9	9	107	0.1969
TCGA_LIHC	magenta	146	142	119	2	2	133	0.3115

S32 Table (columns 1 of 2)

Module-internal robustness. Mean intra-modular topological overlap for HF purple and HCC magenta against both the uniform (v1-style) and connectivity-matched nulls, with Z, empirical P, and the observed-to-null mean ratio — the interpretable effect size, since Z is not comparable between nulls. Also reports the module and universe connectivity distributions that motivate the matched null, and jackknife stability as both the CV of the mean intra-modular connectivity and the median per-gene CV.

disease	module	n.genes	power	n.samples	k.universe mean	k.module.mean	obs.mean intra.TOM	unmatched null.mean	unmatched null.sd	obs.over unmatched null	Z.unmatched
HF	purple	153	14	177	15.2	6.6	0.007868	0.002745	0.000372	2.87	13.77
HCC	magenta	146	14	148	74.1	21.1	0.01798	0.01834	0.002483	0.98	-0.14

S32 Table (columns 2 of 2)

Module-internal robustness. Mean intra-modular topological overlap for HF purple and HCC magenta against both the uniform (v1-style) and connectivity-matched nulls, with Z, empirical P, and the observed-to-null mean ratio — the interpretable effect size, since Z is not comparable between nulls. Also reports the module and universe connectivity distributions that motivate the matched null, and jackknife stability as both the CV of the mean intra-modular connectivity and the median per-gene CV.

disease	P.unmatched	matched.null mean	matched.null sd	obs.over matched.null	Z.matched	P.matched	n.perm	n.bins	jackknife.CV of.mean.kIM pct	jackknife median pergene.CV	jackknife.n degenerate
HF	0.0001	0.0007144	1.591e-05	11.01	449.7	0.0001	10000	20	2.66	53.28	0
HCC	0.544	0.003839	0.000161	4.68	87.83	0.0001	10000	20	2.48	49.68	0

S33 Table

Cross-disease module coherence. Mean pairwise correlation of each designated module's genes measured in the other disease's expression matrix, against a uniform null, for both modules.

module disease	module	tested.in	n.genes module	n.genes present	obs.mean pairwise.r	null.mean.r	null.sd.r	Z	p	n.perm
HF	purple	HCC	153	153	0.3256	0.02396	0.01357	22.22	0.000999	1000
HCC	magenta	HF	146	146	0.14	0.0355	0.007714	13.54	0.000999	1000

S34 Table (columns 1 of 2) (page 1 of 3)

GSE89377 stage-spectrum effect sizes. Hedges g with bootstrap confidence intervals for every scored set at every stage of the hepatocarcinogenesis series (normal, chronic hepatitis low and high grade, cirrhosis, dysplastic nodule low and high grade, early HCC, HCC) against the normal arm, with stage index and group sizes.

cohort	set	tier	stage	stage.index	n.stage	n.control	mean.stage	mean.control	hedges.g	ci.lo	ci.hi
GSE89377	KEGG_RIBOSOME	A	CH_low	2	8	13	0.7163	0.7098	0.6077	-0.08557	1.916
GSE89377	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	CH_low	2	8	13	0.06561	0.05359	0.3383	-0.5753	1.033
GSE89377	REACTOME_TRANSLATION	A	CH_low	2	8	13	0.5199	0.5105	0.7938	-0.004214	1.721
GSE89377	...	A	CH_low	2	8	13	0.6606	0.6541	0.6228	-0.06794	1.948
GSE89377	...	A	CH_low	2	8	13	0.7053	0.6994	0.5482	-0.1416	2.034
GSE89377	REACTOME_TRNA_AMINOACYLATION	A	CH_low	2	8	13	0.2042	0.1922	0.5245	-0.3154	1.288
GSE89377	...	A	CH_low	2	8	13	0.4885	0.4743	0.8748	0.1561	2.049
GSE89377	REACTOME_RRNA_PROCESSING	A	CH_low	2	8	13	0.4501	0.4465	0.263	-0.5074	1.314
GSE89377	...	A	CH_low	2	8	13	0.6718	0.6646	0.7225	-0.009904	1.879
GSE89377	...	A	CH_low	2	8	13	0.6202	0.6143	0.5784	-0.1496	1.849
GSE89377	...	A	CH_low	2	8	13	0.5656	0.5604	0.4891	-0.2639	1.826
GSE89377	REACTOME_MITOCHONDRIAL_TRANSLATION	A	CH_low	2	8	13	0.3839	0.3679	0.5962	-0.1771	1.313
GSE89377	...	A	CH_low	2	8	13	0.3891	0.3729	0.5872	-0.2008	1.308
GSE89377	...	A	CH_low	2	8	13	0.1115	0.1006	0.3482	-0.6116	1.033
GSE89377	...	A	CH_low	2	8	13	0.07167	0.05214	0.4909	-0.1715	1.164
GSE89377	REACTOME_TRNA_PROCESSING	B_extra	CH_low	2	8	13	0.03944	0.02821	0.8166	0.03496	2.489
GSE89377	...	B_extra	CH_low	2	8	13	0.07017	0.06093	0.6498	-0.08597	1.742
GSE89377	...	B_extra	CH_low	2	8	13	-0.01782	-0.02846	0.5359	-0.2434	1.771
GSE89377	...	B_extra	CH_low	2	8	13	0.3194	0.3212	-0.07267	-0.8216	0.8662
GSE89377	...	B_extra	CH_low	2	8	13	0.1534	0.1304	0.8239	0.06907	1.785
GSE89377	...	C_v1	CH_low	2	8	13	-0.01304	0.004659	-0.5278	-1.368	0.2648
GSE89377	HALLMARK_MYC_TARGETS_V1	C_v1	CH_low	2	8	13	0.442	0.4352	0.6546	-0.09191	1.762

S34 Table (columns 1 of 2) (page 2 of 3)

GSE89377 stage-spectrum effect sizes. Hedges g with bootstrap confidence intervals for every scored set at every stage of the hepatocarcinogenesis series (normal, chronic hepatitis low and high grade, cirrhosis, dysplastic nodule low and high grade, early HCC, HCC) against the normal arm, with stage index and group sizes.

cohort	set	tier	stage	stage.index	n.stage	n.control	mean.stage	mean.control	hedges.g	ci.lo	ci.hi
GSE89377	...	C_v1	CH_low	2	8	13	0.4653	0.4553	0.7333	-0.2268	2.046
GSE89377	...	C_v1	CH_low	2	8	13	0.5097	0.5061	0.4735	-0.5467	1.658
GSE89377	...	C_v1	CH_low	2	8	13	0.5325	0.5162	1.075	0.3651	1.938
GSE89377	HALLMARK_MYC_TARGETS_V2	C_v1	CH_low	2	8	13	0.2288	0.2351	-0.1527	-0.9061	0.7241
GSE89377	...	C_v1	CH_low	2	8	13	0.6635	0.6551	0.6815	-0.009102	1.608
GSE89377	...	C_v1	CH_low	2	8	13	0.3627	0.3538	0.5305	-0.2597	1.404
GSE89377	...	C_v1	CH_low	2	8	13	0.3054	0.3122	-0.7148	-1.808	0.1692
GSE89377	...	C_v1	CH_low	2	8	13	0.7769	0.7696	0.6993	0.03967	1.639
GSE89377	HALLMARK_MTORC1_SIGNALING	C_v1	CH_low	2	8	13	0.3825	0.3812	0.1061	-0.5902	1.053
GSE89377	...	C_v1	CH_low	2	8	13	0.08902	0.1023	-0.7476	-1.644	0.02793
GSE89377	...	C_v1	CH_low	2	8	13	0.1513	0.1628	-0.4784	-1.473	0.3458
GSE89377	...	C_v1	CH_low	2	8	13	0.06075	0.0494	0.6033	-0.074	1.498
GSE89377	...	C_v1	CH_low	2	8	13	0.2137	0.2156	-0.06783	-0.6652	1.288
GSE89377	...	C_v1	CH_low	2	8	13	-0.07943	-0.04806	-1.193	-2.285	-0.448
GSE89377	...	C_v1	CH_low	2	8	13	0.2756	0.2852	-0.6065	-1.672	0.2027
GSE89377	...	C_v1	CH_low	2	8	13	0.07729	0.09846	-0.8277	-1.904	-0.0694
GSE89377	...	C_v1	CH_low	2	8	13	-0.03449	-0.01004	-0.7832	-1.977	-0.04909
GSE89377	...	C_v1	CH_low	2	8	13	-0.03449	-0.01004	-0.7832	-1.977	-0.04909
GSE89377	KEGG_RIBOSOME	A	CH_high	3	12	13	0.7264	0.7098	1.401	0.6805	2.517
GSE89377	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	CH_high	3	12	13	0.0596	0.05359	0.1583	-0.7667	0.8021
GSE89377	REACTOME_TRANSLATION	A	CH_high	3	12	13	0.5288	0.5105	1.41	0.7902	2.252
GSE89377	...	A	CH_high	3	12	13	0.6713	0.6541	1.546	0.7983	2.819

S34 Table (columns 1 of 2) (page 3 of 3)

GSE89377 stage-spectrum effect sizes. Hedges g with bootstrap confidence intervals for every scored set at every stage of the hepatocarcinogenesis series (normal, chronic hepatitis low and high grade, cirrhosis, dysplastic nodule low and high grade, early HCC, HCC) against the normal arm, with stage index and group sizes.

cohort	set	tier	stage	stage.index	n.stage	n.control	mean.stage	mean.control	hedges.g	ci.lo	ci.hi
GSE89377	...	A	CH_high	3	12	13	0.7163	0.6994	1.329	0.619	2.403
GSE89377	REACTOME_TRNA_AMINOACYLATION	A	CH_high	3	12	13	0.2045	0.1922	0.4906	-0.2999	1.222
GSE89377	...	A	CH_high	3	12	13	0.5097	0.4743	1.599	0.9212	2.75
GSE89377	REACTOME_RRNA_PROCESSING	A	CH_high	3	12	13	0.4645	0.4465	1.285	0.5735	2.349
GSE89377	...	A	CH_high	3	12	13	0.6826	0.6646	1.57	0.8939	2.594
GSE89377	...	A	CH_high	3	12	13	0.6296	0.6143	1.487	0.7	2.865
GSE89377	...	A	CH_high	3	12	13	0.5759	0.5604	1.352	0.5839	2.58
GSE89377	REACTOME_MITOCHONDRIAL_TRANSLATION	A	CH_high	3	12	13	0.3921	0.3679	0.8414	0.1638	1.523
GSE89377	...	A	CH_high	3	12	13	0.3985	0.3729	0.8517	0.173	1.538
GSE89377	...	A	CH_high	3	12	13	0.1039	0.1006	0.09756	-0.9189	0.7432
GSE89377	...	A	CH_high	3	12	13	0.07404	0.05214	0.5692	-0.1747	1.301
GSE89377	REACTOME_TRNA_PROCESSING	B_extra	CH_high	3	12	13	0.04294	0.02821	0.7528	-0.004462	1.885
GSE89377	...	B_extra	CH_high	3	12	13	0.07004	0.06093	0.4546	-0.2991	1.276
GSE89377	...	B_extra	CH_high	3	12	13	-0.009736	-0.02846	0.7989	0.02786	2.137
GSE89377	...	B_extra	CH_high	3	12	13	0.3407	0.3212	0.7461	0.02111	1.864
GSE89377	...	B_extra	CH_high	3	12	13	0.1613	0.1304	0.9141	0.1307	2.083

S34 Table (columns 2 of 2) (page 1 of 3)

GSE89377 stage-spectrum effect sizes. Hedges g with bootstrap confidence intervals for every scored set at every stage of the hepatocarcinogenesis series (normal, chronic hepatitis low and high grade, cirrhosis, dysplastic nodule low and high grade, early HCC, HCC) against the normal arm, with stage index and group sizes.

cohort	p.ttest	ci.excludes.0	meets.min effect	p.fdr
GSE89377	0.1136	False	True	0.2587
GSE89377	0.3935	False	True	0.4497
GSE89377	0.08365	False	True	0.2587
GSE89377	0.1032	False	True	0.2587
GSE89377	0.1511	False	True	0.2587
GSE89377	0.2025	False	True	0.2822
GSE89377	0.04588	True	True	0.2587
GSE89377	0.5138	False	True	0.5709
GSE89377	0.07258	False	True	0.2587
GSE89377	0.1539	False	True	0.2587
GSE89377	0.2402	False	True	0.3099
GSE89377	0.1588	False	True	0.2587
GSE89377	0.1617	False	True	0.2587
GSE89377	0.3794	False	True	0.4497
GSE89377	0.194	False	True	0.2822
GSE89377	0.06312	True	True	0.2587
GSE89377	0.1297	False	True	0.2587
GSE89377	0.208	False	True	0.2822
GSE89377	0.8581	False	False	0.861
GSE89377	0.05394	True	True	0.2587
GSE89377	0.2117	False	True	0.2822
GSE89377	0.129	False	True	0.2587

S34 Table (columns 2 of 2) (page 2 of 3)

GSE89377 stage-spectrum effect sizes. Hedges g with bootstrap confidence intervals for every scored set at every stage of the hepatocarcinogenesis series (normal, chronic hepatitis low and high grade, cirrhosis, dysplastic nodule low and high grade, early HCC, HCC) against the normal arm, with stage index and group sizes.

cohort	p.ttest	ci.excludes.0	meets.min effect	p.fdr
GSE89377	0.1562	False	True	0.2587
GSE89377	0.3828	False	True	0.4497
GSE89377	0.02	True	True	0.2587
GSE89377	0.7044	False	False	0.7616
GSE89377	0.08591	False	True	0.2587
GSE89377	0.1845	False	True	0.2822
GSE89377	0.1303	False	True	0.2587
GSE89377	0.07899	True	True	0.2587
GSE89377	0.7912	False	False	0.8329
GSE89377	0.0885	False	True	0.2587
GSE89377	0.2744	False	True	0.343
GSE89377	0.112	False	True	0.2587
GSE89377	0.861	False	False	0.861
GSE89377	0.01884	True	True	0.2587
GSE89377	0.1566	False	True	0.2587
GSE89377	0.07703	True	True	0.2587
GSE89377	0.08451	True	True	0.2587
GSE89377	0.08451	True	True	0.2587
GSE89377	0.001378	True	True	0.006124
GSE89377	0.6849	False	False	0.7296
GSE89377	0.001561	True	True	0.006243
GSE89377	0.0005421	True	True	0.004498

S34 Table (columns 2 of 2) (page 3 of 3)

GSE89377 stage-spectrum effect sizes. Hedges g with bootstrap confidence intervals for every scored set at every stage of the hepatocarcinogenesis series (normal, chronic hepatitis low and high grade, cirrhosis, dysplastic nodule low and high grade, early HCC, HCC) against the normal arm, with stage index and group sizes.

Showing the first 60 of 280 rows. The complete table is provided as the machine-readable file Table_S34_GSE89377_stage_spectrum_effect_sizes.csv.

cohort	p.ttest	ci.excludes.0	meets.min effect	p.fdr
GSE89377	0.00231	True	True	0.0077
GSE89377	0.2178	False	True	0.2722
GSE89377	0.0006737	True	True	0.004498
GSE89377	0.00285	True	True	0.008143
GSE89377	0.0005007	True	True	0.004498
GSE89377	0.0007871	True	True	0.004498
GSE89377	0.00198	True	True	0.007199
GSE89377	0.04025	True	True	0.07604
GSE89377	0.03829	True	True	0.07604
GSE89377	0.8026	False	False	0.8232
GSE89377	0.1492	False	True	0.2058
GSE89377	0.07262	False	True	0.1076
GSE89377	0.2639	False	True	0.3105
GSE89377	0.05321	True	True	0.09254
GSE89377	0.06486	True	True	0.09978
GSE89377	0.02898	True	True	0.06439

S35 Table (page 1 of 2)

GSE89377 cirrhosis versus HCC specificity. Per-set effect sizes for the cirrhosis and HCC arms of GSE89377, the direction-agreement flag, and the cirrhosis-to-HCC magnitude ratio, for all scored sets with the tier assignment. The Tier A summary gives 12 of 15 concordant sets (rho = +0.668), with the three discordant sets being the aminoacyl-tRNA and mitochondrial ones.

set	tier	CH.low	CH.high	Cirrhosis	DN.low	DN.high	eHCC	HCC	cirrhosis same direction.as HCC	cirrhosis magnitude ratio
KEGG_RIBOSOME	A	0.6077	1.401	2.026	1.069	0.1532	0.1955	0.473	True	4.28
KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	0.3383	0.1583	-0.7455	0.2835	1.056	0.5937	0.2163	False	3.45
REACTOME_TRANSLATION	A	0.7938	1.41	1.048	1.271	1.245	1.047	0.6593	True	1.59
...	A	0.6228	1.546	1.87	1.166	0.387	0.3176	0.5826	True	3.21
...	A	0.5482	1.329	1.984	1.067	0.2267	0.3449	0.6257	True	3.17
REACTOME_TRNA_AMINOACYLATION	A	0.5245	0.4906	-0.6031	0.4622	1.316	0.645	0.4745	False	1.27
...	A	0.8748	1.599	0.9596	0.5227	1.457	0.4667	0.63	True	1.52
REACTOME_RRNA_PROCESSING	A	0.263	1.285	0.4994	0.1009	0.2918	-0.05144	0.4584	True	1.09
...	A	0.7225	1.57	2.135	1.323	0.457	0.6338	0.8066	True	2.65
...	A	0.5784	1.487	1.759	1.106	0.7364	1.025	0.9308	True	1.89
...	A	0.4891	1.352	1.587	0.9361	0.5535	0.821	0.8586	True	1.85
REACTOME_MITOCHONDRIAL_TRANSLATION	A	0.5962	0.8414	0.333	0.9075	1.064	0.6626	0.2156	True	1.54
...	A	0.5872	0.8517	0.4116	0.9415	1.067	0.7103	0.2528	True	1.63
...	A	0.3482	0.09756	-0.8501	0.3185	0.9819	0.5673	0.3475	False	2.45
...	A	0.4909	0.5692	-0.1247	0.4445	1.334	0.4034	-0.1284	True	0.97
REACTOME_TRNA_PROCESSING	B_extra	0.8166	0.7528	-0.6661	-0.04026	1.42	0.9236	0.9264	False	0.72
...	B_extra	0.6498	0.4546	-0.2671	-0.5052	1.016	0.7574	1.07	False	0.25
...	B_extra	0.5359	0.7989	-0.7863	0.1382	0.9917	0.8682	0.6945	False	1.13
...	B_extra	-0.07267	0.7461	-0.7096	-0.3476	-0.06897	-0.3248	0.2054	False	3.45
...	B_extra	0.8239	0.9141	-0.8385	0.8644	1.114	0.3061	-0.08342	True	10.05
...	C_v1	-0.5278	0.4688	0.4579	0.8314	0.7157	0.607	1.259	True	0.36
HALLMARK_MYC_TARGETS_V1	C_v1	0.6546	1.262	0.8349	0.2166	0.7321	0.6522	1.086	True	0.77

S35 Table (page 2 of 2)

GSE89377 cirrhosis versus HCC specificity. Per-set effect sizes for the cirrhosis and HCC arms of GSE89377, the direction-agreement flag, and the cirrhosis-to-HCC magnitude ratio, for all scored sets with the tier assignment. The Tier A summary gives 12 of 15 concordant sets (rho = +0.668), with the three discordant sets being the aminoacyl-tRNA and mitochondrial ones.

set	tier	CH.low	CH.high	Cirrhosis	DN.low	DN.high	eHCC	HCC	cirrhosis same direction.as HCC	cirrhosis magnitude ratio
...	C_v1	0.7333	0.8672	0.7617	0.7991	1.313	1.778	0.5512	True	1.38
...	C_v1	0.4735	0.9398	0.7481	1.377	1.996	2.038	0.8045	True	0.93
...	C_v1	1.075	1.25	2.447	1.442	1.281	2.402	1.512	True	1.62
HALLMARK_MYC_TARGETS_V2	C_v1	-0.1527	0.3537	-0.9576	-0.5578	0.07352	-0.3329	0.0639	False	14.99
...	C_v1	0.6815	0.5576	1.361	0.2749	0.1921	0.2325	0.8582	True	1.59
...	C_v1	0.5305	0.7123	0.7774	1.614	1.129	1.564	0.8789	True	0.88
...	C_v1	-0.7148	-0.2335	-1.175	0.1381	0.7672	0.583	0.1248	False	9.42
...	C_v1	0.6993	0.9831	1.773	1.046	0.634	1.017	0.7561	True	2.34
HALLMARK_MTORC1_SIGNALING	C_v1	0.1061	0.7573	0.2183	0.3418	0.1831	0.4828	0.1424	True	1.53
...	C_v1	-0.7476	-0.1563	0.5314	-0.1604	-0.5909	-0.8128	-0.3968	False	1.34
...	C_v1	-0.4784	-1.339	-1.694	-0.7458	-0.5584	-0.9661	-0.5042	True	3.36
...	C_v1	0.6033	-0.5445	0.4805	0.7025	-0.2016	0.1729	0.1385	True	3.47
...	C_v1	-0.06783	-0.06755	0.7144	-0.4492	-0.5026	-0.4036	-0.0294	False	24.3
...	C_v1	-1.193	-1.534	-1.328	-0.4481	-0.2499	0.14	0.5933	False	2.24
...	C_v1	-0.6065	-0.7608	-1.276	-0.2282	0.7444	0.5299	0.005175	False	246.5
...	C_v1	-0.8277	-1.623	-1.765	-0.4436	-0.2199	-0.213	0.2767	False	6.38
...	C_v1	-0.7832	-1.526	-1.743	-0.3871	0.0005128	-0.05953	0.446	False	3.91
...	C_v1	-0.7832	-1.526	-1.743	-0.3871	0.0005128	-0.05953	0.446	False	3.91

S36 Table (columns 1 of 2)

Within-disease connectivity-free module cohesion. Mean pairwise Pearson correlation among each designated module's genes measured in that module's own expression matrix, against a uniform random gene-set null, with Z and empirical P. Reported alongside the module-to-universe connectivity ratio, which is the confound this measure is free of. This is the within-disease counterpart to S33, and it is the quantity that shows the connectivity-matched null to be the correct one: under the connectivity-normalised TOM null the HCC module scores Z = -0.14 (P = 0.54), whereas this unnormalised measure of the same property scores Z = 26.00.

disease	module	n.genes	n.samples	n.genes universe	obs.mean pairwise.r	null.mean.r	null.sd.r	Z	p	n.perm	k.module.mean
HF	purple	153	177	10213	0.3403	0.03552	0.007442	40.96	0.000999	1000	6.6
HCC	magenta	146	148	10213	0.3916	0.02393	0.01414	26	0.000999	1000	21.1

S36 Table (columns 2 of 2)

Within-disease connectivity-free module cohesion. Mean pairwise Pearson correlation among each designated module's genes measured in that module's own expression matrix, against a uniform random gene-set null, with Z and empirical P. Reported alongside the module-to-universe connectivity ratio, which is the confound this measure is free of. This is the within-disease counterpart to S33, and it is the quantity that shows the connectivity-matched null to be the correct one: under the connectivity-normalised TOM null the HCC module scores Z = -0.14 (P = 0.54), whereas this unnormalised measure of the same property scores Z = 26.00.

disease	k.universe mean	k.module.over universe
HF	15.2	0.434
HCC	74.1	0.285

S37 Table (columns 1 of 3) (page 1 of 8)

Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	disease	set	tier	n	n.female	g.unadjusted	ci.lo.unadj	ci.hi.unadj	p.unadjusted	g.adjusted	ci.lo.adj
GSE116250	HF	...	A	64	13	-1.046	-1.796	-0.3373	0.0009839	-1.033	-1.892
GSE116250	HF	REACTOME_MITOCHONDRIAL_TRANSLATION	A	64	13	-0.9057	-1.501	-0.3431	0.003938	-0.8895	-1.575
GSE116250	HF	...	A	64	13	-0.8454	-1.42	-0.3584	0.006881	-0.8515	-1.452
GSE116250	HF	...	A	64	13	-0.7373	-1.309	-0.1937	0.01763	-0.714	-1.381
GSE116250	HF	REACTOME_TRANSLATION	A	64	13	-0.4471	-1.133	0.1822	0.1443	-0.435	-1.151
GSE116250	HF	...	A	64	13	0.4269	-0.1239	1.003	0.163	0.448	-0.1154
GSE116250	HF	...	A	64	13	0.4125	-0.3233	1.189	0.1774	0.4064	-0.3108
GSE116250	HF	...	A	64	13	0.3671	-0.3753	1.184	0.2293	0.3535	-0.3966
GSE116250	HF	...	A	64	13	0.3085	-0.3891	1.114	0.3116	0.3024	-0.389
GSE116250	HF	KEGG_RIBOSOME	A	64	13	0.2264	-0.4749	0.9276	0.4569	0.2188	-0.4635
GSE116250	HF	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	64	13	0.2204	-0.322	0.759	0.4688	0.2538	-0.3037
GSE116250	HF	...	A	64	13	0.2126	-0.4667	0.9869	0.4847	0.2093	-0.4839
GSE116250	HF	REACTOME_TRNA_AMINOACYLATION	A	64	13	0.1057	-0.4327	0.6548	0.7278	0.1367	-0.411
GSE116250	HF	...	A	64	13	-0.04968	-0.6832	0.6319	0.87	-0.06322	-0.7098
GSE116250	HF	REACTOME_RRNA_PROCESSING	A	64	13	-0.02782	-0.6897	0.5708	0.927	-0.02653	-0.6809
GSE116250	HF	...	B_extra	64	13	-1.102	-1.891	-0.4133	0.0005475	-1.096	-1.872
GSE116250	HF	...	B_extra	64	13	-0.6463	-1.261	-0.09488	0.03652	-0.6283	-1.238
GSE116250	HF	REACTOME_TRNA_PROCESSING	B_extra	64	13	-0.5042	-1.112	0.1308	0.1005	-0.4817	-1.094
GSE116250	HF	...	B_extra	64	13	0.1654	-0.4324	0.8224	0.5864	0.1956	-0.3811
GSE116250	HF	...	B_extra	64	13	0.0256	-0.6499	0.5807	0.9328	0.03955	-0.6049
GSE116250	HF	...	C_v1	64	13	-1.287	-1.907	-0.799	7.187e-05	-1.263	-1.929
GSE116250	HF	...	C_v1	64	13	-1.163	-1.768	-0.6257	0.0002862	-1.133	-1.808

S37 Table (columns 1 of 3) (page 2 of 8)

Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	disease	set	tier	n	n.female	g.unadjusted	ci.lo.unadj	ci.hi.unadj	p.unadjusted	g.adjusted	ci.lo.adj
GSE116250	HF	...	C_v1	64	13	-1.064	-1.611	-0.5943	0.00082	-1.035	-1.617
GSE116250	HF	...	C_v1	64	13	-1.001	-1.516	-0.5507	0.001552	-0.9747	-1.542
GSE116250	HF	...	C_v1	64	13	-0.9163	-1.539	-0.3779	0.003561	-0.8981	-1.568
GSE116250	HF	...	C_v1	64	13	-0.8974	-1.471	-0.4432	0.004256	-0.8755	-1.443
GSE116250	HF	...	C_v1	64	13	-0.8458	-1.535	-0.2258	0.006858	-0.8314	-1.527
GSE116250	HF	...	C_v1	64	13	-0.8246	-1.423	-0.3089	0.008301	-0.8146	-1.42
GSE116250	HF	...	C_v1	64	13	0.5476	-0.1147	1.313	0.07497	0.5789	-0.06649
GSE116250	HF	...	C_v1	64	13	-0.5036	-0.9367	-0.09362	0.1009	-0.4888	-0.9344
GSE116250	HF	...	C_v1	64	13	0.5005	-0.1879	1.305	0.1029	0.5055	-0.1589
GSE116250	HF	...	C_v1	64	13	-0.2979	-0.7162	0.09987	0.3283	-0.2672	-0.7196
GSE116250	HF	...	C_v1	64	13	0.2779	-0.4423	1.046	0.3617	0.2624	-0.4679
GSE116250	HF	HALLMARK_MYC_TARGETS_V1	C_v1	64	13	-0.2688	-0.7905	0.2913	0.3774	-0.2679	-0.8186
GSE116250	HF	HALLMARK_MYC_TARGETS_V2	C_v1	64	13	0.2571	-0.3421	0.8522	0.3984	0.2649	-0.3267
GSE116250	HF	...	C_v1	64	13	-0.1694	-0.7958	0.525	0.5773	-0.1743	-0.847
GSE116250	HF	...	C_v1	64	13	-0.104	-0.6743	0.4412	0.7321	-0.1178	-0.7012
GSE116250	HF	HALLMARK_MTORC1_SIGNALING	C_v1	64	13	0.07849	-0.6353	0.7507	0.7961	0.07701	-0.6727
GSE116250	HF	...	C_v1	64	13	0.0404	-0.4871	0.6389	0.8941	0.03404	-0.5226
GSE116250	HF	...	C_v1	64	13	0.02663	-0.4981	0.5472	0.9301	0.0426	-0.4751
GSE141910	HF	...	A	366	172	0.5115	0.3103	0.7315	1.654e-06	0.5163	0.2928
GSE141910	HF	...	A	366	172	-0.4446	-0.6387	-0.2473	2.905e-05	-0.4473	-0.6477
GSE141910	HF	...	A	366	172	0.3338	0.1225	0.5425	0.001601	0.3185	0.1052
GSE141910	HF	KEGG_RIBOSOME	A	366	172	0.3292	0.1181	0.5393	0.001854	0.3136	0.09978

S37 Table (columns 1 of 3) (page 3 of 8)

Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	disease	set	tier	n	n.female	g.unadjusted	ci.lo.unadj	ci.hi.unadj	p.unadjusted	g.adjusted	ci.lo.adj
GSE141910	HF	...	A	366	172	0.2276	0.02338	0.4368	0.03086	0.2176	0.006651
GSE141910	HF	...	A	366	172	-0.2201	-0.4121	-0.02926	0.03677	-0.2072	-0.4093
GSE141910	HF	...	A	366	172	0.2038	-0.004436	0.4133	0.05299	0.1929	-0.01789
GSE141910	HF	...	A	366	172	0.1844	-0.01966	0.3937	0.07993	0.1789	-0.02825
GSE141910	HF	...	A	366	172	0.1155	-0.0952	0.3136	0.2719	0.1434	-0.06917
GSE141910	HF	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	366	172	0.1064	-0.09129	0.3128	0.3115	0.1039	-0.1063
GSE141910	HF	...	A	366	172	0.08926	-0.1202	0.2975	0.3958	0.1115	-0.09922
GSE141910	HF	REACTOME_TRANSLATION	A	366	172	0.06832	-0.141	0.264	0.5157	0.0853	-0.1246
GSE141910	HF	REACTOME_MITOCHONDRIAL_TRANSLATION	A	366	172	0.06534	-0.1455	0.2647	0.5341	0.09329	-0.1172
GSE141910	HF	REACTOME_TRNA_AMINOACYLATION	A	366	172	0.02325	-0.1788	0.2221	0.8249	0.02391	-0.1841
GSE141910	HF	REACTOME_RRNA_PROCESSING	A	366	172	-0.01464	-0.2301	0.1897	0.8892	-0.01653	-0.2354
GSE141910	HF	...	B_extra	366	172	-0.8326	-1.041	-0.6396	2.715e-14	-0.817	-1.037
GSE141910	HF	REACTOME_TRNA_PROCESSING	B_extra	366	172	-0.6719	-0.8875	-0.478	4.809e-10	-0.6516	-0.8856
GSE141910	HF	...	B_extra	366	172	-0.5684	-0.7803	-0.3737	1.124e-07	-0.543	-0.7718
GSE141910	HF	...	B_extra	366	172	-0.4657	-0.6802	-0.2613	1.221e-05	-0.4589	-0.6813
GSE141910	HF	...	B_extra	366	172	-0.2827	-0.4927	-0.08625	0.007424	-0.259	-0.4754
GSE141910	HF	...	B_extra	366	172	-0.2537	-0.4589	-0.0547	0.01617	-0.237	-0.4475
GSE141910	HF	...	C_v1	366	172	-1.128	-1.39	-0.8919	1.465e-23	-1.164	-1.444
GSE141910	HF	...	C_v1	366	172	-1.081	-1.289	-0.8861	5.418e-22	-1.086	-1.327
GSE141910	HF	HALLMARK_MTORC1_SIGNALING	C_v1	366	172	-0.9338	-1.116	-0.755	2.801e-17	-0.9215	-1.118
GSE141910	HF	...	C_v1	366	172	-0.6545	-0.8609	-0.448	1.266e-09	-0.6275	-0.8461
GSE141910	HF	...	C_v1	366	172	-0.5205	-0.7197	-0.317	1.099e-06	-0.5572	-0.7589

S37 Table (columns 1 of 3) (page 4 of 8)

Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	disease	set	tier	n	n.female	g.unadjusted	ci.lo.unadj	ci.hi.unadj	p.unadjusted	g.adjusted	ci.lo.adj
GSE141910	HF	...	C_v1	366	172	-0.4175	-0.6245	-0.2139	8.452e-05	-0.4031	-0.6175
GSE141910	HF	...	C_v1	366	172	0.4156	0.2057	0.6254	9.086e-05	0.3721	0.1601
GSE141910	HF	...	C_v1	366	172	0.3207	0.1248	0.5251	0.002422	0.2714	0.07112
GSE141910	HF	HALLMARK_MYC_TARGETS_V2	C_v1	366	172	-0.3012	-0.5081	-0.1029	0.004364	-0.2849	-0.5048
GSE141910	HF	...	C_v1	366	172	-0.2954	-0.5126	-0.09045	0.00517	-0.281	-0.519
GSE141910	HF	...	C_v1	366	172	0.2915	0.08424	0.4998	0.00579	0.2663	0.05286
GSE141910	HF	...	C_v1	366	172	-0.2448	-0.4507	-0.03648	0.02026	-0.2764	-0.4938
GSE141910	HF	...	C_v1	366	172	-0.225	-0.4287	-0.02237	0.03277	-0.1895	-0.404
GSE141910	HF	...	C_v1	366	172	-0.2201	-0.427	-0.01936	0.03679	-0.191	-0.3993
GSE141910	HF	HALLMARK_MYC_TARGETS_V1	C_v1	366	172	-0.2099	-0.4142	-0.01038	0.0463	-0.1716	-0.3808
GSE141910	HF	...	C_v1	366	172	-0.1505	-0.3544	0.05868	0.1525	-0.1902	-0.4074
GSE141910	HF	...	C_v1	366	172	-0.1238	-0.3316	0.08039	0.2391	-0.1659	-0.385
GSE141910	HF	...	C_v1	366	172	0.07226	-0.1329	0.2833	0.4917	0.03135	-0.176
GSE141910	HF	...	C_v1	366	172	-0.04541	-0.2526	0.1594	0.6656	-0.08829	-0.3077
GSE141910	HF	...	C_v1	366	172	-0.02192	-0.228	0.1739	0.8347	-0.01699	-0.2289
GSE57338	HF	...	A	313	96	-0.9199	-1.14	-0.7121	1.575e-14	-0.8337	-1.076
GSE57338	HF	REACTOME_RRNA_PROCESSING	A	313	96	-0.8114	-1.009	-0.6257	7.726e-12	-0.6811	-0.8929
GSE57338	HF	...	A	313	96	-0.6511	-0.8899	-0.426	2.644e-08	-0.4672	-0.707
GSE57338	HF	...	A	313	96	-0.6071	-0.8506	-0.38	1.944e-07	-0.4302	-0.6747
GSE57338	HF	...	A	313	96	-0.5993	-0.832	-0.3842	2.741e-07	-0.4254	-0.6642
GSE57338	HF	...	A	313	96	0.5582	0.3478	0.8013	1.578e-06	0.4634	0.2289
GSE57338	HF	...	A	313	96	-0.5521	-0.7836	-0.3376	2.034e-06	-0.4165	-0.6543

S37 Table (columns 1 of 3) (page 5 of 8)

Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	disease	set	tier	n	n.female	g.unadjusted	ci.lo.unadj	ci.hi.unadj	p.unadjusted	g.adjusted	ci.lo.adj
GSE57338	HF	REACTOME_TRANSLATION	A	313	96	-0.539	-0.7481	-0.3261	3.462e-06	-0.5068	-0.7254
GSE57338	HF	...	A	313	96	-0.4757	-0.719	-0.2495	3.924e-05	-0.2831	-0.5226
GSE57338	HF	KEGG_RIBOSOME	A	313	96	-0.4473	-0.6842	-0.2198	0.000108	-0.2565	-0.4971
GSE57338	HF	REACTOME_TRNA_AMINOACYLATION	A	313	96	-0.3621	-0.5795	-0.1384	0.001648	-0.3498	-0.5755
GSE57338	HF	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	313	96	-0.2085	-0.4259	0.02248	0.06849	-0.2122	-0.4401
GSE57338	HF	...	A	313	96	-0.1005	-0.3021	0.1271	0.3786	-0.2159	-0.4391
GSE57338	HF	REACTOME_MITOCHONDRIAL_TRANSLATION	A	313	96	-0.08824	-0.2918	0.1404	0.4396	-0.1916	-0.4095
GSE57338	HF	...	A	313	96	-0.08121	-0.2871	0.1501	0.4769	-0.1826	-0.4028
GSE57338	HF	...	B_extra	313	96	-0.8652	-1.053	-0.6916	3.821e-13	-0.7351	-0.9298
GSE57338	HF	...	B_extra	313	96	-0.5808	-0.7872	-0.3723	6.096e-07	-0.4577	-0.6684
GSE57338	HF	...	B_extra	313	96	-0.4652	-0.6919	-0.2383	5.745e-05	-0.506	-0.7443
GSE57338	HF	REACTOME_TRNA_PROCESSING	B_extra	313	96	-0.4411	-0.6482	-0.2144	0.0001333	-0.3677	-0.5794
GSE57338	HF	...	B_extra	313	96	-0.3714	-0.5924	-0.1432	0.001249	-0.3984	-0.636
GSE57338	HF	...	B_extra	313	96	-0.1418	-0.3572	0.1075	0.2147	-0.09715	-0.3092
GSE57338	HF	...	C_v1	313	96	-1.759	-2.069	-1.483	2.86e-40	-1.686	-2.003
GSE57338	HF	...	C_v1	313	96	-1.22	-1.499	-0.9801	5.855e-23	-1.119	-1.418
GSE57338	HF	HALLMARK_MYC_TARGETS_V2	C_v1	313	96	-1.208	-1.4	-1.043	1.38e-22	-1.115	-1.328
GSE57338	HF	...	C_v1	313	96	-1.198	-1.44	-0.9898	2.731e-22	-1.195	-1.467
GSE57338	HF	...	C_v1	313	96	-1.099	-1.337	-0.8843	2.049e-19	-1.003	-1.273
GSE57338	HF	...	C_v1	313	96	-1.097	-1.406	-0.8258	2.408e-19	-1.045	-1.392
GSE57338	HF	HALLMARK_MTORC1_SIGNALING	C_v1	313	96	-1.001	-1.24	-0.7778	1.159e-16	-0.8567	-1.105
GSE57338	HF	HALLMARK_MYC_TARGETS_V1	C_v1	313	96	-0.8521	-1.051	-0.6616	8.017e-13	-0.7047	-0.9192

S37 Table (columns 1 of 3) (page 6 of 8)

Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	disease	set	tier	n	n.female	g.unadjusted	ci.lo.unadj	ci.hi.unadj	p.unadjusted	g.adjusted	ci.lo.adj
GSE57338	HF	...	C_v1	313	96	-0.7772	-1.036	-0.5531	4.879e-11	-0.6984	-0.9654
GSE57338	HF	...	C_v1	313	96	-0.6502	-0.9106	-0.4242	2.754e-08	-0.6592	-0.9212
GSE57338	HF	...	C_v1	313	96	-0.6302	-0.8678	-0.4065	6.9e-08	-0.4577	-0.6915
GSE57338	HF	...	C_v1	313	96	-0.5934	-0.8034	-0.3864	3.556e-07	-0.5269	-0.7467
GSE57338	HF	...	C_v1	313	96	-0.5334	-0.7762	-0.3027	4.339e-06	-0.3231	-0.5665
GSE57338	HF	...	C_v1	313	96	-0.5249	-0.7714	-0.2913	6.066e-06	-0.5239	-0.7762
GSE57338	HF	...	C_v1	313	96	0.3116	0.08879	0.5287	0.006643	0.3423	0.1024
GSE57338	HF	...	C_v1	313	96	-0.2876	-0.516	-0.07708	0.01215	-0.2306	-0.4826
GSE57338	HF	...	C_v1	313	96	0.2075	-0.02197	0.42	0.06983	0.1864	-0.06528
GSE57338	HF	...	C_v1	313	96	0.1559	-0.07905	0.3744	0.1727	0.1388	-0.115
GSE57338	HF	...	C_v1	313	96	-0.08337	-0.3295	0.1254	0.4652	-0.001181	-0.2532
GSE57338	HF	...	C_v1	313	96	0.04598	-0.1794	0.2756	0.687	-0.1191	-0.3661
TCGA_LIHC	HCC	...	A	420	143	1.173	0.9568	1.414	5.445e-14	1.16	0.9311
TCGA_LIHC	HCC	REACTOME_TRNA_AMINOACYLATION	A	420	143	1.119	0.9258	1.337	6.435e-13	1.116	0.9039
TCGA_LIHC	HCC	...	A	420	143	1.101	0.8983	1.325	1.424e-12	1.069	0.8573
TCGA_LIHC	HCC	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	420	143	1.024	0.8292	1.244	3.756e-11	1.023	0.8104
TCGA_LIHC	HCC	...	A	420	143	0.9665	0.7595	1.203	3.825e-10	0.9262	0.7155
TCGA_LIHC	HCC	...	A	420	143	0.8786	0.6629	1.117	1.103e-08	0.8433	0.6298
TCGA_LIHC	HCC	...	A	420	143	0.8552	0.6515	1.085	2.585e-08	0.8176	0.602
TCGA_LIHC	HCC	REACTOME_RRNA_PROCESSING	A	420	143	0.8332	0.6455	1.043	5.649e-08	0.7992	0.6024
TCGA_LIHC	HCC	...	A	420	143	0.7321	0.5319	0.9522	1.671e-06	0.695	0.486
TCGA_LIHC	HCC	REACTOME_TRANSLATION	A	420	143	0.6849	0.4885	0.8949	7.173e-06	0.6595	0.4637

S37 Table (columns 1 of 3) (page 7 of 8)

Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	disease	set	tier	n	n.female	g.unadjusted	ci.lo.unadj	ci.hi.unadj	p.unadjusted	g.adjusted	ci.lo.adj
TCGA_LIHC	HCC	...	A	420	143	0.6592	0.4426	0.8836	1.534e-05	0.661	0.4295
TCGA_LIHC	HCC	KEGG_RIBOSOME	A	420	143	0.6375	0.4276	0.8654	2.858e-05	0.5961	0.3808
TCGA_LIHC	HCC	...	A	420	143	-0.2566	-0.428	-0.07849	0.0893	-0.2902	-0.482
TCGA_LIHC	HCC	...	A	420	143	0.1763	-0.02324	0.3882	0.2426	0.1604	-0.04234
TCGA_LIHC	HCC	REACTOME_MITOCHONDRIAL_TRANSLATION	A	420	143	0.1235	-0.07109	0.3299	0.4129	0.107	-0.09925
TCGA_LIHC	HCC	...	B_extra	420	143	1.458	1.298	1.635	4.057e-20	1.44	1.275
TCGA_LIHC	HCC	REACTOME_TRNA_PROCESSING	B_extra	420	143	1.268	1.071	1.477	6.315e-16	1.244	1.038
TCGA_LIHC	HCC	...	B_extra	420	143	0.9677	0.7466	1.196	3.653e-10	0.9465	0.7115
TCGA_LIHC	HCC	...	B_extra	420	143	0.7823	0.5712	0.9971	3.26e-07	0.7678	0.5412
TCGA_LIHC	HCC	...	B_extra	420	143	-0.5114	-0.6714	-0.3526	0.0007543	-0.5443	-0.7244
TCGA_LIHC	HCC	...	B_extra	420	143	0.4894	0.2941	0.7024	0.001255	0.4607	0.248
TCGA_LIHC	HCC	...	C_v1	420	143	1.684	1.48	1.927	1.451e-25	1.682	1.475
TCGA_LIHC	HCC	HALLMARK_MYC_TARGETS_V1	C_v1	420	143	1.46	1.285	1.654	3.618e-20	1.436	1.255
TCGA_LIHC	HCC	...	C_v1	420	143	1.349	1.169	1.563	1.139e-17	1.343	1.152
TCGA_LIHC	HCC	...	C_v1	420	143	1.065	0.8506	1.298	6.734e-12	1.029	0.8177
TCGA_LIHC	HCC	...	C_v1	420	143	1.03	0.829	1.259	2.919e-11	1.005	0.8001
TCGA_LIHC	HCC	HALLMARK_MYC_TARGETS_V2	C_v1	420	143	0.9586	0.7425	1.187	5.233e-10	0.9323	0.7025
TCGA_LIHC	HCC	...	C_v1	420	143	0.819	0.618	1.042	9.301e-08	0.7778	0.5815
TCGA_LIHC	HCC	...	C_v1	420	143	0.7578	0.5591	0.9668	7.311e-07	0.7428	0.5361
TCGA_LIHC	HCC	...	C_v1	420	143	0.6055	0.4112	0.8306	6.948e-05	0.6367	0.4325
TCGA_LIHC	HCC	...	C_v1	420	143	-0.5975	-0.823	-0.3741	8.617e-05	-0.6016	-0.8361
TCGA_LIHC	HCC	HALLMARK_MTORC1_SIGNALING	C_v1	420	143	0.5723	0.409	0.747	0.0001673	0.5395	0.3639

S37 Table (columns 1 of 3) (page 8 of 8)

Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	disease	set	tier	n	n.female	g.unadjusted	ci.lo.unadj	ci.hi.unadj	p.unadjusted	g.adjusted	ci.lo.adj
TCGA_LIHC	HCC	...	C_v1	420	143	0.5455	0.3084	0.8097	0.0003303	0.4846	0.2577
TCGA_LIHC	HCC	...	C_v1	420	143	-0.5128	-0.6781	-0.3523	0.000729	-0.5195	-0.6965
TCGA_LIHC	HCC	...	C_v1	420	143	0.4754	0.2664	0.6808	0.00172	0.4634	0.253
TCGA_LIHC	HCC	...	C_v1	420	143	0.3498	0.1345	0.5654	0.02074	0.353	0.136
TCGA_LIHC	HCC	...	C_v1	420	143	-0.241	-0.4025	-0.08162	0.1105	-0.2341	-0.4004
TCGA_LIHC	HCC	...	C_v1	420	143	0.1553	-0.07463	0.3878	0.3034	0.1632	-0.06595
TCGA_LIHC	HCC	...	C_v1	420	143	0.1445	-0.04662	0.3425	0.338	0.1142	-0.09089
TCGA_LIHC	HCC	...	C_v1	420	143	-0.03548	-0.2598	0.1921	0.8139	-0.03294	-0.2657
TCGA_LIHC	HCC	...	C_v1	420	143	-0.02181	-0.2565	0.2087	0.885	-0.01581	-0.2513

S37 Table (columns 2 of 3) (page 1 of 8)

Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	ci.hi.adj	p.adjusted	b.age	p.age	b.sex.female	p.sex	max.vif	covariates	attenuation	sign.kept	survives
GSE116250	-0.3419	0.001172	-0.2853	0.02975	-0.09066	0.7748	1.03	age+sex	0.01252	True	True
GSE116250	-0.3367	0.004719	-0.2515	0.05438	-0.1723	0.5869	1.03	age+sex	0.0178	True	True
GSE116250	-0.39	0.006688	-0.2665	0.04188	-0.4747	0.1376	1.03	age+sex	-0.007179	True	True
GSE116250	-0.1583	0.02176	-0.229	0.07898	-0.1058	0.7384	1.03	age+sex	0.03163	True	True
GSE116250	0.1674	0.1564	-0.1577	0.2234	-0.3905	0.2206	1.03	age+sex	0.02705	True	False
GSE116250	1.014	0.1446	-0.1536	0.2355	-0.02594	0.9347	1.03	age+sex	-0.04932	True	False
GSE116250	1.227	0.185	0.0777	0.5466	-0.8621	0.008236	1.03	age+sex	0.01484	True	False
GSE116250	1.249	0.248	0.09672	0.4534	-0.7723	0.01729	1.03	age+sex	0.03703	True	False
GSE116250	1.123	0.3223	0.03276	0.7991	-0.7754	0.01688	1.03	age+sex	0.01956	True	False
GSE116250	0.9737	0.473	0.06034	0.6395	-1.073	0.001201	1.03	age+sex	0.03327	True	False
GSE116250	0.8127	0.4056	-0.2146	0.0993	-0.03575	0.9102	1.03	age+sex	-0.1517	True	False
GSE116250	0.9889	0.4924	-0.01452	0.9102	-0.1858	0.5582	1.03	age+sex	0.01526	True	False
GSE116250	0.6846	0.6536	-0.206	0.1131	-0.0446	0.888	1.03	age+sex	-0.2929	True	False
GSE116250	0.6397	0.8355	0.03653	0.7766	-0.5385	0.09299	1.03	age+sex	-0.2727	True	False
GSE116250	0.5731	0.9305	-0.02318	0.8571	-0.1783	0.574	1.03	age+sex	0.04659	True	False
GSE116250	-0.4958	0.0006148	-0.04308	0.738	-0.3406	0.2846	1.03	age+sex	0.005927	True	True
GSE116250	-0.0741	0.04245	-0.05499	0.6694	0.01065	0.9732	1.03	age+sex	0.02776	True	True
GSE116250	0.1574	0.1172	-0.101	0.4338	0.2673	0.4001	1.03	age+sex	0.04461	True	False
GSE116250	0.8808	0.5211	-0.1225	0.343	0.6204	0.05385	1.03	age+sex	-0.1829	True	False
GSE116250	0.6218	0.8966	-0.06388	0.62	0.318	0.3175	1.03	age+sex	-0.5454	True	False
GSE116250	-0.7578	0.0001003	-0.05493	0.6697	-0.1147	0.7175	1.03	age+sex	0.01828	True	True
GSE116250	-0.6119	0.000416	-0.1314	0.3093	0.1022	0.747	1.03	age+sex	0.02573	True	True

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Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	ci.hi.adj	p.adjusted	b.age	p.age	b.sex.female	p.sex	max.vif	covariates	attenuation	sign.kept	survives
GSE116250	-0.5521	0.00115	-0.1717	0.1852	0.2131	0.502	1.03	age+sex	0.02712	True	True
GSE116250	-0.4958	0.002095	-0.1765	0.1736	-0.01159	0.9708	1.03	age+sex	0.02653	True	True
GSE116250	-0.3625	0.004359	-0.03859	0.7644	-0.07766	0.8064	1.03	age+sex	0.01988	True	True
GSE116250	-0.404	0.005375	-0.05612	0.663	0.2115	0.5052	1.03	age+sex	0.02447	True	True
GSE116250	-0.2347	0.008009	-0.05718	0.6571	0.4523	0.1568	1.03	age+sex	0.01702	True	True
GSE116250	-0.3062	0.00929	0.02088	0.8711	0.1981	0.5324	1.03	age+sex	0.01204	True	True
GSE116250	1.441	0.06088	0.1025	0.4271	-1.33	8.54e-05	1.03	age+sex	-0.05719	True	False
GSE116250	-0.04852	0.112	-0.04284	0.7393	0.04144	0.8959	1.03	age+sex	0.02926	True	False
GSE116250	1.4	0.1006	0.08616	0.504	-1.01	0.002184	1.03	age+sex	-0.009898	True	False
GSE116250	0.1373	0.3815	-0.2201	0.09106	0.007127	0.982	1.03	age+sex	0.1031	True	False
GSE116250	1.053	0.39	0.07877	0.5411	0.001716	0.9957	1.03	age+sex	0.05556	True	False
GSE116250	0.318	0.3802	0.0347	0.7875	0.2228	0.4827	1.03	age+sex	0.003332	True	False
GSE116250	0.8669	0.3857	-0.06382	0.6203	0.1134	0.7205	1.03	age+sex	-0.02999	True	False
GSE116250	0.5343	0.5673	0.04757	0.7118	-0.02309	0.9419	1.03	age+sex	-0.02901	True	False
GSE116250	0.4342	0.6989	0.08904	0.4899	-0.1175	0.7109	1.03	age+sex	-0.133	True	False
GSE116250	0.7566	0.8003	0.002472	0.9847	0.01181	0.9702	1.03	age+sex	0.01883	True	False
GSE116250	0.6202	0.9109	0.03629	0.778	-0.01974	0.9503	1.03	age+sex	0.1575	True	False
GSE116250	0.5974	0.8887	-0.05844	0.65	0.4958	0.1213	1.03	age+sex	-0.5998	True	False
GSE141910	0.7533	2.323e-06	0.03466	0.5167	-0.06183	0.5589	1.007	age+sex	-0.009307	True	True
GSE141910	-0.2467	4e-05	0.001232	0.9816	-0.03546	0.7375	1.007	age+sex	-0.006014	True	True
GSE141910	0.5319	0.003262	0.0362	0.4983	-0.2553	0.01621	1.007	age+sex	0.04587	True	True
GSE141910	0.5266	0.003767	0.03782	0.4793	-0.2644	0.01281	1.007	age+sex	0.04746	True	True

S37 Table (columns 2 of 3) (page 3 of 8)

Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	ci.hi.adj	p.adjusted	b.age	p.age	b.sex.female	p.sex	max.vif	covariates	attenuation	sign.kept	survives
GSE141910	0.4309	0.04379	0.03806	0.4765	-0.2071	0.05086	1.007	age+sex	0.04379	True	True
GSE141910	-0.009171	0.05485	0.1138	0.03378	-0.2329	0.0282	1.007	age+sex	0.05859	True	False
GSE141910	0.4071	0.07378	0.03748	0.4833	-0.2136	0.04407	1.007	age+sex	0.05385	True	False
GSE141910	0.383	0.097	0.06159	0.2496	-0.247	0.02	1.007	age+sex	0.02941	True	False
GSE141910	0.3532	0.1834	0.1008	0.05996	-0.08863	0.4023	1.007	age+sex	-0.2409	True	False
GSE141910	0.3208	0.3346	0.007536	0.8879	-0.0421	0.6907	1.007	age+sex	0.02357	True	False
GSE141910	0.3185	0.3006	0.07234	0.1764	-0.04308	0.6838	1.007	age+sex	-0.2489	True	False
GSE141910	0.2907	0.4282	0.08678	0.105	-0.1332	0.2084	1.007	age+sex	-0.2485	True	False
GSE141910	0.299	0.3863	0.1037	0.05287	-0.09591	0.3648	1.007	age+sex	-0.4278	True	False
GSE141910	0.2362	0.8242	0.02067	0.6989	-0.05889	0.5777	1.007	age+sex	-0.02828	True	False
GSE141910	0.1934	0.878	0.04434	0.4069	-0.1543	0.1452	1.007	age+sex	-0.1286	True	False
GSE141910	-0.6205	2.622e-13	0.08289	0.1215	-0.1171	0.2687	1.007	age+sex	0.01876	True	True
GSE141910	-0.4515	3.45e-09	0.02026	0.7046	0.09687	0.36	1.007	age+sex	0.03024	True	True
GSE141910	-0.3338	7.045e-07	0.04438	0.4065	0.06472	0.5407	1.007	age+sex	0.04461	True	True
GSE141910	-0.2492	2.533e-05	-0.03192	0.5504	0.1583	0.135	1.007	age+sex	0.0145	True	True
GSE141910	-0.06135	0.01654	0.08198	0.1257	-0.05934	0.5749	1.007	age+sex	0.0838	True	True
GSE141910	-0.01352	0.02821	0.02479	0.6428	0.05637	0.5941	1.007	age+sex	0.06601	True	True
GSE141910	-0.9254	7.718e-24	0.02181	0.6832	-0.294	0.005699	1.007	age+sex	-0.03217	True	True
GSE141910	-0.8773	2.807e-21	-0.05865	0.2728	0.1627	0.1245	1.007	age+sex	-0.004245	True	True
GSE141910	-0.7324	3.071e-16	-0.02787	0.602	0.2114	0.04622	1.007	age+sex	0.01314	True	True
GSE141910	-0.4099	1.2e-08	0.05681	0.2881	0.03989	0.7061	1.007	age+sex	0.04124	True	True
GSE141910	-0.3451	3.679e-07	-0.1112	0.03807	0.06004	0.5704	1.007	age+sex	-0.07053	True	True

S37 Table (columns 2 of 3) (page 4 of 8)

Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	ci.hi.adj	p.adjusted	b.age	p.age	b.sex.female	p.sex	max.vif	covariates	attenuation	sign.kept	survives
GSE141910	-0.1963	0.000207	0.03153	0.5552	0.01178	0.9113	1.007	age+sex	0.03432	True	True
GSE141910	0.5815	0.0006053	0.03678	0.4914	-0.6284	6.516e-09	1.007	age+sex	0.1047	True	True
GSE141910	0.4803	0.01204	-0.08333	0.1195	-0.1562	0.1403	1.007	age+sex	0.1536	True	True
GSE141910	-0.06885	0.008417	0.03748	0.4832	0.0116	0.9127	1.007	age+sex	0.05393	True	True
GSE141910	-0.06339	0.00935	0.02687	0.6151	0.02879	0.7855	1.007	age+sex	0.0486	True	True
GSE141910	0.4771	0.01373	0.04599	0.3897	-0.3909	0.0002507	1.007	age+sex	0.08616	True	True
GSE141910	-0.0614	0.01057	-0.1161	0.03041	0.1129	0.286	1.007	age+sex	-0.1289	True	True
GSE141910	0.0261	0.07899	0.06688	0.2112	0.08496	0.4221	1.007	age+sex	0.158	True	False
GSE141910	0.009748	0.07655	0.08096	0.1304	-0.01316	0.901	1.007	age+sex	0.1319	True	False
GSE141910	0.03761	0.1115	0.05967	0.2646	0.1319	0.213	1.007	age+sex	0.1826	True	False
GSE141910	0.03005	0.07784	-0.1437	0.007476	0.1335	0.2073	1.007	age+sex	-0.2634	True	False
GSE141910	0.04628	0.1238	-0.132	0.01388	0.07208	0.4957	1.007	age+sex	-0.3401	True	False
GSE141910	0.24	0.7708	-0.1298	0.01553	0.06553	0.5357	1.007	age+sex	0.5662	True	False
GSE141910	0.1247	0.4122	-0.1315	0.0143	0.05994	0.571	1.007	age+sex	-0.9441	True	False
GSE141910	0.188	0.8746	0.04796	0.3697	-0.1095	0.3007	1.007	age+sex	0.225	True	False
GSE57338	-0.6164	5.423e-11	-0.1718	0.003356	0.06952	0.5889	1.001	age+sex	0.09374	True	True
GSE57338	-0.4757	5.983e-08	-0.2101	0.0003508	0.1894	0.1415	1.001	age+sex	0.1605	True	True
GSE57338	-0.2312	0.0001672	-0.2486	2.532e-05	0.3446	0.007729	1.001	age+sex	0.2824	True	True
GSE57338	-0.1895	0.0005164	-0.2436	3.64e-05	0.3159	0.01452	1.001	age+sex	0.2914	True	True
GSE57338	-0.1905	0.0005955	-0.2354	6.514e-05	0.3135	0.01526	1.001	age+sex	0.2902	True	True
GSE57338	0.7363	0.0001884	0.04179	0.4728	-0.2855	0.02707	1.001	age+sex	0.1698	True	True
GSE57338	-0.1891	0.0007706	-0.1907	0.001155	0.2159	0.09404	1.001	age+sex	0.2456	True	True

S37 Table (columns 2 of 3) (page 5 of 8)

Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	ci.hi.adj	p.adjusted	b.age	p.age	b.sex.female	p.sex	max.vif	covariates	attenuation	sign.kept	survives
GSE57338	-0.2749	4.599e-05	-0.1063	0.06838	-0.05138	0.6896	1.001	age+sex	0.05964	True	True
GSE57338	-0.04291	0.02158	-0.2513	2.082e-05	0.3509	0.006684	1.001	age+sex	0.4048	True	True
GSE57338	-0.01401	0.03723	-0.2469	2.873e-05	0.3455	0.007566	1.001	age+sex	0.4264	True	True
GSE57338	-0.1066	0.00462	-0.102	0.08029	-0.1169	0.3637	1.001	age+sex	0.0338	True	True
GSE57338	0.0295	0.08457	-0.05877	0.3128	-0.1089	0.3975	1.001	age+sex	-0.01773	True	False
GSE57338	0.02231	0.07922	0.06955	0.2325	-0.3013	0.01968	1.001	age+sex	-1.148	True	False
GSE57338	0.04733	0.1191	0.05373	0.3561	-0.2845	0.0276	1.001	age+sex	-1.171	True	False
GSE57338	0.05712	0.1373	0.043	0.4601	-0.295	0.02238	1.001	age+sex	-1.249	True	False
GSE57338	-0.5393	5.652e-09	-0.1763	0.002636	0.2341	0.06953	1.001	age+sex	0.1503	True	True
GSE57338	-0.236	0.0002254	-0.07957	0.1721	0.3377	0.009033	1.001	age+sex	0.2121	True	True
GSE57338	-0.2578	4.732e-05	-0.002742	0.9624	-0.1535	0.2333	1.001	age+sex	-0.08779	True	True
GSE57338	-0.1332	0.002929	-0.05705	0.3272	0.1746	0.1753	1.001	age+sex	0.1665	True	True
GSE57338	-0.1544	0.001285	-0.07471	0.1997	-0.2129	0.09865	1.001	age+sex	-0.07251	True	True
GSE57338	0.1537	0.4288	-0.0364	0.5317	0.1014	0.4309	1.001	age+sex	0.3148	True	False
GSE57338	-1.415	6.563e-34	-0.138	0.01822	0.0787	0.5408	1.001	age+sex	0.04136	True	True
GSE57338	-0.8595	9.32e-18	-0.1919	0.001075	0.1206	0.3487	1.001	age+sex	0.08278	True	True
GSE57338	-0.9303	1.216e-17	-0.1473	0.01178	0.1343	0.2969	1.001	age+sex	0.07706	True	True
GSE57338	-0.9645	9.304e-20	-0.1429	0.01454	-0.1794	0.1637	1.001	age+sex	0.001978	True	True
GSE57338	-0.7563	7.466e-15	-0.154	0.008475	0.1359	0.291	1.001	age+sex	0.08768	True	True
GSE57338	-0.757	7.142e-16	-0.1279	0.02852	-0.0004989	0.9969	1.001	age+sex	0.04765	True	True
GSE57338	-0.6195	1.725e-11	-0.1416	0.0154	0.3524	0.006459	1.001	age+sex	0.1437	True	True
GSE57338	-0.4964	2.167e-08	-0.1563	0.007571	0.335	0.009591	1.001	age+sex	0.173	True	True

S37 Table (columns 2 of 3) (page 6 of 8)

Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	ci.hi.adj	p.adjusted	b.age	p.age	b.sex.female	p.sex	max.vif	covariates	attenuation	sign.kept	survives
GSE57338	-0.4612	2.857e-08	-0.1008	0.08389	0.1297	0.3139	1.001	age+sex	0.1014	True	True
GSE57338	-0.4171	1.5e-07	-0.1685	0.004021	-0.264	0.04083	1.001	age+sex	-0.01386	True	True
GSE57338	-0.2267	0.0002254	-0.2441	3.505e-05	0.3006	0.01998	1.001	age+sex	0.2738	True	True
GSE57338	-0.3027	2.321e-05	-0.1141	0.05058	0.06213	0.6291	1.001	age+sex	0.1121	True	True
GSE57338	-0.07874	0.008825	-0.26	1.087e-05	0.4211	0.001169	1.001	age+sex	0.3942	True	True
GSE57338	-0.2798	2.57e-05	-0.04459	0.4437	-0.07196	0.576	1.001	age+sex	0.001892	True	True
GSE57338	0.5714	0.005565	-0.01664	0.7748	0.08667	0.5006	1.001	age+sex	-0.09859	True	True
GSE57338	0.005255	0.06098	-0.08776	0.1322	0.06464	0.6154	1.001	age+sex	0.1984	True	False
GSE57338	0.4179	0.1295	0.001015	0.9861	-0.07237	0.5738	1.001	age+sex	0.1015	True	False
GSE57338	0.3726	0.2585	-0.00257	0.9648	-0.06424	0.6176	1.001	age+sex	0.1095	True	False
GSE57338	0.2276	0.9923	-0.07729	0.1847	0.1723	0.1811	1.001	age+sex	0.9858	True	False
GSE57338	0.1226	0.3319	0.19	0.001203	-0.2919	0.02382	1.001	age+sex	3.591	False	False
TCGA_LIHC	1.406	1.254e-13	0.04466	0.3634	-0.2231	0.03192	1.007	age+sex	0.01137	True	True
TCGA_LIHC	1.35	8.846e-13	0.1056	0.03211	-0.2877	0.00574	1.007	age+sex	0.002185	True	True
TCGA_LIHC	1.289	6.862e-12	-0.05992	0.2229	-0.279	0.007384	1.007	age+sex	0.02893	True	True
TCGA_LIHC	1.268	4.622e-11	0.1162	0.01843	-0.2878	0.005738	1.007	age+sex	0.0006651	True	True
TCGA_LIHC	1.142	2.146e-09	-0.08447	0.08605	-0.4334	3.517e-05	1.007	age+sex	0.04177	True	True
TCGA_LIHC	1.079	4.48e-08	-0.08892	0.07081	-0.2584	0.01303	1.007	age+sex	0.04013	True	True
TCGA_LIHC	1.05	1.1e-07	-0.1147	0.01989	-0.2628	0.01159	1.007	age+sex	0.04396	True	True
TCGA_LIHC	1.016	2.058e-07	-0.004985	0.9192	-0.4248	4.993e-05	1.007	age+sex	0.04081	True	True
TCGA_LIHC	0.9175	5.777e-06	-0.0887	0.07151	-0.2646	0.01101	1.007	age+sex	0.05069	True	True
TCGA_LIHC	0.8652	1.648e-05	0.033	0.5018	-0.3364	0.001262	1.007	age+sex	0.03714	True	True

S37 Table (columns 2 of 3) (page 7 of 8)

Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	ci.hi.adj	p.adjusted	b.age	p.age	b.sex.female	p.sex	max.vif	covariates	attenuation	sign.kept	survives
TCGA_LIHC	0.8871	1.577e-05	0.1348	0.006302	-0.2554	0.01414	1.007	age+sex	-0.002746	True	True
TCGA_LIHC	0.8269	9.56e-05	-0.09601	0.05116	-0.3019	0.003764	1.007	age+sex	0.06502	True	True
TCGA_LIHC	-0.09485	0.05576	0.1074	0.02929	-0.4093	9.193e-05	1.007	age+sex	-0.131	True	False
TCGA_LIHC	0.3751	0.2896	0.08005	0.1037	-0.2708	0.009292	1.007	age+sex	0.09003	True	False
TCGA_LIHC	0.3223	0.4799	0.07741	0.1156	-0.2683	0.009951	1.007	age+sex	0.1336	True	False
TCGA_LIHC	1.624	1.482e-19	-0.1076	0.02892	-0.05498	0.596	1.007	age+sex	0.01255	True	True
TCGA_LIHC	1.456	2.593e-15	-0.02308	0.6385	-0.2162	0.03759	1.007	age+sex	0.01898	True	True
TCGA_LIHC	1.183	9.818e-10	0.05482	0.2647	-0.4232	5.311e-05	1.007	age+sex	0.02188	True	True
TCGA_LIHC	1.006	5.865e-07	0.07127	0.1473	-0.2961	0.004493	1.007	age+sex	0.01852	True	True
TCGA_LIHC	-0.3657	0.0003598	0.04686	0.3404	-0.3175	0.002324	1.007	age+sex	-0.06434	True	True
TCGA_LIHC	0.6843	0.002475	0.02275	0.6433	-0.3298	0.001569	1.007	age+sex	0.05869	True	True
TCGA_LIHC	1.931	2.512e-25	0.01445	0.7686	-0.005568	0.9572	1.007	age+sex	0.00124	True	True
TCGA_LIHC	1.634	1.784e-19	-0.1007	0.04093	-0.1301	0.21	1.007	age+sex	0.01644	True	True
TCGA_LIHC	1.564	2.075e-17	-0.1059	0.03148	0.06116	0.5554	1.007	age+sex	0.004529	True	True
TCGA_LIHC	1.269	3.629e-11	-0.08967	0.06847	-0.3104	0.002908	1.007	age+sex	0.03349	True	True
TCGA_LIHC	1.236	9.65e-11	0.02234	0.6492	-0.3657	0.0004635	1.007	age+sex	0.02406	True	True
TCGA_LIHC	1.173	1.697e-09	0.009146	0.8523	-0.3261	0.001771	1.007	age+sex	0.02743	True	True
TCGA_LIHC	0.9864	4.213e-07	-0.05789	0.239	-0.4302	4.015e-05	1.007	age+sex	0.05028	True	True
TCGA_LIHC	0.9551	1.313e-06	-0.01996	0.6846	-0.09641	0.3527	1.007	age+sex	0.01982	True	True
TCGA_LIHC	0.8784	3.153e-05	-0.02962	0.5466	0.2839	0.006418	1.007	age+sex	-0.05159	True	True
TCGA_LIHC	-0.3744	8.264e-05	-0.1358	0.00593	0.2224	0.03247	1.007	age+sex	-0.006793	True	True
TCGA_LIHC	0.7203	0.0004049	0.0182	0.711	-0.3907	0.0001867	1.007	age+sex	0.05734	True	True

S37 Table (columns 2 of 3) (page 8 of 8)

Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	ci.hi.adj	p.adjusted	b.age	p.age	b.sex.female	p.sex	max.vif	covariates	attenuation	sign.kept	survives
TCGA_LIHC	0.7241	0.001466	-0.106	0.03139	-0.6031	1.176e-08	1.007	age+sex	0.1117	True	True
TCGA_LIHC	-0.3484	0.000656	0.006058	0.9019	-0.07534	0.4676	1.007	age+sex	-0.01288	True	True
TCGA_LIHC	0.6758	0.002333	-0.05634	0.2518	-0.0202	0.8456	1.007	age+sex	0.02518	True	True
TCGA_LIHC	0.5943	0.02012	-0.1843	0.0001987	0.2395	0.0213	1.007	age+sex	-0.009058	True	True
TCGA_LIHC	-0.06698	0.1225	-0.003605	0.9415	0.06167	0.5521	1.007	age+sex	0.0284	True	False
TCGA_LIHC	0.401	0.2814	-0.106	0.03147	0.2155	0.03819	1.007	age+sex	-0.05108	True	False
TCGA_LIHC	0.3273	0.4506	0.1071	0.02973	-0.4567	1.336e-05	1.007	age+sex	0.2095	True	False
TCGA_LIHC	0.1971	0.8278	-0.1131	0.02176	0.1908	0.06627	1.007	age+sex	0.07164	True	False
TCGA_LIHC	0.2215	0.9168	-0.1181	0.01655	0.2284	0.02805	1.007	age+sex	0.2752	True	False

S37 Table (columns 3 of 3) (page 1 of 8)

Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	p.fdr.unadj	p.fdr.adj
GSE116250	0.007871	0.009378
GSE116250	0.01892	0.0236
GSE116250	0.02502	0.02675
GSE116250	0.05423	0.06695
GSE116250	0.3038	0.3128
GSE116250	0.3259	0.3044
GSE116250	0.338	0.3524
GSE116250	0.4169	0.451
GSE116250	0.5419	0.5606
GSE116250	0.6463	0.6525
GSE116250	0.6463	0.5794
GSE116250	0.6463	0.6566
GSE116250	0.8613	0.7922
GSE116250	0.9328	0.9283
GSE116250	0.9328	0.9305
GSE116250	0.0073	0.008198
GSE116250	0.1043	0.1213
GSE116250	0.2287	0.2605
GSE116250	0.733	0.6723
GSE116250	0.9328	0.9305
GSE116250	0.002875	0.004011
GSE116250	0.005724	0.008198

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Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	p.fdr.unadj	p.fdr.adj
GSE116250	0.007871	0.009378
GSE116250	0.01035	0.01397
GSE116250	0.01892	0.0236
GSE116250	0.01892	0.02389
GSE116250	0.02502	0.02912
GSE116250	0.02767	0.03097
GSE116250	0.1999	0.1623
GSE116250	0.2287	0.2605
GSE116250	0.2287	0.2514
GSE116250	0.5472	0.5777
GSE116250	0.5786	0.5777
GSE116250	0.5807	0.5777
GSE116250	0.5902	0.5777
GSE116250	0.733	0.7091
GSE116250	0.8613	0.8222
GSE116250	0.9098	0.9146
GSE116250	0.9328	0.9305
GSE116250	0.9328	0.9305
GSE141910	7.536e-06	1.058e-05
GSE141910	0.0001083	0.0001491
GSE141910	0.00469	0.009552
GSE141910	0.005068	0.0103

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Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	p.fdr.unadj	p.fdr.adj
GSE141910	0.05502	0.07806
GSE141910	0.05801	0.09369
GSE141910	0.07759	0.1157
GSE141910	0.113	0.1371
GSE141910	0.3483	0.2349
GSE141910	0.387	0.4035
GSE141910	0.4773	0.3735
GSE141910	0.5873	0.4745
GSE141910	0.5919	0.4525
GSE141910	0.8556	0.8665
GSE141910	0.8892	0.878
GSE141910	2.783e-13	2.688e-12
GSE141910	3.944e-09	2.829e-08
GSE141910	6.583e-07	3.61e-06
GSE141910	5.005e-05	0.0001039
GSE141910	0.01522	0.03228
GSE141910	0.03157	0.05256
GSE141910	6.008e-22	3.164e-22
GSE141910	1.111e-20	5.754e-20
GSE141910	3.828e-16	4.197e-15
GSE141910	8.65e-09	8.202e-08
GSE141910	5.632e-06	2.155e-06

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Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	p.fdr.unadj	p.fdr.adj
GSE141910	0.0002866	0.0007074
GSE141910	0.0002866	0.001909
GSE141910	0.006205	0.02597
GSE141910	0.01052	0.02157
GSE141910	0.01178	0.02255
GSE141910	0.0125	0.02814
GSE141910	0.03775	0.02408
GSE141910	0.05598	0.1157
GSE141910	0.05801	0.1157
GSE141910	0.07031	0.1523
GSE141910	0.2084	0.1157
GSE141910	0.3163	0.1637
GSE141910	0.576	0.8317
GSE141910	0.7182	0.4695
GSE141910	0.8556	0.878
GSE57338	8.069e-14	2.779e-10
GSE57338	2.88e-11	2.044e-07
GSE57338	8.067e-08	0.0003809
GSE57338	4.981e-07	0.0009623
GSE57338	6.611e-07	0.001062
GSE57338	3.235e-06	0.0004065
GSE57338	3.971e-06	0.001316

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Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	p.fdr.unadj	p.fdr.adj
GSE57338	6.452e-06	0.0001141
GSE57338	6.435e-05	0.0295
GSE57338	0.000164	0.04923
GSE57338	0.002252	0.007016
GSE57338	0.0842	0.102
GSE57338	0.4196	0.09842
GSE57338	0.4743	0.1395
GSE57338	0.4888	0.1522
GSE57338	1.741e-12	2.575e-08
GSE57338	1.316e-06	0.0004402
GSE57338	9.059e-05	0.0001141
GSE57338	0.0001952	0.004618
GSE57338	0.001765	0.002108
GSE57338	0.2445	0.4395
GSE57338	1.173e-38	2.691e-32
GSE57338	1.2e-21	1.247e-16
GSE57338	1.886e-21	1.247e-16
GSE57338	2.799e-21	1.907e-18
GSE57338	1.646e-18	5.102e-14
GSE57338	1.646e-18	5.857e-15
GSE57338	6.791e-16	1.01e-10
GSE57338	3.287e-12	8.883e-08

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Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	p.fdr.unadj	p.fdr.adj
GSE57338	1.667e-10	1.065e-07
GSE57338	8.067e-08	4.732e-07
GSE57338	1.886e-07	0.0004402
GSE57338	8.099e-07	6.798e-05
GSE57338	7.736e-06	0.01248
GSE57338	1.036e-05	7.023e-05
GSE57338	0.008786	0.008149
GSE57338	0.01557	0.07813
GSE57338	0.0842	0.1474
GSE57338	0.2022	0.2789
GSE57338	0.4888	0.9923
GSE57338	0.687	0.3489
TCGA_LIHC	3.721e-13	8.57e-13
TCGA_LIHC	3.769e-12	5.181e-12
TCGA_LIHC	7.297e-12	3.517e-11
TCGA_LIHC	1.4e-10	1.895e-10
TCGA_LIHC	1.206e-09	6.284e-09
TCGA_LIHC	3.014e-08	1.224e-07
TCGA_LIHC	6.625e-08	2.819e-07
TCGA_LIHC	1.362e-07	4.963e-07
TCGA_LIHC	3.262e-06	1.128e-05
TCGA_LIHC	1.337e-05	2.938e-05

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Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	p.fdr.unadj	p.fdr.adj
TCGA_LIHC	2.734e-05	2.938e-05
TCGA_LIHC	4.883e-05	0.0001507
TCGA_LIHC	0.1077	0.06725
TCGA_LIHC	0.2763	0.3209
TCGA_LIHC	0.4341	0.5045
TCGA_LIHC	5.545e-19	2.438e-18
TCGA_LIHC	5.178e-15	2.126e-14
TCGA_LIHC	1.206e-09	3.355e-09
TCGA_LIHC	7.034e-07	1.266e-06
TCGA_LIHC	0.001031	0.0005463
TCGA_LIHC	0.00166	0.003172
TCGA_LIHC	5.948e-24	1.03e-23
TCGA_LIHC	5.545e-19	2.438e-18
TCGA_LIHC	1.168e-16	2.127e-16
TCGA_LIHC	3.068e-11	1.653e-10
TCGA_LIHC	1.197e-10	3.597e-10
TCGA_LIHC	1.532e-09	5.351e-09
TCGA_LIHC	2.119e-07	9.596e-07
TCGA_LIHC	1.499e-06	2.692e-06
TCGA_LIHC	0.000114	5.387e-05
TCGA_LIHC	0.0001359	0.0001355
TCGA_LIHC	0.0002541	0.0005929

S37 Table (columns 3 of 3) (page 8 of 8)

Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	p.fdr.unadj	p.fdr.adj
TCGA_LIHC	0.0004836	0.002003
TCGA_LIHC	0.001031	0.0009274
TCGA_LIHC	0.002204	0.003086
TCGA_LIHC	0.02576	0.025
TCGA_LIHC	0.1295	0.1435
TCGA_LIHC	0.3362	0.3205
TCGA_LIHC	0.3647	0.4862
TCGA_LIHC	0.8343	0.8485
TCGA_LIHC	0.885	0.9168

S38 Table

Mirror recheck after age and sex adjustment. Cross-disease effect-size correlation and mirror count between TCGA-LIHC and each heart-failure cohort, computed on unadjusted and on age- and sex-adjusted effects under an identical mirror gate, for Tier A sets and for all scored sets. The Tier A mirror count is unchanged in every cohort; across all scored sets it is unchanged everywhere except GSE141910, where it falls from 15 to 12. The discovery cohort's correlation moves from rho = -0.550 to rho = -0.461.

hf.cohort	tier	n.sets	rho unadjusted	mirror unadjusted	rho.adjusted	mirror adjusted	gate
GSE116250	A	15	0.2107	1	0.275	1	opposite signs, |g| >= 0.2, CI...
GSE116250	ALL	40	0.2859	7	0.2938	7	opposite signs, |g| >= 0.2, CI...
GSE141910	A	15	-0.1464	1	-0.1214	1	opposite signs, |g| >= 0.2, CI...
GSE141910	ALL	41	-0.08589	15	-0.07265	12	opposite signs, |g| >= 0.2, CI...
GSE57338	A	15	-0.55	10	-0.4607	10	opposite signs, |g| >= 0.2, CI...
GSE57338	ALL	41	-0.4563	26	-0.3929	26	opposite signs, |g| >= 0.2, CI...

S39 Table (columns 1 of 2) (page 1 of 3)

Translation score and overall survival. For every HCC cohort with usable outcome annotation: the Cox hazard ratio per IQR of the score and as a median split, with 95% confidence interval, Cox and log-rank P-values, the proportional-hazards test, and BH FDR, before and after adjustment for tissue composition. Each row names the covariates used — hepatocyte (the tumour-purity proxy), proliferation, and immune content — and reports the covariate hazard ratios and a flag for whether the proportional-hazards assumption holds. Models are fitted only where at least ten events per covariate are available, and the sample count and event count are given for every row, so a cohort absent from a given covariate set can be identified as underpowered rather than omitted.

cohort	stratum	score	model	covariates	n.covariates	n	n.events	hr	ci.lo	ci.hi	p.cox
GSE141198	HBV	TCS	continuous_per_IQR	none	0	30	21	1.241	0.5807	2.65	0.5777
GSE141198	HCV	TCS	continuous_per_IQR	none	0	99	63	0.8908	0.6141	1.292	0.5425
GSE141198	NBNC	TCS	continuous_per_IQR	none	0	19	10	1.303	0.6893	2.464	0.4151
GSE141198		ME_HCC_weighted	continuous_per_IQR	none	0	148	94	0.8801	0.6557	1.181	0.3951
GSE141198		ME_HCC_weighted	median_split	none	0	148	94	0.9453	0.6301	1.418	0.786
GSE141198		ME_HF_weighted	continuous_per_IQR	none	0	148	94	0.8675	0.6406	1.175	0.3583
GSE141198		ME_HF_weighted	median_split	none	0	148	94	0.7681	0.5114	1.153	0.2034
GSE141198		TCS	continuous_per_IQR	none	0	148	94	1.038	0.7774	1.385	0.8014
GSE141198		TCS	continuous_per_IQR	proliferation	1	148	94	1.011	0.7551	1.353	0.943
GSE141198		TCS	continuous_per_IQR	purity	1	148	94	1.026	0.7655	1.376	0.8616
GSE141198		TCS	continuous_per_IQR	purity+proliferation+immune	3	148	94	1.016	0.7531	1.372	0.9155
GSE141198		TCS	median_split	none	0	148	94	1.156	0.7701	1.734	0.4849
GSE141198		TCS	median_split	proliferation	1	148	94	1.122	0.7433	1.695	0.5831
GSE141198		TCS	median_split	purity	1	148	94	1.148	0.7632	1.726	0.5084
GSE141198		TCS	median_split	purity+proliferation+immune	3	148	94	1.143	0.7476	1.747	0.5378
GSE14520_GPL3921		ME_HCC_weighted	continuous_per_IQR	none	0	221	85	0.9962	0.7681	1.292	0.9768
GSE14520_GPL3921		ME_HCC_weighted	median_split	none	0	221	85	0.995	0.6503	1.522	0.9816
GSE14520_GPL3921		ME_HF_weighted	continuous_per_IQR	none	0	221	85	1.351	0.993	1.838	0.05543
GSE14520_GPL3921		ME_HF_weighted	median_split	none	0	221	85	1.69	1.098	2.599	0.01698
GSE14520_GPL3921		TCS	continuous_per_IQR	none	0	221	85	1.548	1.174	2.041	0.001971
GSE14520_GPL3921		TCS	continuous_per_IQR	proliferation	1	221	85	1.465	1.094	1.963	0.01038
GSE14520_GPL3921		TCS	continuous_per_IQR	purity	1	221	85	1.519	1.145	2.017	0.003809

S39 Table (columns 1 of 2) (page 2 of 3)

Translation score and overall survival. For every HCC cohort with usable outcome annotation: the Cox hazard ratio per IQR of the score and as a median split, with 95% confidence interval, Cox and log-rank P-values, the proportional-hazards test, and BH FDR, before and after adjustment for tissue composition. Each row names the covariates used — hepatocyte (the tumour-purity proxy), proliferation, and immune content — and reports the covariate hazard ratios and a flag for whether the proportional-hazards assumption holds. Models are fitted only where at least ten events per covariate are available, and the sample count and event count are given for every row, so a cohort absent from a given covariate set can be identified as underpowered rather than omitted.

cohort	stratum	score	model	covariates	n.covariates	n	n.events	hr	ci.lo	ci.hi	p.cox
GSE14520_GPL3921		TCS	continuous_per_IQR	purity+proliferation+immune	3	221	85	1.431	1.062	1.928	0.01846
GSE14520_GPL3921		TCS	median_split	none	0	221	85	1.771	1.147	2.735	0.009941
GSE14520_GPL3921		TCS	median_split	proliferation	1	221	85	1.627	1.034	2.56	0.03514
GSE14520_GPL3921		TCS	median_split	purity	1	221	85	1.758	1.138	2.715	0.01095
GSE14520_GPL3921		TCS	median_split	purity+proliferation+immune	3	221	85	1.557	0.9872	2.456	0.05683
GSE76427		ME_HCC_weighted	continuous_per_IQR	none	0	115	23	1.193	0.6322	2.252	0.5857
GSE76427		ME_HCC_weighted	median_split	none	0	115	23	1.283	0.5529	2.975	0.5621
GSE76427		ME_HF_weighted	continuous_per_IQR	none	0	115	23	1.177	0.6345	2.184	0.6049
GSE76427		ME_HF_weighted	median_split	none	0	115	23	1.37	0.5985	3.137	0.456
GSE76427		TCS	continuous_per_IQR	none	0	115	23	1.353	0.761	2.405	0.3032
GSE76427		TCS	continuous_per_IQR	proliferation	1	115	23	1.366	0.7673	2.432	0.2891
GSE76427		TCS	continuous_per_IQR	purity	1	115	23	1.35	0.7619	2.391	0.304
GSE76427		TCS	median_split	none	0	115	23	2.025	0.8562	4.791	0.1082
GSE76427		TCS	median_split	proliferation	1	115	23	2.062	0.8673	4.903	0.1015
GSE76427		TCS	median_split	purity	1	115	23	2.073	0.8739	4.917	0.09809
TCGA_LIHC		ME_HCC_weighted	continuous_per_IQR	none	0	366	130	1.53	1.231	1.902	0.0001278
TCGA_LIHC		ME_HCC_weighted	median_split	none	0	366	130	1.818	1.276	2.59	0.0009433
TCGA_LIHC		ME_HF_weighted	continuous_per_IQR	none	0	366	130	1.422	1.158	1.747	0.0007956
TCGA_LIHC		ME_HF_weighted	median_split	none	0	366	130	1.734	1.222	2.461	0.002051
TCGA_LIHC		TCS	continuous_per_IQR	none	0	366	130	1.13	0.9011	1.417	0.2897
TCGA_LIHC		TCS	continuous_per_IQR	proliferation	1	366	130	1.127	0.8988	1.413	0.3007
TCGA_LIHC		TCS	continuous_per_IQR	purity	1	366	130	1.146	0.9153	1.434	0.2351

S39 Table (columns 1 of 2) (page 3 of 3)

Translation score and overall survival. For every HCC cohort with usable outcome annotation: the Cox hazard ratio per IQR of the score and as a median split, with 95% confidence interval, Cox and log-rank P-values, the proportional-hazards test, and BH FDR, before and after adjustment for tissue composition. Each row names the covariates used — hepatocyte (the tumour-purity proxy), proliferation, and immune content — and reports the covariate hazard ratios and a flag for whether the proportional-hazards assumption holds. Models are fitted only where at least ten events per covariate are available, and the sample count and event count are given for every row, so a cohort absent from a given covariate set can be identified as underpowered rather than omitted.

cohort	stratum	score	model	covariates	n.covariates	n	n.events	hr	ci.lo	ci.hi	p.cox
TCGA_LIHC		TCS	continuous_per_IQR	purity+proliferation+immune	3	366	130	1.131	0.9027	1.418	0.2841
TCGA_LIHC		TCS	median_split	none	0	366	130	1.376	0.9731	1.946	0.07095
TCGA_LIHC		TCS	median_split	proliferation	1	366	130	1.504	1.059	2.135	0.0225
TCGA_LIHC		TCS	median_split	purity	1	366	130	1.398	0.9874	1.979	0.05899
TCGA_LIHC		TCS	median_split	purity+proliferation+immune	3	366	130	1.533	1.079	2.178	0.017

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Translation score and overall survival. For every HCC cohort with usable outcome annotation: the Cox hazard ratio per IQR of the score and as a median split, with 95% confidence interval, Cox and log-rank P-values, the proportional-hazards test, and BH FDR, before and after adjustment for tissue composition. Each row names the covariates used — hepatocyte (the tumour-purity proxy), proliferation, and immune content — and reports the covariate hazard ratios and a flag for whether the proportional-hazards assumption holds. Models are fitted only where at least ten events per covariate are available, and the sample count and event count are given for every row, so a cohort absent from a given covariate set can be identified as underpowered rather than omitted.

cohort	p.logrank	p.ph.global	ph.ok	max.vif	hr.covariates	time.unit	p.fdr
GSE141198		0.166	True	1		days	0.8746
GSE141198		0.9022	True	1		days	0.8746
GSE141198		0.2061	True	1		days	0.8746
GSE141198		0.3492	True	1		days	0.8746
GSE141198	0.7875	0.2122	True	1		days	0.943
GSE141198		0.3955	True	1		days	0.8746
GSE141198	0.2015	0.9521	True	1		days	0.8746
GSE141198		0.1317	True	1		days	0.943
GSE141198		0.3329	True	1.012	proliferation 1.09	days	0.943
GSE141198		0.3133	True	1.008	hepatocyte 0.97	days	0.943
GSE141198		0.6645	True	1.15	hepatocyte 1.00; proliferation...	days	0.943
GSE141198	0.4843	0.4277	True	1		days	0.8746
GSE141198	0.4843	0.6612	True	1.005	proliferation 1.09	days	0.8746
GSE141198	0.4843	0.616	True	1.001	hepatocyte 0.97	days	0.8746
GSE141198	0.4843	0.8924	True	1.142	hepatocyte 1.00; proliferation...	days	0.8746
GSE14520_GPL3921		0.4742	True	1		months	0.9816
GSE14520_GPL3921	0.9816	0.5405	True	1		months	0.9816
GSE14520_GPL3921		0.02332	False	1		months	0.0682
GSE14520_GPL3921	0.01577	0.03754	False	1		months	0.03165
GSE14520_GPL3921		0.4857	True	1		months	0.02285
GSE14520_GPL3921		0.7262	True	1.13	proliferation 1.14	months	0.02628
GSE14520_GPL3921		0.7029	True	1.03	hepatocyte 0.95	months	0.02285

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Translation score and overall survival. For every HCC cohort with usable outcome annotation: the Cox hazard ratio per IQR of the score and as a median split, with 95% confidence interval, Cox and log-rank P-values, the proportional-hazards test, and BH FDR, before and after adjustment for tissue composition. Each row names the covariates used — hepatocyte (the tumour-purity proxy), proliferation, and immune content — and reports the covariate hazard ratios and a flag for whether the proportional-hazards assumption holds. Models are fitted only where at least ten events per covariate are available, and the sample count and event count are given for every row, so a cohort absent from a given covariate set can be identified as underpowered rather than omitted.

cohort	p.logrank	p.ph.global	ph.ok	max.vif	hr.covariates	time.unit	p.fdr
GSE14520_GPL3921		0.1974	True	1.151	hepatocyte 0.96; proliferation...	months	0.03165
GSE14520_GPL3921	0.008994	0.6426	True	1		months	0.02628
GSE14520_GPL3921	0.008994	0.8734	True	1.124	proliferation 1.17	months	0.05272
GSE14520_GPL3921	0.008994	0.7793	True	1.009	hepatocyte 0.91	months	0.02628
GSE14520_GPL3921	0.008994	0.1898	True	1.144	hepatocyte 0.91; proliferation...	months	0.0682
GSE76427		0.9217	True	1		years	0.6049
GSE76427	0.5652	0.4487	True	1		years	0.6049
GSE76427		0.9873	True	1		years	0.6049
GSE76427	0.4524	0.8531	True	1		years	0.6049
GSE76427		0.7987	True	1		years	0.5067
GSE76427		0.9041	True	1.024	proliferation 0.93	years	0.5067
GSE76427		0.2229	True	1.002	hepatocyte 1.11	years	0.5067
GSE76427	0.1011	0.4957	True	1		years	0.3606
GSE76427	0.1011	0.7342	True	1.014	proliferation 0.92	years	0.3606
GSE76427	0.1011	0.1764	True	1.013	hepatocyte 1.16	years	0.3606
TCGA_LIHC		0.01407	False	1		days	0.001533
TCGA_LIHC	0.0007972	0.05342	True	1		days	0.003773
TCGA_LIHC		0.0769	True	1		days	0.003773
TCGA_LIHC	0.00181	0.2332	True	1		days	0.006153
TCGA_LIHC		0.433	True	1		days	0.3007
TCGA_LIHC		0.000949	False	1.004	proliferation 1.47	days	0.3007
TCGA_LIHC		0.771	True	1	hepatocyte 0.86	days	0.3007

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Translation score and overall survival. For every HCC cohort with usable outcome annotation: the Cox hazard ratio per IQR of the score and as a median split, with 95% confidence interval, Cox and log-rank P-values, the proportional-hazards test, and BH FDR, before and after adjustment for tissue composition. Each row names the covariates used — hepatocyte (the tumour-purity proxy), proliferation, and immune content — and reports the covariate hazard ratios and a flag for whether the proportional-hazards assumption holds. Models are fitted only where at least ten events per covariate are available, and the sample count and event count are given for every row, so a cohort absent from a given covariate set can be identified as underpowered rather than omitted.

cohort	p.logrank	p.ph.global	ph.ok	max.vif	hr.covariates	time.unit	p.fdr
TCGA_LIHC		0.01151	False	1.082	hepatocyte 0.89; proliferation...	days	0.3007
TCGA_LIHC	0.06978	0.8949	True	1		days	0.1064
TCGA_LIHC	0.06978	0.00217	False	1	proliferation 1.50	days	0.04499
TCGA_LIHC	0.06978	0.9353	True	1	hepatocyte 0.86	days	0.1011
TCGA_LIHC	0.06978	0.02034	False	1.08	hepatocyte 0.89; proliferation...	days	0.0408

S39b Table

Survival annotation coverage. For every cohort in the pipeline, whether a survival field is deposited, how many samples join it, how many are dropped for a missing or non-positive time, and how many events remain, with the source of each annotation. The three heart-failure cohorts deposit no survival field and are therefore not analysed, and GSE14520_GPL571 carries annotation whose sample identifiers do not intersect that platform's cohort, so it is reported as not analysed rather than as covered.

cohort	disease	analysed	n.samples	n.joined	n.dropped time	n.analysed	n.events	time.unit	note	source
GSE57338	HF	False	313	0	0	0	0		no survival field deposited in...	none deposited
GSE141198	HCC	True	148	148	0	148	94	days		GSE141198_clinical.rds:...
TCGA_LIHC	HCC	True	421	371	5	366	130	days		TCGA_LIHC_se.rds colData:...
GSE116250	HF	False	64	0	0	0	0		no survival field deposited in...	none deposited
GSE14520_GPL3921	HCC	True	445	225	4	221	85	months		...
GSE14520_GPL571	HCC	False	41	0	0	0	0	months	annotation present but no...	...
GSE76427	HCC	True	167	167	52	115	23	years		GSE76427_series_matrix.txt.gz:...
GSE141910	HF	False	366	0	0	0	0		no survival field deposited in...	none deposited